EXHALE v1.10: User Manual

Scope. *This manual describes how to set up, run, and analyze models with* `EXHALE`*, the in-house extension of the ATES (ATmospheric EScape) code to trace metals, metal-line cooling, a runtime opacity dispatcher, the Roche tidal potential, and excited-hydrogen / Lyα physics. It documents every input parameter (with a brief physical meaning), every output file (column by column, with units), the analysis tools, and a set of worked examples. The equations and algorithms as the code applies them are in* `EXHALE_physics_and_algorithms.pdf`*; the detailed derivations live in the companion notes (the dated update logs* `Update_EXHALE_stage0.pdf` *to* `Update_EXHALE_stage3.pdf`*, whose Markdown sources are* `md/Update_EXHALE_stage1.md` *to* `stage3.md`*,* `transmission_spectrum.tex`*,* `cooling_formulas.tex` *(the analytic cooling coefficients),* `EXHALE_BC_and_IC.tex` *(the boundary- and initial-condition reference), and* `wind_ae_solver.tex`*, the Wind-AE steady-wind solver used by* `IC mode: windae`*); this document is the practical user guide.*

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1. What changed in v1.10

The chapters following this section were copied from the v1.00 manual. They describe the physical options and output schemas of that version. Their planet-specific timings, residuals, and convergence examples are historical v1.00 measurements; they are not renewed v1.10 results. This section gives the current v1.10 behavior where the versions differ. In particular, the older use of the word "converged" for a marching snapshot does not certify a stationary solution.

1.1 Two levels of convergence

Regional stabilization is the finite-duration test implemented for atomic global physical-time marching. It checks whether upper-atmosphere profiles have small variations while lower-atmosphere variations remain within the configured bounds. It ends a successful regional run without launching a Newton–Krylov solve. **Stationary convergence** requires the enabled fluid, chemical, radiative, and boundary equations to satisfy specified stationary residual and consistency tolerances. Regional stabilization alone does not imply a stationary root. The numerical method does not define either physical criterion.

Enable the regional test with `Regional convergence: True`. The main program accepts it only with `Run mode: phys` in an atomic calculation using global physical time steps, with Newton solving disabled. A local-time step, molecular chemistry, stationary restart, or post-processing-only run with this option stops at startup. The absence of the key leaves the regional monitor disabled. The reconstruction must be fixed from the first step: a PLM stage, a PLM–WENO3 ramp, or an update map is refused. If secondary ionization is enabled, it must also be active from the first step; the default immediate secondary-ionization setting meets this condition. The Huang and Salz settings under `examples/regional_convergence/` are fragments; merge one into a complete planetary `input.inp`, replacing duplicate keys.

Quantity checked	Upper region	Lower/transition regions
Window-mean change and span	0.01	0.10
Full value range	0.05	0.20
Mass and internal-energy inventory	0.01	0.05
Mean actual face mass flow	0.01	0.05

These are source defaults, not universal observational accuracy bounds. The three configured regions use their own limits. The inventory limit applies to both the change and the full range. Mass flow is checked through its window means and their span; its instantaneous range is recorded but is not a stop gate. All criteria must hold over four validation windows times a confirmation factor of two, for eight consecutive windows, with at least 40 accepted endpoints in each. Rejected time steps do not advance the confirmation clock. When a candidate fails, the test starts a new candidate with a new fixed reference; it does not combine favorable windows from different candidates.

The radial cuts, window duration, settling time, total time budget, thresholds, and profile cadence are editable with `Regional ...` input lines. The complete v1.10 set is:

```
Regional convergence: True
Regional lower radius: 1.3
```

```
Regional upper radius: 1.85
Regional window seconds: 87965.04804471003
Regional maximum seconds: 0
Regional settling seconds: 0
Regional validation windows: 4
Regional confirmation factor: 2
Regional minimum samples: 40
Regional lower mean limit: 0.10
Regional transition mean limit: 0.10
Regional upper mean limit: 0.01
Regional lower range limit: 0.20
Regional transition range limit: 0.20
Regional upper range limit: 0.05
Regional lower inventory limit: 0.05
Regional transition inventory limit: 0.05
Regional upper inventory limit: 0.01
Regional lower mass flow limit: 0.05
Regional transition mass flow limit: 0.05
Regional upper mass flow limit: 0.01
Regional profile interval: 40
```

This fragment shows the Huang radial cuts and window; it is not a complete input file. A common `Regional inventory limit` or `Regional mass flow limit` may replace the three corresponding regional limits. An explicit regional value takes precedence over the common value regardless of line order. Unknown `Regional ...` labels stop the run. If `Regional window seconds: 0`, the duration is twice the initial outward acoustic travel time. If `Regional maximum seconds: 0`, the budget is the settling time plus two complete confirmation intervals; zero does not mean unlimited time. The initial lower and upper cuts default to $1.2R_{\mathrm{p}}$ when no cuts are supplied, so an empty transition region is allowed. The cuts and duration should be stated when comparing calculations.

1.2 Progress records and plotting

The regional monitor writes three files:

- `output/regional_convergence.csv`: criterion history;
- `output/regional_convergence_status.txt`: effective settings and current outcome;
- `output/regional_profile_history.txt`: profiles with region labels.

The monitor evaluates every accepted step. Profiles and the collecting status are updated every 40 accepted steps by default, with initial and terminal profile records. The status file reports `convergence_level=regional_stabilization` only after a complete successful candidate and `stationary_root_claim=False` in that case. An incomplete run reports `convergence_level=none`. When regional monitoring is enabled but the specified criteria are not met by normal

finalization, the program exits with status 4; a separate startup or runtime failure can end earlier.
Run the plotter from the calculation directory after output appears:

```
python3 /path/to/EXHALE_v1.10/EXHALE_plots.py --regional --live 60
python3 /path/to/EXHALE_v1.10/EXHALE_plots.py --regional --save output/regional
```

The first invocation refreshes the display every 60 seconds. The second writes PDF figures. The regional view shows physical profiles and progress separately for the lower, transition, and upper regions. The plotter reads existing output; it does not advance a calculation. The old profile view is available with `--profiles`.

1.3 Stationary lower-boundary closure and current scope

The v1.10 stationary residual closes the lower ghost composition and thermodynamics at a fixed column. Certification of the stored state also checks the boundary result; an unclosed boundary cannot establish a certified stationary state. The relevant modules are `src/modules/states/stationary_boundary_chemistry.f90` and `reservoir_composition.f90`. This implementation does not by itself establish stationary convergence for the historical Yan, Huang, or Salz states. No such claim follows from the new regional test.
The older `Max steps` input and `EXHALE_MAXSTEPS` environment control set separate caps. Similarly, `EXHALE_MARCH_FLUX_TH` changes the marching stop, while `Flux spread tol` belongs to stationary acceptance. These values must not be substituted for one another. `EXHALE_MONITOR_STEPS` must be a positive integer in the current source. The copied v1.00 descriptions of experimental environment settings do not establish that they are documented user inputs in v1.10. The v1.10 tree lacks the v1.00 regression harness. Its `make check` target reports the missing harness as a failure. The README in this tree describes source-build requirements and the status of the paused calculation sequence. No model run was made to produce this documentation update.

2. Introduction

2.1 What the code does

`EXHALE` solves the time-dependent, spherically-symmetric (1-D) radiation–hydrodynamics of an irradiated planetary upper atmosphere and the hydrodynamic wind it drives ("atmospheric escape"). It integrates the Euler equations for mass, momentum, and energy of a single fluid,

$$\partial_t \rho + \tfrac{1}{r^2}\partial_r(r^2\rho v) = 0, \quad \partial_t(\rho v) + \tfrac{1}{r^2}\partial_r(r^2\rho v^2) + \partial_r p = -\rho\,\partial_r\Phi,$$

$$\partial_t E + \tfrac{1}{r^2}\partial_r\big(r^2 v(E+p)\big) = -\rho v\,\partial_r\Phi + \mathcal{H} - \mathcal{C},$$

through a time-dependent calculation and, when enabled, toward a stationary state, where Φ is the gravitational (optionally Roche) potential (§2.3), $\mathcal{H}$ is the XUV photoionization heating and $\mathcal{C}$ the radiative cooling. Coupled to the hydrodynamics, at every step it solves the *ionization equilibrium* of H, He, and the trace metals (C, N, O, Mg, Si, Ca, Na, K, S, Fe) with a Newton (MINPACK) solver, including photoionization (of the inner shells of the metals too, with their Auger electrons), collisional ionization, radiative + dielectronic recombination, and charge exchange. The

photoelectron energy budget carries secondary ionization (a fast photoelectron drives further H I / He I / H_2 ionizations instead of thermalizing fully) on the Shull & van Steenberg (1985) ionization budget with the Dalgarno, Yan & Liu (1999) heating fraction and molecular channels. The ionizing recombination radiation of H II, He II and He III is absorbed on the spot, so that the ground captures count as case A where the gas is thin and case B where it is thick, and the He I 584 Å resonance line is scattered by He I and partly converted into the $2\,^1S$ two-photon continuum (§9.). The He $\leftrightarrow$ H charge exchange ($He^+ + H$, its reverse, and the radiative $He^{2+} + H$ capture) acts in every helium-bearing ionization system. When the metastable He $2\,^3S$ triplet is included the code adds its explicit collisional-excitation cooling (the 10830 Å channel) and the He $2\,^3S + H$ / H_2 Penning ionization that removes the metastable. Optional terms, each off by default except where stated, add to the energy equation heat conduction with the conductivity of the mixture (`Conduction`), viscosity, and, when the elements diffuse, the enthalpy that the element, molecular-carrier and ionization-stage fluxes carry (`Interdiffusion enthalpy flux`, on by default); and the composition can be carried with the flow instead of being the local root of each cell: the molecular carriers (`Molecular carrier transport`, on by default with the oxygen chemistry), the ionization stages of H and He (`Ionization transport`) and the He $2\,^3S$ population (`He 2^3S transport`). The converged run yields the radial profiles of density, velocity, temperature, and the ionization state, plus the steady-state *mass-loss rate* $\dot{M}$. A post-processing pass adds advection to the ionization balance, and the Python tool `EXHALE_transit.py` turns a converged model into transit *transmission spectra* (He 10830, Lyα, Hα/Hβ, the metal resonance lines Mg II, Ca II, Na I, and the O I 1302/1304/1306 triplet).

2.2 The lower boundary, radii, and normalization

The computational domain starts at a **lower boundary** placed at the 1 μbar pressure level, which defines $r = 1$ and the base radius R_p (the input `Planet radius`). The base number density n_0 and temperature T_0 (the equilibrium temperature) are set there. That level is the lower *face* of the first cell, $r_{\rm edg}(0) = 1$, in every grid type: the first cell center lies half a cell above it and the two lower ghost cells lie below it (§3.6). Internally the code is non-dimensional: lengths in R_p, velocities in $v_0 = \sqrt{k_B T_0/\mu}$, densities in n_0, etc. *All output files are written back in cgs* (see §5. for the exact normalization of each column). The key dimensionless number is the *Jeans / escape parameter* $b_0 = GM_p\mu/(k_B T_0 R_p)$, the depth of the gravitational well in units of the thermal energy; small b_0 means an easily-escaping (or tidally unbound) atmosphere.

2.3 The gravitational potential

The potential Φ in the momentum and energy equations is selected by `Domain mode` (Table 3). The code writes it in the dimensionless units introduced above (lengths in R_p, potential in $v_0^2 = k_B T_0/\mu$), so it is fixed by just three numbers: the Jeans parameter $b_0 = GM_p\mu/(k_B T_0 R_p)$, the stellar-to-planet mass ratio $q = M_\star/M_p$, and the orbital separation $\tilde{a} = a/R_p$.

Spherical mode (`Domain mode: Spherical`). Only the planet's own gravity is kept, so the domain can extend far past the Hill radius:

$$\phi(r) = -\frac{b_0}{r}.$$

Roche mode (default). The full co-rotating Roche potential, evaluated along the star–planet (substellar) line $x = r$, adds the stellar tidal attraction and the centrifugal term of the synchronously-rotating frame:

$$\phi(r) = -\frac{b_0}{r} - \frac{b_0\, q}{\tilde{a} - r} - \frac{b_0\,(1+q)}{2\tilde{a}^3}\left(\frac{q\tilde{a}}{1+q} - r\right)^2,$$

where the second term is the star's gravity at distance $\tilde{a} - r$ and the third is the centrifugal potential about the center of mass at $x_{\rm cm} = q\tilde{a}/(1+q)$ with orbital frequency $\Omega^2 = G(M_p + M_\star)/a^3$ (synchronous rotation). The effective gravity used by the hydrodynamics is $g(r) = -\mathrm{d}\phi/\mathrm{d}r$, with

$$\frac{\mathrm{d}\phi}{\mathrm{d}r} = \frac{b_0}{r^2} - \frac{b_0\, q}{(\tilde{a} - r)^2} + \frac{b_0\,(1+q)}{\tilde{a}^3}\left(\frac{q\tilde{a}}{1+q} - r\right).$$

The inner Lagrange point L_1 is the root of $\mathrm{d}\phi/\mathrm{d}r = 0$, where planetary gravity, the stellar tide, and the centrifugal force cancel; in Roche mode the outer boundary is placed at this L_1 radius by default, beyond which gas is no longer bound to the planet.

Roche mode with an extended outer boundary (`Outer radius` in Roche mode). By default Roche mode truncates at L_1. Supplying an explicit `Outer radius [R_p]:` > 1 *without* `Domain mode: Spherical` now keeps the full tidal Roche potential above *and* pushes the outer boundary out to the requested radius (e.g. $10\,R_p$), reproducing the extended-domain treatment of Yan et al. (2022) / Huang et al. (2023), whose escape models retain the tidal force but integrate well past L_1. This differs from `Domain mode: Spherical`, which also extends the domain but *drops* the tidal and centrifugal terms (pure $-b_0/r$). Two caveats apply. First, beyond L_1 the Roche potential is past its maximum, so the net force is outward and the 1-D radial flow there is a spherical approximation to what is physically an L_1 funnel: the same approximation Yan/Huang make. Second, the cold hydrostatic IC stands on the running maximum of the potential, so past L_1 it holds the density at its value on the ridge (in practice the 10^{-8} floor) rather than letting it rise outward; `IC mode: auto` (or `transonic`) seeds a transonic wind through the interior sonic point and is the faster start. Third, where the outermost cell is supersonic its two ghosts are the steady isothermal wind through it (the supersonic Bernoulli root, $\rho v r^2$ constant), not the hydrostatic continuation, which past the ridge would rise without bound; a subsonic outermost cell keeps the hydrostatic continuation. The absorbing columns start at the outer face $r_{\rm edg}(N)$ in every domain; the outer ghosts carry no column (update log stage 3 section 138). The tidal singularity of the $-b_0\, q/(\tilde{a} - r)$ term sits at the star ($r = \tilde{a} \gg 10$), so a $10\,R_p$ domain is numerically safe.

Full 3-D form (transmission post-processing). The 3-D Roche-equipotential reconstruction in `EXHALE_transit.py` / `roche_recon.py` (§6.2) needs the potential off-axis as well. With the substellar direction $\hat{x}$ pointing toward the star and (y, z) the sky plane,

$$\phi(x,y,z) = -\frac{b_0}{\sqrt{x^2+y^2+z^2}} - \frac{b_0\, q}{\sqrt{(\tilde{a}-x)^2+y^2+z^2}} - \frac{b_0\,(1+q)}{2\tilde{a}^3}\left[\left(x_{\rm cm} - x\right)^2 + y^2\right],$$

which reduces to the substellar $\phi(r)$ above on the $+x$ axis. The centrifugal term involves only the rotation-plane coordinates (x, y), not z, so surfaces of constant ϕ are triaxial (most extended toward L_1, least along the spin axis z).

2.4 Workflow

A model run consists of:

1. Prepare `input.inp` (planet/star parameters, numerics, physics switches), by hand or via the GUI; optionally `metals.inp` (trace metal abundances) and `opacity.inp` (opacity model).
2. Build and run `./EXHALE.x`; it reads the input files from the current directory and writes profiles to `output/`.
3. Inspect with `EXHALE_plots.py` (live profiles) and the example notebook; compute transmission spectra with `EXHALE_transit.py`.

3. Installation and running

3.1 Requirements

A Fortran compiler (`gfortran` by default; Intel `ifort`/`ifx` supported), `make`, and (for the GUI and analysis) Python 3 with `tkinter`, `numpy`, `scipy`, `matplotlib`, and `astropy` (the last is used by `EXHALE_transit.py` for `astropy.convolution`).

3.2 Build

`EXHALE` uses an incremental `Makefile`; module dependencies are auto-generated, so only edited sources and their dependents recompile.

```
make                # build ./EXHALE.x with gfortran
make FC=ifort       # Intel ifort   (or FC=ifx for the LLVM compiler)
make wind_ae_ic     # standalone Wind-AE IC generator -> ./wind_ae_ic.x
make check          # regression against the stored goldens
make clean          # remove .o/.mod (keeps EXHALE.x)
make distclean      # wipe build/ and EXHALE.x
```

`make check` rebuilds, re-runs each regression case single-threaded (`OMP_NUM_THREADS=1`) and compares `output/Hydro_ioniz.txt` and `output/Ion_species.txt`, and the `_adv` pair where the case writes one, against the stored reference outputs, *bitwise* first and otherwise within `REGRESSION_REL_TOL` (default 10^{-3} relative), the verdict line saying which. The standing rule it encodes: a refactor with no intended physics change must be byte-identical, and a new physics option must default to off. The harness and its reference outputs live in `backup/regression/`; the target reports that it found no harness, without failing, in a checkout that does not carry it. `make wind_ae_ic` builds the out-of-process IC generator described in §3.10.

Thread count. The cell sweeps (ionization equilibrium, radiation field, rate coefficients, carrier operator) are OpenMP loops over cells, each cell solved from its own inputs, and the only reductions in them are integer counts, maxima and wall-clock times, none of which enters the state. A run is therefore meant to be the same function of its input at every thread count, and `src/tests/thread_count_identity` checks it: 200 marching steps of the metals-on regression case `wasp_full` at 1 and at 8 threads must write bitwise identical data lines. That suite found one data race, fixed on 2026-09-26: a variable of the metal sweep (the inner-shell Auger share of neutral photoionization) was shared

between threads, so every metals-on run at more than one thread between 2026-09-25 and that fix depended on the thread count (`md/metal_sweep_openmp_race_20260926.md`). The regression matrix still runs single-threaded, which is why it could not see that race. The line order of the `EXHALE_STAGE_CHANNELS` diagnostic (Table 1) is the one output that depends on the thread count.

3.3 Running

The driver script `run_EXHALE.sh` (1) launches a Python/Tk GUI that writes `input.inp`, (2) builds, and (3) runs `./EXHALE.x`:

```
chmod +x run_EXHALE.sh
./run_EXHALE.sh           # gfortran (GUI -> build -> run)
./run_EXHALE.sh --ifort   # Intel ifort   (--ifx for ifx)
```

The GUI writes the legacy core block only: it offers `Load from file`, `Power-law` and `Monochromatic` as spectrum types, and every optional key of Table 3 (`Spectrum type: Planck` and the two stellar quantities it needs among them) is added to the written file by hand or inherited by copying an `examples/` configuration.

You may equally skip the GUI: place a hand-written `input.inp` (and optional `metals.inp`/`opacity.inp`) in a run directory, ensure an `output/` subdirectory exists, and run `./EXHALE.x` there. Each modeled planet typically lives in its own subdirectory holding its `input.inp`, its `EXHALE.x`, and its `output*/` folders. Progress (iteration counter, the cell-centered mass-flux spread `du`, and the time-derivative norm `dtu`) is printed to the terminal, and every monitor step prints a `[stop]` line with the two numbers the marching stop reads (§3.4).

3.4 Convergence control

The marching stop (since 2026-10-09). The marching loop stops when the *numerical* face mass flux is flat *and* the state has stopped changing, both judged at a monitor step (every `N_resid` = 500 steps), out of the PLM stage of a two-stage run (defaults in `src/modules/init/parameters.f90`; the thresholds are parameters, not input keys):

Parameter	Default	Condition
`march_flux_spread_th`	4×10^{-4}	face mass-flux spread over $r \geq r_{\rm flux}$ below it
`march_change_th`	1×10^{-4}	AND the state change over one monitor window below it
`stall_tol`, `N_stall`	10^{-6}, 2000	stall: settled, face flux no longer falling, for `N_stall` steps
`count_max` (`Max steps:`)	10^6	hard iteration cap

The face spread is that of $r_{j+1/2}^2 F_\rho$, the Riemann mass flux the update itself uses, over the faces at $r \geq r_{\rm flux} = 1.2 R_p$ (the quantity the steady solvers gate on, §3.7). The state change is

$$\Delta_{\rm win} = \max_{j=1\ldots N} \left(\left| \frac{\rho_j}{\rho_j^{\rm win}} - 1 \right|, \left| \frac{E_j}{E_j^{\rm win}} - 1 \right|, \frac{|(\rho v)_j - (\rho v)_j^{\rm win}|}{\sqrt{(\rho E)_j^{\rm win}}} \right),$$

with the superscript "win" the state `N_resid` steps earlier. Every monitor step prints `[stop] step N face mass-flux spread ... (stop below 4.000E-04) state change over the window ... at cell`

`j (stop below 1.000E-04)`, and the final `-> stopped:` line names the condition that fired. A state that has settled ($\Delta_{\rm win} <$ `march_change_th`) while its face spread has stopped falling (less than 1% drop from one window to the next) for `N_stall` steps ends as "stall, target not met". The environment variables `EXHALE_MARCH_FLUX_TH`, `EXHALE_MARCH_CHANGE_TH` and `EXHALE_MONITOR_STEPS` replace the two thresholds and the window, for measurement only (`EXHALE_MARCH_FLUX_TH` ≤ 0 stops on the state change alone).

Why these values (update log stage 3 section 127): marching `wasp_full` on from a face spread of 4.0×10^{-4} to 1.9×10^{-5} (27 500 further steps) changed ρ by at most 1.45×10^{-3}, v by 1.24×10^{-3}, T by 3.3×10^{-4} and $\log_{10}\dot{M}$ by less than 0.005, below the 1% profile budget of the acceptance by observable precision (§3.7). A marching state stopped this way is a relaxation snapshot at that precision, not a solution of the stationary equations; a quantitative $\dot{M}$ comes from the Newton finish. The stop therefore makes no stationarity claim: the state is certified as written, and when the certification refuses it the file carries `certified=F cert_reason=no_stationary_claim` and the run exits 0 (update log stage 3 section 135; the exit statuses are listed in §3.7).

The cell-centered *du*, a diagnostic. *du* is the radial spread of the cell-centered mass flux $F = \rho v r^2$ over the window $r \geq$ `Escape radius` (cells $j_{\rm min} \ldots N$),

$$du = \frac{\max_j F_j - \min_j F_j}{|\,\overline{F}\,|}, \qquad \overline{F} = \frac{1}{N - j_{\rm min} + 1} \sum_{j=j_{\rm min}}^{N} F_j,$$

with F taken *signed* and the normalization floored at 10^{-30} (`mass_flux_spread` in `EXHALE_main.f90`). Until 2026-10-09 the run stopped at $du <$ `du_th`. That was a property of the cell averages, not of stationarity: a certified 500-cell `wasp_full` state with a face spread of 5.3×10^{-13} has $du = 3.6 \times 10^{-3}$ (1.7×10^{-3} on 1000 cells), set by the two cells next to the outer ghost, so the stop could never fire on it. *du* is now printed on the `[diag]` line and read only by the PLM $\rightarrow$ WENO3 switch (the first `du_th` value of `du_th [PLM,WENO3]:`, or a *du* plateau); `du_th` itself stops nothing, and the one-step `dtu` stop is retired too (`dtu` is still printed). An input written for the old rule runs unchanged and stops on the new one. The optional `Level tol:` gate and `Force start` (the first 1000 iterations forced) still apply.

Caveat: a flat mass flux is not an energy balance. The face spread measures only the mass row; it is blind to an operator-split energy imbalance. A WASP-121b reference stopped on the old *du* rule was low by $\sim$0.08 dex in $\dot{M}$ with an energy residual of $\sim$30. For production answers use the Newton finish (`Solver: Newton`, §3.7) or the residual-based stop (`Resid tol:`); details in `steady_solver_memo`.

Secondary ionization from step 0 (since 2026-10-09). With `Secondary_ionization: True` (the default) the photoelectron coupling is applied from the first step. Until 2026-10-09 the default was *staged*: the wind first relaxed without the coupling, because from a cold initial condition the coupling had been seen to amplify the base acoustic mode into a NaN runaway on high-gravity cases. That runaway did not recur (MEASURED: `wasp_full` and `wasp_full_newton` from their cold initial conditions with the coupling on from step 0, 20 000 steps, no NaN; the Newton finish at $\log_{10}\dot{M} = 13.29$ as with staging; update log stage 3 section 127), and the staged switch, keyed on a *du* stop, never fired in a march whose *du* sat at its floor. `Secondary_ionization: Staged` keeps the staged behavior: the coupling switches on, out of the PLM stage, the first time $\Delta_{\rm win} <$ `march_change_th` (the terminal prints `secondary`

`ionization activated at step N`), the stops are held for `N_stall` steps, and the run re-converges; a run that never settles never switches it on, and its answer carries no secondary ionization. `Immediate` is a synonym of `True`; `False` disables the coupling. The startup report (`EXHALE_setup.out`) says which applies.

3.5 Base grid resolution (`Base grid [dr,cells]`)

The `Mixed` grid stacks `cells` uniform cells of size `dr` (in R_p) on the lower boundary and fills the rest of the domain with $N-$`cells` geometrically stretched cells; the total cell count N is set by the optional key `Grid cells:` (default 500). The two numbers share one input line because they are not independent: their product is the radial extent of the uniform region ($0.01\,R_p$ with the default `2.0e-4 50`).

The quantity that matters physically is how many uniform cells span the base density scale height, H/dr with $H = kT/(\mu g)$. Where H/dr is of order a few cells the discretization supports a *stationary* $2\,dr$ entropy (contact) mode that nothing in the scheme removes: HLLC resolves a contact of zero speed exactly and so adds no dissipation to it, the gravity source is cell-local, the WENO3 pressure gradient sees only interface pressures, and there is no conduction or physical viscosity unless they are switched on. The measured dependence is an alternating $\ln\rho$ amplitude of $\sim 10^{-2}$ for $H/dr < 5$ and $\sim 10^{-4}$ for $H/dr > 100$. High-gravity planets have the least margin: at $T_{\rm eq}$ on the default grid HD 189733 b spans only $\sim$27 cells per H, against $\sim$60–100 for HD 209458 b, WASP-52 b and WASP-121 b, and a cool, shielded base shrinks H further. The value in effect is echoed in `EXHALE_setup.out` as `Basescale-heightresolution:H(T_eq)/dr`, with a warning below 10 cells.

Refine at fixed extent, i.e. divide `dr` and multiply `cells` by the same factor: `Base grid [dr,cells]: 5.0e-5 200` is the $4\times$ refinement. Two costs follow. The CFL time step is set by the smallest cell, so a $4\times$ finer base means $\sim 4\times$ more steps for the same physical time; and at fixed N the stretched region loses those cells and coarsens (for a $4.43\,R_p$ domain, the cell size at $1.5\,R_p$ grows from 6.1×10^{-3} to $1.3\times10^{-2}\,R_p$). Raising `Grid cells:` alongside the refinement keeps the upper stretch fixed instead: the pairs `1.0e-4 100`/500, `5.0e-5 200`/658, `2.5e-5 400`/916 and `1.25e-5 800`/1375 hold the stretch ratio at its default-grid value, at the price of proportionally more work in every sweep. In practice the $2\times$ step (`1.0e-4 100`) is the one to try first: it is where the mode goes, and $4\times$ is not reachable to convergence on HD 189733 b within the step cap (`Max steps:`) because its time step is $4\times$ smaller.

The exact default width, and inputs written before it was pinned. The default `dr` is the double 2.0×10^{-4} written in the source with a `d` exponent, so the value is the one a reader of the key would spell: an `input.inp` without the key and one stating `Base grid [dr,cells]: 2.0e-4 50` build the same grid to the bit, the list-directed read of `2.0e-4` into a `real*8` being that double. This matters because the grid is part of a stored state: a restart is refused when the file's cell centers differ from the run's by more than 10^{-10} relative. An input written before 2026-09-19 was run on the default-real literal, the width $1.9999999494757503\times10^{-4}$, whose centers differ from the present grid's by 3.8×10^{-9} to 6.6×10^{-9} relative, which exceeds that tolerance. Every such input standing beside stored results carries that width explicitly as `Base grid [dr,cells]: 1.9999999494757503e-4 50` (`src/utils/pin_base_grid.py`), and each preserved tree whose inputs were not edited has a `GRID_DEFAULT_NOTE.md` at its root stating the same line. The width a run used is recorded at round-trip precision in the resolved-configuration record, and the refusal itself prints it with where it came from, the key or the default. That $2\times$ setting is what `HD189733b/input.inp` carries; the other three

production planets run on the default.

3.6 The lower boundary: a characteristic condition at the face

The lower boundary is imposed at the first cell *face* $r_{\rm edg}(0)$, not in a ghost cell. At that face the characteristic speeds are $v-c$, v and $v+c$, and a wave enters the domain when its speed is positive; in EXHALE the radius increases outward, so gas entering the base moves outward and inflow is $v > 0$. For the subsonic inflow a wind rests on, two characteristics enter and one leaves, so the lower atmosphere states *two* conditions and the interior supplies *one*.

The two the lower atmosphere states are its **pressure** and its **specific entropy** at the base level $r = 1$, the entropy carried as the isentrope through (p_0, T_0) at the base composition. **The face is the level**: every grid type (`Uniform`, `Stretched`, `Mixed`) places its first face exactly at $r_{\rm edg}(0) = 1$ (`define_grid.f90` sets the first ghost center to the mirror of the first cell center, $r(0) = 2 - r(1)$, and stops the run if the face is not 1 or a ghost cell reaches $r \leq 0$), so the reservoir's (p, s) is the state of the face side it owns, with no transport along any isentrope, and the contact between the reservoir and the domain does not move with the cell width. The one the interior supplies is the outgoing acoustic invariant of the first cell,

$$p_b - \rho_i c_i v_b \;=\; p_i - \rho_i c_i v_i, \tag{1}$$

where the subscript i is cell 1 carried to the face and c_i uses cell 1's own $\gamma_{\rm eff}$. The closed-form invariant $v - 2c/(\gamma-1)$ is deliberately not used: the caloric equation of state gives every cell its own $\gamma_{\rm eff}(x_{\rm H_2}, T)$, and that invariant assumes one constant γ.

With `Well balanced: True` (boundary model v5, `face_at_level_discrete_equilibrium_v5`) cell 1 is carried to the face by its own constant-density hydrostatic equilibrium, $p_i = p_1 + \rho_1[\Phi(r_1) - \Phi(r_{\rm edg}(0))]$, $\rho_i = \rho_1$, the pressure the well-balanced reconstruction extrapolates cell 1 to at the same face, and the ghost cells continue that same discrete equilibrium below the level: each ghost's constant-density equilibrium meets the face state's pressure (ghost 0) or the ghost above it (ghost −1) at its upper face, at the reservoir-side temperature. The boundary and the reconstruction then share one discrete mechanical equilibrium (Käppeli & Mishra 2016, A&A 587, A94, section 2.1): a column at rest in it, meeting the reservoir's pressure at the level, carries no flux through the base face and no momentum defect in the base cells, to rounding, whatever its entropy stratification, and the face state carries the interior's own mass flux $\rho_1 v_1 r_1^2$ at the density of the face state. The reservoir still states the pressure and the entropy of the gas it supplies. What this removes is an $O[(\Delta r/H)^2]$ truncation difference between two second-order constructions: on a column built in the discrete equilibrium the previous (v4) boundary left a base flux 1.47×10^4 times a Mach 10^{-8} wind; with v5 the base face carries that wind to 2×10^{-3} (`src/tests/grid_and_gates`, `base_reservoir_equilibrium` and `low_mach_face_flux`). A seed mapped from another grid or model is not in this discrete equilibrium, and its lower layer is not at rest until it is projected onto it.

Without it (boundary model v4) cell 1 is carried to the face along its own hydrostatic isentrope, and the ghost cells are the volume averages of the reservoir's hydrostatic isentrope through the face state continued *below* the face, i.e. below the level. In both models the velocity carries a constant $\rho v r^2$. The ghosts are not copies of the face state: the two differ by half a cell of stratification, and that half cell is what the previous ghost-cell closure left as a resolution-independent momentum imbalance of 25% of the weight of cell 1.

The face at the level (boundary model v4, 2026-09-27). Until 2026-09-27 the level was a ghost cell *center* ($r(0) = 1$ on the `Mixed` grid, $r(1-N_g) = 1$ on the other two), the face lay half a cell (`Mixed`) or one and a half cells above it, and the reservoir was carried up to the face along its own hydrostatic isentrope. That put a slab of reservoir gas, whose thickness scaled with the cell width, between the level and the domain, and made the base pressure of the domain $p_0[1-\Delta r/(2H_{\rm res})]$, a first-order error in Δr: MEASURED, the molecular lower layer of the LHS 1140 b He/H = 1.8 state sat 2.2% / 1.15% low in density on 500 / 1000 cells (`md/ew_grid_order_20260926.md`, section 3). With the face at the level the pressure gap between the interior at the level and the reservoir of that state falls from 2.93 / 1.44 / 0.70% to 0.114 / 0.067 / 0.038% on 500 / 1000 / 2000 cells (`md/Update_EXHALE_stage3.md` section 29.11). Two first-order terms remain at the base and are recorded rather than changed: the Dirichlet He/H of the element-diffusion operator is imposed at the cell-1 *center*, half a cell above the level (`binary_element_diffusion.f90`), and the conduction bath at the ghost center $r(0)$ (`viscous_conduction.f90`; conduction is off by default).

States written before 2026-09-27 do not load. Every cell center moved (by half a cell on the `Mixed` grid, by one and a half on the other two), so the restart grid check (`Grid cells:`, Table 3) refuses every such state. One enters a run only as an *initialization seed*, mapped onto the new grid with

```
python3 src/utils/map_state_to_grid.py <old_dir> <target_grid_file> <out_dir> --ic
```

where `<target_grid_file>` is any `Hydro_ioniz*.txt` written on the target grid (a one-step cold run of the target `input.inp` gives one), and `--ic` names the outputs `Hydro_ioniz_IC.txt` / `Ion_species_IC.txt`. The mapped pair states `mode=init t_phys=0 certified=F cert_reason=mapped_seed` whatever the source said, because an interpolated state is neither the certified state nor the physical clock of the run that wrote it; the new run has to be solved again. A physical cell of the target is interpolated between the source's physical cells only: a target center between the base face and the source's first center (the first cell of a refined base block) takes the source's first cell, never a value interpolated against the lower ghost rows, which are the reservoir (boundary data that the loader does not read). Two further options of the tool change the composition on purpose: `--reservoir El/H <value>` multiplies every row of that element by one factor, keeping the shape of its profile, with the hydrogen-nucleus count of every row held: HeH^+ is multiplied with helium and the hydrogen nucleus it holds is taken back out of H I, H II, H_2, H_2^+ and H_3^+ in proportion, so every element ratio of the row other than the one named is left exactly as it was (left in, that hydrogen put C/H 1.5×10^{-6} off the value in the LHS 1140 b He/H ascent from 4×10^4 to 7×10^4). The carry is refused when a base row comes out more than 10^{-6} relative from the value, the `heh_dev_tol` within which `load_IC` accepts the loaded He/H (an oxygen carrier OH, H_2O or CO, which holds nuclei of two elements and is not multiplied, is what produces such a departure), or when the hydrogen-only species do not hold the hydrogen a HeH^+ carry takes, and `--uniform` (with `--reservoir`) sets the element to that ratio in every row instead, which is the seed of a run whose element does not separate (helium with `He_diffusion` off, for which `load_IC` refuses any row whose He/H is not the input's) made from a state solved with separation. The header of `src/utils/map_state_to_grid.py` states the checks the tool makes before it writes anything.

Where the flow reverses ($-c < v < 0$) only one characteristic enters, so the reservoir states the pressure alone and the entropy at the face is the interior's, advected out. That choice is discrete and has no width: a contact carrying different entropies on its two sides has direction-dependent traces, and a face density between them is an interpolation rather

than a law of thermal contact. The side that owns the face is the side the level's mass flux comes from, read from the wind window $r \geq r_{\mathrm{flux}}$ where that window carries one flux (its relative standard deviation at or below 2×10^{-3} and its face-mapped Mach number at or above 10^{-12}), and from the matched face velocity where it does not, a velocity below the rounding of the cancellation it is formed from counting as zero. At rest and at inflow the reservoir owns the level, the physical assumption being that the lower atmosphere below it is a radiatively controlled heat bath. Supersonic outflow prescribes nothing, and that transition alone carries a declared numerical width, a smoothstep of half width 10^{-8} in M_i about -1, where the number of outgoing characteristics changes. Supersonic *inflow* would need three reservoir conditions and a (p, s) reservoir has two, so it is reported as a configuration outside the model rather than repaired.

The composition of the two lower ghost cells is part of the boundary model as well, and is what distinguishes model v3 from v2 in the state file's `# boundary_model` line. The ghost composition is not prescribed but solved by the ionization sweep, and the sweep is a map from the composition it is entered at to the one it returns, so under v2 two entry compositions could return ghosts 4.7×10^{-3} to 5.5×10^{-2} apart in their trace ions, both accepted by every test the sweep applies, and the base continuity row of cell 1 read 8.335×10^{-8} in one and 2.923×10^{-9} in the other against the same tolerance. v3 states both halves of the fix: the ghost solve starts from the reservoir row at the first sweep and from the ghost the previous sweep returned after it, and the ghost it returns is held to a fixed point of that same map, the largest absolute change of any ghost species density in the cell's own units at most 10^{-11} and the ghost's heavy-particle plus electron count moving by at most 10^{-6} of itself. A boundary that cannot reach that in twenty applications stops the run by name with both moves. Nothing here is a key: the seed and what the fixed point reached are printed in the boundary report, and the diagnostics that replace the seed for a measurement are the `EXHALE_GHOST_*` variables of Table 1, all default off.

Nothing in this is selectable. There is no ghost velocity, pressure or temperature closure to choose, and the four keys that used to choose them (`Valve eps`, `Hydrostatic base`, `Base ghost temperature`, `Base velocity`) are refused at startup with a message naming the replacement, rather than ignored. The one key that remains is `Base BC:`, which states *where* the boundary is, not what it does.

3.7 The steady-state Newton solver (`Solver: Newton`)

The finite-volume steady state can be solved directly ($F(Y) = 0$ for the hydro state, with the ionization equilibrium eliminated locally inside the residual) by a preconditioned Jacobian-free Newton–Krylov method (banded LAPACK preconditioner, frozen WENO weights, diagonal scaling, non-monotone line search; the boundary's one branch of nonzero width, the supersonic-outflow transition, sits decades from any operating point, so nothing has to be smoothed for it). Each unknown is scaled by its own cell's state (ρ, $\rho(|v| + c_s)$, E), so that no cell sets another's scale; the momentum scale uses the fastest signal speed because ρv itself vanishes wherever the flow reverses. This reaches residual levels unattainable by marching at a small fraction of the cost: WASP-121b converges in 8 Newton iterations ($\sim$200 marching-steps of cost, vs. $>$35,000 insufficient marching steps), HD189733b in 33 s where marching spent $>$3 h without converging.

Two ways to use it:

- **Production**: add `Solver: Newton` to `input.inp`. The run marches (including the two-stage PLM→WENO3

when `Reconstruction scheme: PLM+WENO3` is set) until the *face* mass-flux spread over $r \geq r_{\rm flux}$ (§3.4) drops below the hand-off threshold `newton_flux_switch` (default 0.1; the marching stop is suspended), then the Newton solve finishes to `Resid tol` (default 10^{-5}), and the standard outputs and post-processing follow. The hand-off is armed the first time a monitor step sees the spread at or above the switch, fires out of the PLM stage at a monitor step below it, and also fires on a spread that has settled within 5 times the switch. The threshold is the optional third token of the key, `Solver: Newton <switch>` (e.g. `Solver: Newton 3e-2` hands off later); a bare `Solver: Newton` keeps 0.1. **Changed meaning (2026-10-10, update log stage 3 section 131):** the token was a threshold on the cell-centered *du* (`newton_du_switch`, default 10^{-2}); an input that states a value now hands off at that *face* spread. With the outer face at $2R_p$, *du* never fell below its floor of 1.04×10^{-2} on `wasp_full_newton`, and the *du* hand-off came only through its stall clause after 45 118 steps; MEASURED there, the face-spread switch at 0.2, 0.1 and 0.03 hands off at steps 11 500, 16 500 and 21 500, every JFNK solve reaching `info` = 0 at $\log_{10}\dot{M} = 13.28$ (0.1 at the L_1 face: step 7000, 13.29). `EXHALE_resolved.out` records the value as `newton_flux_switch`.

- **From an existing state** (recommended for re-converging old runs): put the state in `output/*_IC.txt`, set `Load IC? True` and `Resid tol: 1.0e-5`, and run with the environment hook

```
EXHALE_PTC=1 EXHALE_PTC_JFNK=1 EXHALE_PTC_DTAU0=1.0 ./EXHALE.x
```

  which solves from the loaded state, writes `output/`, and stops.

The warm start matters: Newton has a finite basin, so the hand-off must wait until the marching warm-up has flattened the wind. Handing off earlier (a larger `newton_flux_switch`) risks starting Newton outside it: 0.2 still converged on `wasp_full_newton` but needed more non-monotone accepts than 0.1, whereas a cold-IC start (no warm-up at all) stalls outright. Earlier revisions keyed the hand-off on the residual itself ($\|R\| < 5 \times 10^{-2}$), then on *du*; the residual remains the *target* of the JFNK finish. Convergence is linear (frozen-weight refresh and the chemistry noise floor). Planets with a breathing base (HD189733b) used to stop at a practical floor $\|R\| \sim 3 \times 10^{-4}$; the 2026-08-11 line-search and base-energy fixes removed it, and a warm re-convergence of HD189733b now reaches $\|R\| = 1.2 \times 10^{-5}$ in 11 outer iterations (re-measured 2026-08-15). The Newton finish is also *safe to fail*: the solver tracks the best iterate seen, aborts once the line search has failed to find any acceptable step in 12 consecutive outer iterations, and on failure the run prints `JFNK failed`, keeps the best iterate, and falls back to plain time-marching with the marching stop re-armed: an unconverged Newton state is never accepted as the final answer. The associated diagnostic hooks (`EXHALE_RESIDUAL`, `EXHALE_JAC_TEST`, `EXHALE_NEWTON_TEST`, ...) are environment variables, collected in §3.8 (Table 1). Full development record and validation: `steady_solver_memo.pdf`. Figure 1 shows the whole route end-to-end on WASP-121b (Roche): the monitored residual during the marching warm-up (left; note that the former *du*-based stop would have fired long before the residual is actually small) and the JFNK finish (right). The figure predates the face-spread hand-off of 2026-10-10 and the outer face at $2R_p$.

What certifies a converged state, and what the exit status says. Reaching `Resid tol` is not the same as being certified. At the marching stop, at the JFNK final and at a pseudo-transient declaration, one evaluator builds an inventory of every balance the configuration activates and returns, for each, a status (`not_applicable` / `evaluated` /

unavailable), the row measure (the maximum over physical cells of $|\text{res}|/\max(\text{scale}, \text{floor})$), the volume average, the worst cell and the scale it was divided by. The scale of a carrier row is the sum of the magnitudes of its own terms: the two face fluxes, the advective terms and the reaction production and loss as the row hands them out (the gross channels of H_2, of the oxygen carriers, of the He $2\,^3$S level, and of the proton and the two ionized helium stages, the latter formed from the gross rate of their rows), with an absolute floor of 10^{-20} of the element. The temperature, the sources and the lower ghost cells a certificate reads are rebuilt from the state being certified (a sweep that holds its composition), in the solve's passes, at the write and in a stationary evaluation alike, so a written state and its reload are certified by the same numbers. The tolerances are 3×10^{-12} for the mass row, 10^{-8} for momentum and 10^{-6} for energy, the level balances and the eliminated-species closures.

The species rows are gated by regime. An element row and a carrier row are judged against 10^{-5} for cells at $r \geq 1.20\,R_\mathrm{p}$ only; below that radius the row is measured and printed and does *not* gate. The gate is anchored on the element operator's own stated floor and sits a decade above the discretization error a smooth manufactured column shows at the production spacing (1.5×10^{-6}); the layer is excluded because its element fluxes are not conserved there at all (their radial spread over $r < 1.10$ is 9.8 times the median), so no threshold means anything against them yet. The band $1.10 \leq r < 1.20$ is likewise reported and does not gate. A state that passes with species rows present reports that it is certified *in the wind*, and its state file records `cert_reason=certified_in_wind`. The two radii are properties of the flow and are constants of `certification.f90`, not input keys.

Exit statuses.

Status	Meaning
0	the written state is certified, or the run made no stationary claim (the marching stop of §3.4, a step cap, a stall, a NaN, `Do only PP`): a state the certification refuses is written `certified=F cert_reason=no_stationary_claim`
2	the run DECLARED a stationary state (the residual gate met while marching, or a JFNK finish with `info` = 0; `certification_note_stationarity_claim`) and it is not certified. The state is written in full and the failing rows named; status 2 is a *flag*, callers continue with it. The run's last line is `(certification) the state was written in full; the run exits with status 2 because it is not certified.` A gfortran runtime error also exits 2 and lacks that line. The marching stop is not a claim: its face threshold 4×10^{-4} is twenty times the steady gate, so a marching state stopped there exits 0 (update log stage 3 section 135).
3	the stationary solve of a steady route (a stationary restart, the direct steady route) returned `info` $\neq 0$: the solver's last iterate is written with `certified=F cert_reason=stationary_solve_failed` (until 2026-10-10 it was written as a relaxation snapshot and exited 0, which hid the failure from scripts; update log stage 3 section 127.5)

The regression harness never tests the status. The closure driver (`element_flux_closure.py`), `LHS1140b/models/run_case.sh` and `ew_of_state.sh` read 0 and 2 as written states (2 carried as uncertified), stop on 3, and treat a 2 without the closing line as fatal.

Acceptance by the precision of the observable; the certification is a diagnostic. Since 2026-10-09 (user decision; `md/acceptance_by_observable_precision_20261009.md`) a state is accepted as a result for an observable O when

(A1) continuing the solve (more outer passes, a tighter tolerance, more marching) changes O by less than its budget, a third of its target precision (proposed: 0.3% in $\dot{M}$; 0.002 %Å in the width-matched He 10830 EW of LHS 1140 b against an observed σ of 0.0289 %Å); (A2) the mass and energy budgets close over the region that forms O (proposed: 1% of the heating); and (A3) T, v, n_e and the absorber density change by less than 1% there under the same continuation, shown as overlaid profiles. The certification above is still computed and reported for every state; a failure no longer refuses a result but is carried as "not certified, O stable to X", and a row far above its scale (a factor 100 is proposed) is first investigated as a possible defect.

The outer face and the Mach number reported with $\dot{M}$. Every run prints, with its $\dot{M}$, `---> Mach number of the outermost cell = M (r = ... R_p)` (also in `EXHALE_setup.out`), and below $M = 1.1$ a note: at a subsonic outer face one acoustic characteristic enters the domain, the outer ghost condition (a hydrostatic continuation with the velocity copied, `Apply_BC.f90`) supplies it, and it can select $\dot{M}$. For a quantitative $\dot{M}$ of a transonic wind put the outer face past the sonic point with `Outer radius [R_p]:` (§2.3). MEASURED on `wasp_full_newton` (`md/outer_boundary_subsonic_review_20261009.md` section 5, 500 cells, JFNK `info` = 0 in each, with the binary of before update log stage 3 section 138, whose columns still counted the outer ghosts): the default L_1 face ($1.574\,R_p$, Mach 0.950) gives $\log_{10}\dot{M} = 13.29$, an outer face at $2\,R_p$ (Mach 1.50) 13.28 and at $3\,R_p$ (Mach 2.88) 13.27; ρ and v of the L_1 run differ from the $2\,R_p$ run by up to 15% and 17% near $1.31\,R_p$, five times the grid effect between the two supersonic domains. The `wasp_*` regression cases run at $2\,R_p$ for this reason (update log stage 3 section 127.6): `wasp_full` and `wasp_he23off` as 15 000-step snapshots, `wasp_full_newton` uncapped; `wasp_full_converged`, outside the default matrix and run by name, marches `wasp_full` uncapped to the marching stop (about 74 000 steps from the cold initial condition). `iontrans_metals`, a stationary restart of a stored L_1 state, keeps its subsonic L_1 face.

A subsonic outer face: what the $\dot{M}$ means (decision 2026-10-12). The code has no other treatment of a subsonic outer face. The escape models it is compared with put the outer face past the sonic point and state no incoming condition (Salz et al. 2016, sect. 3.2; Murray-Clay et al. 2009, sect. 2.2; Caldiroli et al. 2021, sect. 5); a relaxed characteristic condition (Poinsot & Lele 1992) would make $\dot{M}$ conditional on an exterior pressure none of them states, and is not implemented. Below Mach 1.1 at the outermost cell the run log therefore prints `NOT A QUANTITATIVE MDOT` under the Mach number, and `EXHALE_setup.out` carries the same line: that $\dot{M}$ is set in part by the outer ghost condition (a gas at rest at the p/ρ of the outermost cell) and is not a result to quote. Put the outer face past the sonic point instead (`md/outer_boundary_subsonic_review_20261009.md` section 6 lists the stored states of record that end subsonic).

Supersonic outer cells. Where the outermost cell flows out faster than its adiabatic sound speed, its two ghosts are the steady isothermal wind through it (§2.3), and where a cell is cold and highly supersonic ($\rho e < 10^{-2}\,\mathcal{E}$, about $M \gtrsim 13$) and expanding, with the constant γ of an atomic gas, its internal energy is taken from the adiabat p/ρ^{γ} advected with the gas rather than from the small difference $\mathcal{E} - \rho v^2/2$, whose truncation error otherwise drives the temperature to the floor (the dual-energy method of Ryu et al. 1993 and Bryan et al. 1995; update log stage 3 section 139). The run log prints the count after the run, `internal energy from the adiabat (cold supersonic expanding cells), cell-steps: n`; zero means the method never acted.

Transported species: the outer passes of the stationary route. A species the run moves by a transport step instead of a Newton row (the element fractions of `He_diffusion` always, the molecular carriers unless `Coupled carrier solve: True`) has a balance no hydrodynamic solve touches, so the stationary route alternates: solve the wind at fixed transported composition, relax the composition to its own steady state in that wind, solve again (`steady_wind_with_element_diffusion`, `EXHALE_main.f90`). Each such round is an *outer pass*; the cap is 20 passes (`EXHALE_OUTER_PASSES` overrides it), and a run with no transported species has one pass, the wind solve alone. The element update is under-relaxed by a factor ω that starts at 0.5 (`EXHALE_DIFF_OMEGA`) with floor 0.125, and the carrier update is limited by a *movement bound*, the largest relative change of one cell's particle count $n_{\rm tot} + n_e$ a pass may make, which starts at 10^{-2} (`EXHALE_CARRIER_TRUST`) with floor 10^{-3} and ceiling 0.5. The iteration ends when the joint certification passes, when a row is not finite, when the element update is refused, when three consecutive passes make no progress, when the pass budget ends, when no solve has converged (the first solve is repeated once from its own iterate before that is concluded), or when a failed update cannot be shortened.

A failed hydrodynamic solve hands on nothing (2026-10-09). Its returned state is an iterate, not a wind: the last converged wind is restored *together with its composition* (a wind is a solution for its composition, and an old wind is never relabeled as the solution of a newer one), the pass keeps its failure flag, and the next composition update is taken from the restored wind with a shorter step. The log line "the wind converged on pass k is restored WITH ITS COMPOSITION" marks such a pass; if the run ends after it, the state written is that restored wind, not accepted. Inside every stationary solve a trial, probe or Jacobian column whose conserved state is not a gas in some cell ($\rho \leq 0$ or $\rho e \leq 0$) is refused before anything is evaluated at it; the solve's refusal census reports them as "refused for a conserved state that is not a gas" (`docs/EXHALE_physics_and_algorithms.pdf`, the stationary-state chapter).

The progress control (revised 2026-09-27). A pass makes progress when the joint distance of the state falls, or, where that distance is carried by a row that is not hydrodynamic, when the composition's own residual falls. A pass whose carrier update ended on its movement bound, whose solve converged and whose joint distance rose by no more than a factor 1.1 is *not* judged by those two measures: the state moved by the whole bound, the relative measures fall together with their scales and cannot show it, and such a pass is neither counted toward the ending nor allowed to shorten a step. The two step lengths answer to the hydrodynamic solve that consumes the update: the movement bound doubles, up to 0.5, after such a pass; it halves when a bound-limited update raised the joint distance of a solved wind by more than 1.1, or when it preceded a failed solve (whether or not the update ended on the bound), or when a pass without progress moved the carriers short of the bound on a solved wind. ω halves on a pass without progress whose wind was solved and on a failed solve that consumed an element update, and doubles back toward its starting value after a converged solve on a pass with progress. `EXHALE_CARRIER_TRUST_HOLD` freezes the bound at a stated value for every pass. Where the helium settles against a slow wind the alternation can oscillate between a helium-retaining and a helium-depleted wind: the open-base-face LHS 1140 b He/H = 2.09 wind did so from $\omega = 1$ with the growing bound (refused at pass 37), and certified at pass 44 with $\omega = 0.5$ and the bound held at 0.04 (`md/Update_EXHALE_stage3.md` section 121). What the revision measured, and what it leaves open (the starting bound for transported ionization stages; the overshoot of the H_2 row at the outer cells of the molecular He/H = 0.083 continuation, which neither the old nor the new control certifies), is in `md/Update_EXHALE_stage3.md` section 29.7.

A restartable state after every pass (2026-09-28; published as generations since 2026-09-29). Every outer pass after

which the iteration goes on publishes the state the next pass starts from as one generation `output/pass_state/<state_id>/`, named by `output/pass_state/current`, so a run stopped from outside (a wall-clock limit, a kill) keeps its last completed pass; §5.4 describes the files and the restart from them.

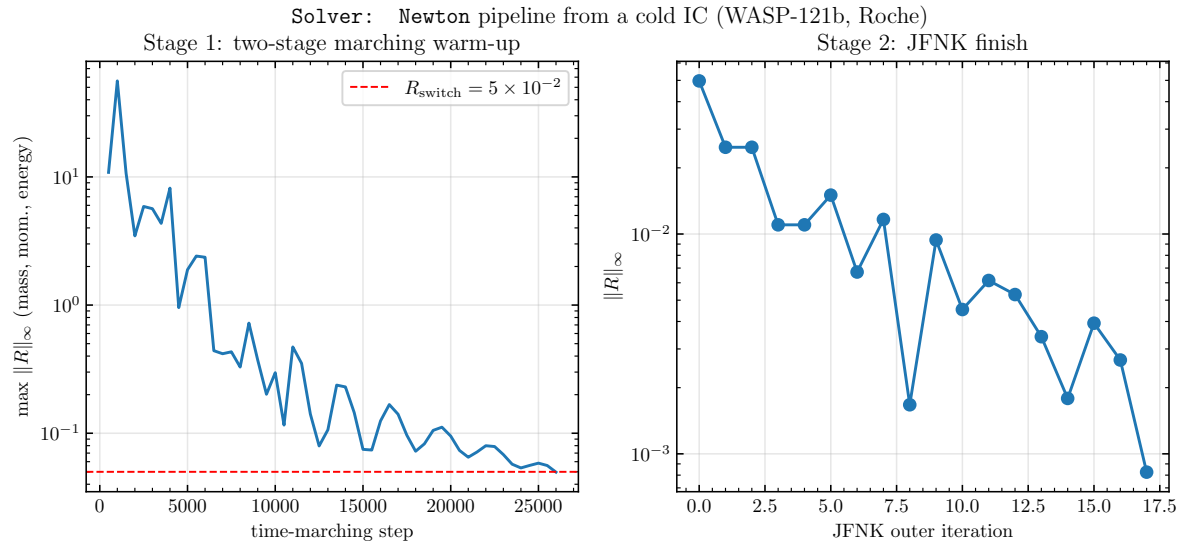


Figure 1: The `Solver: Newton` route from a cold IC (WASP-121b, Roche). Left: maximum steady residual max $\|R\|_\infty$ over the mass, momentum, and energy equations during the two-stage (`Reconstruction scheme: PLM+WENO3`) marching warm-up; the dashed line is the legacy residual hand-off threshold $R_{\mathrm{switch}} = 5 \times 10^{-2}$ used when this figure was produced (the current default keys the hand-off on $du < 10^{-2}$ instead, at a comparable point on the warm-up). Right: residual vs. JFNK outer iteration during the Newton finish, down to `Resid tol` (10^{-5}).

3.8 Environment-variable hooks (diagnostics and expert runs)

A small set of *environment variables* expose diagnostic and expert entry points without touching `input.inp` (Table 1). They are read once at startup; when unset they have strictly zero effect, so normal runs are unaffected. All of them are diagnostics or validation aids, apart from `EXHALE_ROOT`, which the lower-atmosphere pre-step needs: the others used in a production workflow are the `EXHALE_PTC` family, which implements the "Newton from an existing state" recipe of §3.7 (and `examples/04_newton_from_state/`), and `EXHALE_CARRIER_TRUST` / `EXHALE_DIFF_OMEGA`, which set the starting step lengths of the stationary outer passes and are how a run continued from a pass snapshot takes over the step lengths it had reached (§5.4), `EXHALE_PTC_SER`, which changes how the pseudo-transient solve accepts its steps, and the two continuation factors `EXHALE_CONDUCTION_SCALE` / `EXHALE_INTERDIFF_ENTH_SCALE`, which change the equations and are therefore written into the state identity (§4.1). The table is not exhaustive: the measurement keys a single investigation needed are documented at their code sites (a `get_environment_variable` call with a comment stating the key). Typical usage:

```
EXHALE_RESIDUAL=1 ./EXHALE.x                          # rank a "converged" state
EXHALE_PTC=1 EXHALE_PTC_JFNK=1 EXHALE_PTC_DTAU0=1.0 ./EXHALE.x   # re-converge it
```

Table 1: Environment-variable hooks. All are read once at startup and have zero effect when unset. The four diagnostic tests (top block), `EXHALE_PARSE_DUMP` and `EXHALE_PTC` runs stop after printing/writing; they do not march.

Variable	Effect
`EXHALE_DUMP_IC=1`	After `Load IC? True`, write the restored state to `output/Hydro_ioniz.txt` and `output/Ion_species.txt` and stop, *before* the first ionization-equilibrium solve re-equilibrates the species fractions (verifies a snapshot/restart round-trip).
`EXHALE_RESIDUAL=1`	Evaluate the steady residual R of the loaded state (WENO3, boundary conditions applied first), print the norms for each equation (mass, momentum, energy), and stop. The criterion-independent way to rank candidate "converged" states (a premature-dip state shows $\|R\|_E \sim O(10)$ despite $du < 10^{-3}$).
`EXHALE_NEWTON_TEST=1`	Self-test of the Newton infrastructure (`pack_U`/`unpack_U` round-trip and the frozen residual against the `EXHALE_RESIDUAL` values); prints and stops.
`EXHALE_JAC_TEST=1`	Verify the colored-FD banded Jacobian against a directional finite difference $Jr \simeq [F(Y+\epsilon r)-F(Y)]/\epsilon$; a correct band layout matches to $\sim 10^{-6}$, a wrong one is $O(1)$ off. Prints and stops.
`EXHALE_PTC=1`	Solve the *loaded* state directly with the steady solver (no marching), write `output/`, and stop. Requires `Load IC? True`; set `Resid tol: 1.0e-5` (§3.7). Default engine: banded PTC.
`EXHALE_PTC_JFNK=1`	With `EXHALE_PTC=1`: use the full JFNK solver (GMRES on the matrix-free J) instead of the banded PTC, the recommended engine.
`EXHALE_PTC_DTAU0=`$\langle v \rangle$	With `EXHALE_PTC=1`: override the initial pseudo-time step $\Delta\tau_0$ (default: the CFL Δt of the loaded state). A large value (e.g. `1.0`) probes the pure-Newton regime directly. The stationary restart (`Restart intent: stationary`) starts at $\Delta\tau_0 = 1$ instead, as the marching hand-off to the Newton finish does, and this key replaces that start too (the marching hand-off does not read it).
`EXHALE_PTC_NFIX=`$\langle n \rangle$	With `EXHALE_PTC=1`: anchor the first n physical cells (frozen-base experiment; solves the wind on top of a fixed base).
`EXHALE_DIAG_BASE=1`	Write `output/base_diag.txt` (r, n, v, T, heating, cooling of the first few cells over the first steps) to trace IC-startup transients at the base.
`EXHALE_FORCE_HYBRD1=1`	Force every ionization-equilibrium solve to use MINPACK `hybrd1`, skipping the analytic-Jacobian Newton and the Newton that otherwise continues from `hybrd1`'s result: lets the same binary produce both solvers' outputs for an A/B validation (`Update_EXHALE_solver.pdf` §6).
`EXHALE_IEQ_TOL=`$\langle v \rangle$	Replace the tolerance τ of the cell solve ($\sqrt{\epsilon_{\text{mach}}} = 1.5 \times 10^{-8}$): the solve stops when the Newton correction of every unknown is below $\tau\|x_i\| + 2.2 \times 10^{-16}$, and τ is also `hybrd1`'s step tolerance. A diagnostic.

continued on the next page

Variable	Effect
EXHALE_DIFFUSION_CHECK=1	Write the face-by-face elemental fluxes of every element-diffusion step to output/element_flux_profile.txt (§5.), and the composition profile of each direct-steady outer pass to output/diffusion_pass_profiles.txt. With a lower-atmosphere profile in use the flux file is written anyway.
EXHALE_DIFF_OMEGA=⟨v⟩	Set the starting under-relaxation factor ω of the element update of the stationary outer passes (default 0.5; the progress control halves it down to 0.125 and doubles it back up to this starting value, §3.7), so the undamped loop (1.0) can be compared with the damped one without a rebuild. A restart does not inherit the ω a run had reached: to continue from a pass snapshot (§5.4) at the value its # pass_snapshot line records, pass it here.
EXHALE_CARRIER_TRUST=⟨v⟩	Set the starting movement bound of the carrier update of the stationary outer passes, the largest relative change of one cell's particle count one pass may make (default 10^{-2}; the progress control moves it between 10^{-3} and 0.5, §3.7). A state that has to move far, such as a restart that turns Ionization transport on, can need a large start (MEASURED: the LHS 1140 b molecular He/H = 1.8 state with transported stages certified in 14 passes from 0.5 and not from 10^{-2}). Also the way to continue from a pass snapshot at the bound its # pass_snapshot line records, which a restart does not inherit.
EXHALE_OUTER_PASSES=⟨n⟩	The pass budget of the stationary outer iteration (default 20).
EXHALE_MAXSTEPS=⟨n⟩	Deterministic step cap: leave the marching loop after n steps and write the outputs. Separate from the Max steps: key (§3.4) and only ever earlier; it is what the regression harness uses to pin a relaxation-snapshot case.
EXHALE_PROFILE=1	Print a timing breakdown: the ioniz_eq share of the step every 500 steps, and the totals at the end of the run.
EXHALE_PARSE_DUMP=1	Write every value input_read derived from input.inp (plus any base.inp override) to parse_dump.txt and stop, before any grid/IC/output work. Feeds the parser regression corpus.
EXHALE_ROOT=⟨path⟩	Install directory for the lower-atmosphere pre-step generators (§4.7); required when Lower atmosphere: is set on a relocated install. Not a diagnostic.
EXHALE_JFNK_MAXIT=⟨n⟩	Cap a stationary solve at n outer iterations in place of the number its caller passed (the default is 3000 at all three call sites). It also caps the *run*: under this variable the run takes its FIRST stationary solve and refuses later entries by name, so that a bounded measurement is one solve of one state rather than the sum of several from different states. Nothing about the step depends on it.
EXHALE_INVENTORY_REPORT=1	Print the elemental inventory of every cell (what the carriers hold, what is locked, what the reservoir specifies, what the closure supplies, and the budget each side spends) at the carrier operator's entry.

continued on the next page

Variable	Effect
EXHALE_ELEMENT_ASSERT=⟨1|2⟩	Verify the element census after each write-back and report (1) or stop on (2) a breach; off by default, and the only check that sees a write-back that creates nuclei.
Measurement options of the stationary solve. Each was implemented, measured against both species-row fixtures and left in the default the measurement supported; none changes a default run.	
EXHALE_ELEMENT_CONSTRAINT_ROWS=0	Limit a species trial by the five coordinate-box corners instead of the shared element constraint rows (the state before 2026-09-09; the corners let two carriers of one element each spend the whole budget).
EXHALE_ELEMENT_WRITE_BACK	The element-conserving write-back of each Newton iterate.
EXHALE_TR_RESTART_STALL=⟨n⟩	Restart the trust-region state after n outer iterations without improvement. Delivered disarmed: no reset measured helped either fixture.
EXHALE_KRYLOV_ON_THE_BALL=1	Steihaug–Toint truncation of the Krylov cycle at the trust-region radius. Much cheaper in products, worse on the carrier fixture; default off.
EXHALE_GM_REORTHO=1	Reorthogonalize the Arnoldi basis. The measured orthogonality loss is 10^{-13} to 3×10^{-12}, so it is not the limiter, and the option is worse on both fixtures; default off.
EXHALE_COLUMN_EQUIL=1	Equilibrate the columns of the model as well as of the factored band. It moves the trust-region coordinates and trials ascend; measured and not adopted.
EXHALE_CAUCHY_LEG_BY_IMAGE=1	Let the image-share test alone decide whether the approximate-gradient leg is dropped. Worse on both fixtures; default off.
EXHALE_CERT_ANCHOR=1	Run and print the five anchors behind the species-row certification tolerances (reproducibility, closure seed, state perturbation, derivative, manufactured column). The default certification report is byte-identical without it.
EXHALE_ELEM_DIAG	The diagnostic line the species-row step prints at every trial.
The lower boundary's composition. All default off; with none of them set the seed is the boundary model's own (§3.6) and not one number of the run moves.	
EXHALE_GHOST_COMPOSITION_SEED=⟨s⟩	Enter the two lower ghost cells into the composition sweep at a stated composition instead of the model's seed. `the_state_file_ghost_rows` takes the ghost rows the restart pair carries, which the loader otherwise drops; `the_previous_sweep_ghost` takes the ghost the previous sweep of this run returned; a path takes a file of rows in the format EXHALE_GHOST_SEED_WRITE writes. A run with it set is entered at a composition the boundary model does not state, so its result is a measurement and never a certificate, and the boundary report says so.
EXHALE_GHOST_SEED_WRITE=⟨path⟩	Write the lower ghost composition as it was installed, in the format the seed key reads back, so that a perturbed seed is made from a measured one rather than assembled by hand.

continued on the next page

Variable	Effect
`EXHALE_GHOST_RECORD=1`	Append one record per evaluation to `output/ghost_record.txt`: per lower ghost cell the thermodynamic row the boundary left there, the heavy-particle and electron counts of the returned composition, its molecular and ionization fractions, the charge and elemental budgets, the reaction residual the cell was accepted at, the residual of its imposed molecular partition and the pass the closure ended on; and, in the same record, the base cell's continuity row with the two face mass fluxes it is built from, their signed difference, the row scale and the row's rounding floor.
`EXHALE_INTERIOR_COMPOSITION_HELD=1`	Return the composition the sweep was handed on every cell above the two lower ghosts, and the solved composition on the ghosts themselves, so that a base row measured after the call carries the ghost's own refresh and nothing else. The heating and cooling returned are those of the composition the sweep solved on every cell, so a run with the key set is not a state anything may be certified on.
`EXHALE_BOUNDARY_TRACE=1`	Write the lower boundary at named points of an evaluation to `output/boundary_trace.txt`, so that the order in which the boundary and the composition are built can be read off a run instead of argued.
`EXHALE_CARRIER_TRUST_HOLD=`$\langle v \rangle$	Hold the bound on how far one outer pass may move the molecular carriers at v for every pass, taking every change the progress control would make to it (growth and shortening alike) out. It separates a relaxation the bound throttles, which decays at a rate proportional to the bound, from a mode the alternation cannot damp, which decays at the same rate whatever the bound.
`EXHALE_STAGE_CHANNELS=`$\langle$list$\rangle$	Up to eight grid cell indices, comma separated (`107,248`). Every evaluation of the ionization stage rows at one of those cells appends one line to `output/stage_channels.txt` carrying the gas branch, the cell, a call counter, the kernel's own arguments and the channels, so the rows can be re-evaluated afterwards at the very state the run gave them. A line holds the state entries 1–29 of both gas branches (temperature, densities, photoionization, recombination and collision coefficients; 26, 27 and 29 the He $\leftrightarrow$ H charge-exchange coefficients of $He^0 + H^+$, $He^+ + H^0$ and $He^{2+} + H^0$, 28 the $2\,^3S \rightarrow 1\,^1S$ de-excitation q_{31g}), entries 30–59 of the molecular network in a molecular branch, then the 35 gross channels of the H II, He II and He III rows; the file's header names every entry and channel (`ion_residual_core.f90`). The rows run inside the cell sweep, so the order of the lines is deterministic only with one thread.
`EXHALE_CONDUCTION_SCALE=`$\langle s \rangle$	The continuation factor $s \in [0, 1]$ on the heat conductivity (1 when unset). It changes the equations, so it is part of the state identity: the `cond` token of the `# options` line carries it (§4.1).
`EXHALE_INTERDIFF_ENTH_SCALE=`$\langle s \rangle$	The same for the interdiffusion enthalpy flux, carried by the `interdiff_enth` token.

continued on the next page

Variable	Effect
`EXHALE_IEQ_DUMP_CELL=`$\langle j \rangle$	Measurement only. Write every He 2^3S-branch composition solve of cell j to `ieq_cell_dump.txt` (temperature, densities, every rate coefficient, the start and returned vectors, the dimensional rows, the normalized residual and the scale each row is divided by, which is the chemical turnover of the row's element group at the returned state since 2026-10-07 and was an upper bound of the row's rates before, the forward-difference Jacobian) and the cell's photoionization rates at perturbed entry states to `ieq_cell_rate_probe.txt`. `EXHALE_IEQ_REPORT_CELL=`$\langle j \rangle$ prints the accepted state of cell j at every sweep; its second fraction is unknown 4 of the system solved and is named by it (He 2^3S in the atomic triplet system, the H_2 nuclei in the molecular one).
`EXHALE_IEQ_CLASS_RECORD=1`	Measurement only. After every composition sweep, rewrite `output/ieq_acceptance_classes.txt`: for each cell its acceptance class, its normalized reaction residual, and row by row the share of its element group's turnover each kinetic row carries and the row's imbalance over its own gross rate (so the file holds the last sweep of the run).
`EXHALE_RESID_SC_TRACE=`$\langle n \rangle$	Measurement only; stops the run. Trace the composition elimination of the steady residual for n passes from two seeds in lockstep. `EXHALE_RESID_SC_TRACE_SEED=shift|h2|partition` selects seed B: the composition shifted by five cells (`shift`, the default without a scale), the H_2 fraction scaled by 1+`EXHALE_RESID_SC_TRACE_SCALE` (`h2`), or the elemental inventory and the charge closure kept and a fraction of H I and He I moved into H II and He II (`partition`).
`EXHALE_JV_FRESH_WEIGHTS=1`	Measurement only. Take the products of the Newton model with the WENO3 weights recomputed instead of frozen.
`EXHALE_LINEAR_ROWS=1`	Measurement only. Print, row by row, the linear residual the Krylov solve left against the rows of F, and cell by cell over the band `EXHALE_LINEAR_ROWS_CELLS=`lo,hi (default 290,312) the energy row, the model's $F + J\,\delta Y$, the row found at $Y + \delta Y$ and difference quotients along the step.
`EXHALE_ADV_COLUMN_FLOOR_IT=`$\langle n \rangle$	Measurement only. Take n further Newton steps of the column energy solve past its step test, so that the residual reported is the rounding floor the margin of a_E is anchored on (§5.).
`EXHALE_ADV_COLUMN_EXCLUDE_UNROOTED=1`	Measurement only. Make the cells whose marching energy step found no root fixed values of the column energy solve (written `failed`) instead of unknowns of it, the set the column had until 2026-10-01.

continued on the next page

Variable	Effect
EXHALE_PTC_SER=⟨1|2⟩	Accept the steps of the pseudo-transient solve by the pseudo-time control instead of by merit descent (steady_newton.f90, ptc_ser_on; Kelley & Keyes 1998, SIAM J. Numer. Anal. 35, 508). A trial is accepted when it is admissible (positive, finite, a chemical root in every cell) and no cell's density or total energy moves by more than EXHALE_PTC_SER_MAX_CHANGE (default 0.3) of itself; otherwise λ is halved. 1: $\Delta\tau$ follows the residual reduction, $\Delta\tau \leftarrow \Delta\tau \min(2, \max(1/2, f_2/f_{2,\mathrm{try}}))$; 2: $\Delta\tau$ doubles on a full step; both multiply it by λ on a shortened one. The returns to the best iterate, the stagnation stop and the damped Gauss–Newton escape are not taken, and the solve hands back its *last* iterate unless the best one met the gates. Once $\|R\|$ falls below EXHALE_PTC_SER_RNORM (default 10^{-3}) the merit line search accepts the steps again (the terminal Newton phase). Unset: the merit line search throughout. The LHS 1140 b continuation chains run with 2 (md/Update_EXHALE_stage3.md sections 42 and 50).
EXHALE_CARRIER_ROW_TERMS=1	Write the carrier rows of the last assembly term by term to output/carrier_row_terms.txt: the diffusive and advective transport, production, loss, photolytic loss, net source, residual, floor, the row's own physical scale and the measure the certification divides by; the H_2 row reaction by reaction; both faces of every row separately. The production and loss columns are the gross channels of every row, the ionized stages and the He 2^3S level included (diffusive_photochemistry.f90).
EXHALE_ELEMENT_ROW_TERMS=1	Write the helium element row of every state the certification measures, term by term, to output/element_row_terms.txt (binary_element_diffusion.f90).
EXHALE_CONSERVATION_BUDGET=⟨n⟩	Export every term of the three finite-volume rows of every cell (face fluxes, geometry, potential, sources, conduction, viscous and enthalpy-flux terms) for the next n assemblies of the stationary residual, so that a reader sharing no arithmetic with the code can rebuild each cell's balance (conservation_budget.f90, whose header states the four momentum branches and the sign of the gravitational work).
EXHALE_EVAL_STATE_DUMP=1	Write the state and the composition-derived module state (particle and electron counts) at named points of the stationary evaluation and of the final certification to output/eval_state_dump.txt, for an in-memory comparison of the loaded and the written state.

3.9 Metal cooling on/off comparison

Metal-line cooling is active *iff* a metals.inp is present (§4.); remove or rename it for a metals-off run. Because the metal abundances are read at run time, the executable is built once and the two cases differ only by whether metals.inp

exists: no source edits or rebuild. To run the *same* planet both ways:

```
make                                        # build once
# metals ON (solar C/N/O from metals.inp):
printf 'CI 2.69e-4\nNI 6.76e-5\nOI 4.90e-4\n' > metals.inp
./EXHALE.x ; mv output output_metals_on
# metals OFF (no metals.inp):
rm -f metals.inp
./EXHALE.x ; mv output output_metals_off
```

The notebooks for each planet `metals_comparison*.ipynb` read the two folders and plot the temperature, velocity / mass-flux, H and metal ionization, and heating/cooling differences (cf. §8.).

3.10 Wind-AE warm-start initial conditions

A 1-D steady-state Parker-wind solver (a Fortran port of *Wind-AE*; Murray-Clay et al. 2009 / Broome et al. 2025) is included under `src/modules/wind_ae/` and can supply the initial condition for a hard-to-launch EXHALE run. There are two ways to use it; the underlying algorithm, ramp, and BC/IC are documented in `docs/wind_ae_solver.tex`.

In-process (recommended): `IC mode: windae`. Add the single line

```
IC mode: windae
```

to `input.inp` (with `Load IC? False`). On startup EXHALE maps the planet's parameters to Wind-AE, ramps a shipped seed solution to the planet, solves the steady wind, and writes the IC onto the EXHALE grid (`output/*_IC.txt`), which is then loaded automatically: the whole input → Wind-AE solve → IC → run sequence is one `./run_EXHALE.sh` (one `EXHALE.x`) invocation. This is a *warm start*: EXHALE re-solves the ionization (and any metals) from the first step, so the Wind-AE IC need not be exactly self-consistent. Always finish with `Solver: Newton` for a quantitative $\dot{M}$ (§3.7). The shipped seed and spectrum live in `inputdata/windae_seed.csv` and `inputdata/windae_spectrum.inp`. The ramp runs in two stages, and the second one is what makes far-from-seed planets reachable. It first tries the *static-BC* ramp, which moves the system parameters (flux, gravity, star) while holding the seed's base boundary conditions fixed: fast, and enough for a seed-adjacent hot Jupiter such as HD 209458 b. If that stalls, the seed is reloaded and the ramp is retried with the *self-consistent-BC* continuation, which re-converges the base $R_{\rm min}/T/\rho$, the sonic-point column density, and the bolometric/molecular layer turn-off as the parameters move (`wae_continuation.f90`). That is the default path, and with it the ramp converges the far-from-seed HD 189733 b as well as the seed-adjacent HD 209458 b (`examples/11_windae_ic` and `examples/12_windae_ic_hd209`). Only if *both* stages fail does the run stop with a message advising `IC mode: auto`. What the Wind-AE IC does not fix is EXHALE's own HD 189733 b base-breathing instability, which limits the warm start there whatever the IC source.

Because a generated IC is an analytic profile that the marching loop has not touched, its *du* can start far below every threshold and would once have tripped a false convergence stop within a few steps. Both the *du* stop and the JFNK hand-off now require *du* to have been seen at or above their own threshold first, so only a descending crossing counts (§3.4). Measured on `examples/12_windae_ic_hd209` (2026-08-15): *du* at step 2 is 1.43×10^{-4}, neither trigger fires,

and the two arm at steps 23 and 222 on the first ascending crossing. The false stop is structurally gone for this IC mode; an $\dot{M}$ read off a *du* stop still carries its usual path-dependent spread, so the Newton finish remains the prescription.

Standalone generator: `wind_ae_ic.x`. The same solver builds an IC out of process:

```
make wind_ae_ic     # builds ./wind_ae_ic.x (separate from EXHALE.x)
./wind_ae_ic.x <input.inp> <seed.csv> <spectrum.inp> \
               <IC_dump_grid> <outdir> [rhoscale]
```

It reads an EXHALE `input.inp`, ramps from the seed, and writes `<outdir>/{Hydro_ioniz,Ion_species}_IC.txt` on the grid given by an `IC_dump.txt` from a prior EXHALE run with the same domain settings. Run EXHALE on the result with `Load IC? True` + `Solver: Newton`. (The standalone tool is independent of the in-process path above; either can be used.)

4. Input files

4.1 `input.inp`: the main parameter file

`input.inp` is read *line by line in a fixed order* for the core block (Table 2), followed by an *optional keyword block* (Table 3) whose lines are matched by keyword and may be omitted (older files without them keep the default behavior). A few lines are *conditional*: the spectrum-property line depends on `Spectrum type`, and the energy-band / X-ray-luminosity lines depend on `Use only EUV?`.

Table 2: `input.inp` core block (fixed order). Conditional lines are marked (∗).

Label	Value	Meaning
`Planet name:`	string	Label for the run (cosmetic).
`Log10 lower boundary number density [cm^-3]:`	real ($\log_{10}$)	$\log_{10} n_0$, the H number density at the base. *Optional since 2026-09-03*: a `base.inp` carrying `p_base` states the base level itself and n_0 follows from it. Give one or the other; if both are given they must agree to 1% or startup refuses the pair.

Table 2, continued

Label	Value	Meaning
`Planet radius [R_J]:`	real (R_J)	Base radius R_p (the 1 μbar level); the grid radius r is in units of this. The unit is the IAU 2015 nominal Jupiter *equatorial* radius, $R_J = 7.1492 \times 10^9$ cm (`parameters.f90`); the Python tools take the same value from `RJ_CM` of `examples/exhale_io.py`.
`Planet mass [M_J]:`	real (M_J)	Planet mass M_p (sets gravity / the Jeans parameter b_0). The unit is the IAU 2015 nominal $M_J = 1.8982 \times 10^{30}$ g; the stellar mass below is in $M_\odot = 1.98842 \times 10^{33}$ g.
`Equilibrium temperature [K]:`	real (K)	Base temperature T_0.
`Orbital distance [AU]:`	real (AU)	Star–planet separation a (sets the XUV flux $\propto a^{-2}$ and the tidal/Roche potential).
`Escape radius [R_p]:`	real (R_p)	Inner radius of the constant-momentum convergence window; must lie *inside* the domain ($< r_{\max}$). For L1-truncated Roche runs use a value below the lobe.
`He/H number ratio:`	real	Helium abundance $n_{\rm He}/n_{\rm H}$; > 0 enables He chemistry.
`2D approximate method:`	see text	Day–night averaging of the XUV flux and $\dot{M}$: `Rate/2 + Mdot/2` (dayside, $\xi = \frac{1}{2}$), `Mdot/4` (full-sphere), `Mdot`, or `alpha` (then a trailing number sets the rate-correction a_τ).
`Parent star mass [M_sun]:`	real ($M_\odot$)	Stellar mass $M_\star$ (tidal potential, Roche-lobe size).
`Spectrum type:`	`Power-law` | `Planck` | `Load` | `Monochromatic`	The incident spectrum. One type builds *every* band of the photon grid, the sub-13.6 eV band and the Balmer continuum included (§4.5); no band is filled from a type the input did not select.

Table 2, continued

Label	Value	Meaning
(∗) `Power-law index: / Spectrum file: / Photon energy [eV]:`	real / path / real	Property line of the chosen type: the power-law slope E^{PLind}, the numerical SED file, or the monochromatic energy. `Planck` has no property line: it is built from `Stellar Teff [K]` and `Stellar radius [R_sun]` of the optional block, and startup refuses `Planck` without both of them. With metals or the He $2\,^3$S triplet, mind the required wavelength coverage: a `Loaded` table that does not reach the band an active absorber needs stops the run (§4.5).
`Use only EUV? True/False:`	logical	`True`: EUV band only. `False`: include X-rays (adds the band's third energy and an X-ray luminosity line).
(∗) `[E_low,E_mid,E_high] = [. . . ]`	reals (eV)	Band edges: two values (EUV only) or three (with X-rays).
(∗) `Log10 of X-ray luminosity [erg/s]:`	real ($\log_{10}$)	Only if X-rays are included; $\log_{10} L_X$.
`Log10 of EUV luminosity [erg/s]:`	real ($\log_{10}$)	$\log_{10} L_{\mathrm{EUV}}$ (with a, sets the incident XUV flux).
`Grid type:`	`Uniform` | `Stretched` | `Mixed`	Radial grid spacing. `Mixed` (fine near the base, stretched outward) is the usual choice. Every type puts the first cell face at the base level, $r_{\mathrm{edg}}(0) = 1$ (§3.6).

Table 2, continued

Label	Value	Meaning
`Numerical flux:`	`HLLC` | `ROE` | `LLF`	Riemann solver for the hydrodynamic fluxes (`Num_Fluxes.f90`). `HLLC` is the usual choice; `LLF` (local Lax–Friedrichs) is the most diffusive and is kept for solver-isolation tests. MEASURED 2026-09-17 (item L25): on a wind whose base is subsonic to $M \sim 10^{-7}$ the HLLC flux can fail to reach the stationary root on a 500-cell grid where it reaches it on 1000 cells, and `ROE` reaches it on 500; where both fluxes solve the same case they differ by –0.3 to –0.6 percent in $\dot{M}$ and by 6 to 8 percent of opposite sign in the first cell's T and ρ.
`Low Mach velocity jump:`	logical (def. `False`)	Scale the normal velocity jump of the `ROE` branch by min(Ma, 1) after Rieper (2011, J. Comput. Phys., 230, 5263); no effect under `HLLC` or `LLF`. Delivered off and NOT recommended on a near-hydrostatic column: the factor vanishes exactly where two neighboring velocities cancel in the Roe average, so it removes the damping of the odd-even velocity mode at the faces that mode lives on, and it was measured to prevent a solve that unmodified `ROE` completes. Rieper's premise, that the local and the global Mach number are comparable, is violated by eight decades on such a column.

Table 2, continued

Label	Value	Meaning
`Reconstruction scheme:`	`PLM` | `WENO3` | `PLM+WENO3`	Spatial reconstruction order and staging. `PLM` or `WENO3` runs a single stage; `PLM+WENO3` selects the two-stage mode (march in PLM to the first `du_th` value, then switch to WENO3). Either way the run stops on the face mass-flux spread and the state change (§3.4), not on a `du_th` value.
`Include He23S?:`	logical	Include the metastable He $2\,^3$S triplet (needed for the He 10830 line). *On by default*, and the only line of the core block that may be omitted: a file without it gets the triplet. `False` is the deliberate opt-out, used to compare against the atomic helium limit. Forced off when the gas carries no helium.
`Load IC?:`	logical	Restart from the previous run's `output/*_IC.txt` (warm start). What the restart *continues* is stated by `Restart intent:` below; the state files also carry the configuration they were produced under, and a restart under another composition, grid, constant set or option set is refused naming the field (`md/input_schema.md` appendix D.2), unless the differing options are named by `Restart option change:` below.

Table 2, continued

Label	Value	Meaning
`Do only PP:`	logical	Skip the time loop and only run the post-processing on existing output. It still takes one time step before it stops, so it is not an identity on the state it reads, it reconstructs with whatever `Reconstruction scheme:` states rather than with the scheme the solution was reached under, and it writes the stepped state back over `Hydro_ioniz.txt`. For the advection-corrected products of a *solved* state use `Restart intent: stationary evaluate` instead, which takes no step at all (item L9, 2026-09-17).
`Force start:`	logical	Force the first 1000 iterations regardless of the convergence test (useful when the IC is close to a fixed point).

The constants these units mean. Every physical constant has one definition, in `src/modules/init/parameters.f90`, and the Python tools take the same numbers from `examples/exhale_io.py`. The ones a user input is quoted in, or that set the scale of a reported number, are

$$R_J = 7.1492 \times 10^9 \text{ cm} \quad M_J = 1.8982 \times 10^{30} \text{ g} \quad M_\odot = 1.98842 \times 10^{33} \text{ g}$$
$$R_\odot = 6.957 \times 10^{10} \text{ cm} \quad \text{AU} = 1.495978707 \times 10^{13} \text{ cm} \quad G = 6.67430 \times 10^{-8} \text{ cgs}$$

R_J, M_J and $M_\odot$ are the IAU 2015 nominal values (Prša et al. 2016, AJ 152, 41, Table 1), R_J being the nominal *equatorial* radius, which is what a transit depth measures. The mass unit of the code is the hydrogen *atom*, $m_\mathrm{H} = 1.67353284 \times 10^{-24}$ g, and one helium nucleus weighs $m_\mathrm{He}/m_\mathrm{H} = 3.9715$: the helium-4 atom mass $6.6464790722 \times 10^{-24}$ g divided by m_H, not the integer 4, so the mass density, the mean molecular weight and the base composition are the helium-4 gas and not a 0.7% heavier one. The ionization thresholds are the NIST ASD values, one definition each and every cross section turning on exactly there: $e_\mathrm{th}(\mathrm{H\,I}) = 13.598434599$, $e_\mathrm{th}(\mathrm{H_2}) = 15.425927$, $e_\mathrm{th}(\mathrm{He\,I}) = 24.587389$, $e_\mathrm{th}(\mathrm{He\,II}) = 54.417765$ and $e_\mathrm{th}(\mathrm{He}\,2\,^3S) = 4.767775$ eV.

Table 3: `input.inp` optional keyword block (any line may be omitted; defaults restore the standard behavior).

Keyword	Value	Meaning
`Domain mode:`	`Spherical`	If present and `Spherical`: pure planetary potential $-b_0/r$, domain extended to `Outer radius`. Absent (default): full Roche (tidal) potential; outer boundary at the Hill/L1 radius unless `Outer radius` is given (§2.3).
`Outer radius [R_p]:`	real > 1 (R_p)	Outer boundary (e.g. 10–15 R_p). In `Spherical` mode it sets the pure-$-b_0/r$ domain; in Roche mode (no `Domain mode` line) it extends the domain past L_1 while *keeping* the tidal potential (Yan/Huang-style; pair with `IC mode: auto`). For a quantitative $\dot{M}$ of a transonic wind the outer face must lie past the sonic point: the run reports the Mach number of the outermost cell with every $\dot{M}$ (§3.7).
`Base grid [dr,cells]:`	real (R_p), int (def. `2.0e-4 50`)	Resolution of the uniform region of the `Mixed` grid: `cells` uniform cells of size `dr` stacked on the base, then $N -$ `cells` stretched cells out to $r_{\max}$. The product `dr` $\times$ `cells` is the extent of the uniform region ($0.01\,R_p$ by default), so refining the base at fixed extent means dividing `dr` and multiplying `cells` by the same factor (`5.0e-5 200` is the $4\times$ refinement). Requirement: `dr` must resolve the base density scale height $H = kT/(\mu g)$; below $\sim$5 cells per H the scheme carries an undamped stationary $2dr$ entropy mode at the base (§3.5). Ignored by `Grid type: Uniform` and `Stretched`. At fixed `Grid cells:` a finer base leaves fewer cells for the stretched region; raise the two together to hold the upper stretch (§3.5).

Table 3, continued

Keyword	Value	Meaning
`Grid cells:`	int ≥ 10 (def. 500)	Total number of computational cells N (ghost cells not counted). The default reproduces the value that used to be compiled in, so an `input.inp` without the key is unchanged. A restart (`Load IC? True`) must load an IC written on *this run's grid*: the row count is checked against $N + 2N_g$, and then the file's cell centers are compared with the ones `define_grid` built over the physical cells $1 \ldots N$. A relative difference above 10^{-10} anywhere stops the run, naming the cell and both radii. The reader does not interpolate, so a state written under a different `Grid type`, `Base grid` or `Outer radius`, or by a binary older than 2026-09-27 (whose first face was not the base level), has to be mapped onto this grid with `src/utils/map_state_to_grid.py --ic` before it is handed back, and then loads as a seed and not as a continuation (§3.6; `load_IC.f90 verify_restart_radii`).
`Outer shells [r_face,cells]:`	real (R_p), int (def. none)	Append `cells` cells beyond the outer face of the grid the keys above construct, the last of them ending on the face `r_face`, which must lie beyond $r_{\max}$; center spacings in geometric progression (`define_grid.f90 append_outer_shells`). The constructed cells, faces and volumes are unchanged bit for bit, so the key moves the outer boundary without moving the interior, where `Outer radius` rebuilds every cell. N becomes `Grid cells` + `cells`; both numbers are required and a missing or non-positive one stops the run.

Table 3, continued

Keyword	Value	Meaning
`Max steps:`	int (def. 10^6)	Hard cap on marching iterations (`count_max`); the run reports that it stopped without converging when the cap is reached. Lower it to bound an exploratory run, raise it for a high-gravity case whose base time step makes 10^6 steps too short (§3.4).
`Coronal cutoff width:`	real (def. 0.1)	Roll-off width w of the coronal-excitation guard below the 10^3 K CHIANTI fit floor: the coronal part of every metal-line coefficient is multiplied by $\exp(-x^2)$ with $x = (T_{\rm floor}/T - 1)/w$, so smaller w cuts the extrapolated coronal cooling off more sharply. Since the ground-term fine-structure floors were solved in statistical equilibrium (§9.) the results no longer depend on w: the base-cell balance temperature of all four paper planets is identical to every digit printed over $w = 0.02$–1.2, against a 500–650 K spread before. The default is kept at 0.1, and the guard is kept because extrapolating a fit below its fitted range is the wrong thing to do, not as a tuning knob (`md/coronal_cutoff_width.md`, section 7).

Table 3, continued

Keyword	Value	Meaning
`Base IR field:`	`True`/`False` (def. `False`)	Let the infrared coolants of the molecular layer see the thermal radiation of the atmosphere below the base instead of emitting into vacuum. The lower atmosphere is taken to be black at those wavelengths and to radiate $B_\nu(T_0)$ over the sky fraction $1-\sqrt{1-(R_\mathrm{p}/r)^2}$, and the cooling becomes the net rate, emission minus absorption of that field, so each band stops cooling at its own radiative-equilibrium temperature (576–642 K for the C I/O I fine-structure lines and 941 K for the H_3^+ bands, at $T_0 = 1140$ K and half-sky coverage). Applies to the eight ground-term fine-structure lines of C I, C II, N II, O I and to the H_3^+ infrared cooling; every other channel keeps the optically thin, no-incident-field limit. Off by default, and with no effect on an atomic run, where those channels carry no cooling (`md/lower_atmosphere_coupling.md`).

Table 3, continued

Keyword	Value	Meaning
`Molecular IR bands:`	`True/False` (def. `False`)	Add the infrared coolants an H_2 atmosphere carries below the H_2 →H front and the atomic channels do not: the H_2 quadrupole and magnetic dipole line spectrum (Roueff et al. 2019, A&A 630, A58) and the H_2O and CO vibration–rotation bands (HITEMP, through the correlated-k coefficients distributed with Photochem). Each is computed in LTE, in the optically thin limit, as the *net* exchange $4\pi n_X \int \sigma_\nu(T)[B_\nu(T) - W B_\nu(T_0)]\, d\nu$ with the same diluted field `Base IR field` supplies, so each stops cooling at its own radiative-equilibrium temperature: 889, 918 and 984 K for H_2O, CO and H_2 at T_0 = 1140 K and half-sky coverage. The closure keeps stimulated emission, so the net rate is identically zero for gas at T_0 with the whole sky black. H_2 needs only `Molecular chemistry`; H_2O and CO exist only with `Oxygen chemistry`, and the key is inert on species the run does not carry. Used together with `Base IR field`: on its own the new bands emit into vacuum, which deepens the collapse of the molecular layer instead of holding it, and `input_read` warns. Writes three net columns (`H2_IR`, `H2O_IR`, `CO_IR`, negative where the band heats) and the Planck-mean optical depths of the H_2O and CO columns to `output/Cooling_breakdown.txt`, and a column-integrated infrared block to `output/FUV_bands.txt`. Cross sections tabulated 50–2000 K and clamped outside it (`src/modules/lower_atmosphere/molecular_infrared_cooling.f90`, `cooling_data/molecular_infrared_bands.py`).

Table 3, continued

Keyword	Value	Meaning
`Molecular reaction heat:`	`True/False` (def. `True`)	Deposit the energy the *collisional* reactions of the H_2/He network release into the gas. **On by default; the key exists to turn it off.** In an atomic gas the ionization energy a photon spends comes back as the Lyman photon of a radiative recombination and leaves, so it is correctly absent from the heating. In a molecular gas it does not: the cycle $H_2 + h\nu \rightarrow H_2^+ + e$ (the photon pays $I(H_2)$ = 15.43 eV and the photoelectron keeps the rest, which the photoheating already deposits), H_2^++H_2 $\rightarrow H_3^+$+H (+1.70 eV), $H_3^+ + e \rightarrow$$H_2$+H (+9.25 eV, *dissociative*) returns to H_2 having turned one H_2 into H + H and left $I(H_2) - D_0(H_2)$ = 10.95 eV in the gas with no photon to carry it away. Measured at 81 percent of the total heating rate at $1.02\,r_{\rm base}$ on the converged He/H = 0.0793 hot Uranus. The heat is built from one table of species enthalpies, so a closed chemical cycle releases exactly zero; the photon-driven reactions, the radiative recombinations, the collisional ionizations, the Penning channels and the H/He charge exchange are excluded because their energy is already in the ledger (the charge exchange in its own channel, `heat_charge_exchange`). Needs `Molecular chemistry`, so an atomic run is unaffected to the bit. Writes the column `heat_mol_chem` to `output/Heating_breakdown.txt` (`src/modules/lower_atmosphere/molecular_reaction_heat.f90`, `Update_EXHALE_stage1` Sec. 141).

Table 3, continued

Keyword	Value	Meaning
`H2 double ionization:`	`chung80` | `yan_rho` | `off` (def. `chung80`)	Resolve $H_2 + h\nu \rightarrow H^+ + H^+ + 2e$ as a channel of its own, vertical threshold 51.4 eV (Yan, Sadeghpour & Dalgarno 1998), two protons an event where the single dissociative channel gives one. `chung80`: 20% of the H^+-producing events at and above 80 eV, the number Chung et al. (1993) state outright, ramped linearly from the threshold (the ramp is not in the source); `yan_rho`: built from the double-to-single ionization ratio of Yan et al. (1998). At 110 eV `chung80` gives 15% more double ionization than `yan_rho`. Running both is how that uncertainty is reported. `off` folds the channel back into the single dissociative one, which reproduces the code before the channel existed and is not a physical statement. Any other word stops the run (`src/modules/functions/h2_photo_channels.f90`).
`H2 neutral dissociation:`	`True`/`False` (def. `True`)	Resolve $H_2 + h\nu \rightarrow H + H$, the absorptions that make no ion, over the 33–41 eV window in which Chung et al. (1993) Table 1 measure a photoionization yield below unity (up to 7.4% of the absorptions at 37.5 eV); zero outside it, where their source assumes unit yield. `False` folds that share back into the ionizing channels, which reproduces the code before the channel existed.

Table 3, continued

Keyword	Value	Meaning
`Stellar LW flux [erg/cm2/s]:`	real (def. 0)	Band-integrated stellar flux in the H_2 Lyman–Werner bands (912–1201 Å, i.e. 10.3–13.6 eV) *at the planet's orbit*. Switches on H_2 photodissociation, $H_2 + h\nu \rightarrow H + H$, in the molecular network. The rate is the incident band photon fluence $F_{LW}/\overline{h\nu}$ contracted with a dissociation cross section $\sigma_{diss}(N_{H_2}, T, n_H)$ read from a level-resolved CLOUDY table (line self-shielding and the trapping of the fluorescent decay photons are both inside it) times $e^{-\tau}$ of the continuum of the same interval, and it is averaged over each cell between its two faces rather than taken at one of them (the cross section falls by four decades across the self-shielding transition). Plus 0.4 eV of heating per dissociation (Black & Dalgarno 1977). The band is a separate input because the photon grid need not carry it: the grid floor is the lowest ionization threshold of an *active* absorber (§4.5), so a molecular run with no metastable, no low-IP metal and no armed excited-H coupling has no grid point in 11.2–13.6 eV at all, and where the grid does reach that far, a power-law type gives only the XUV fit extrapolated downward rather than a stellar FUV spectrum. With `Spectrum type: Load` the key may be omitted: the band is then integrated from the spectrum file over 912–1201 Å (the exact integral of the piecewise-linear table, the flux interpolated at each clipped edge; one read of the file serves the LW, B3 and B4 bands; the file is at the planet already) and the setup report says so. A stated value always wins, and a stated 0 switches the band off even when the file could supply it (the comparison with a model that has no H_2 photodissociation). A file that does not reach both edges of the band stops the run and asks for the key: a positive field is never turned into zero silently. Without a spectrum file, 0 (default) = no photodissociation. The key is inert without `Molecular chemistry: True`. Writes `output/Lyman_Werner.txt`

Table 3, continued

Keyword	Value	Meaning
`Stellar FUV B1 flux [erg/cm2/s]:`	*retired*	Band `B1` was 1110–1201 Å, between the Lyman–Werner interval and Lyα. The H_2 Lyman and Werner lines pump on both sides of 1110 Å at the 700–3200 K of a planetary base, so a band edge there normalized the H_2 pumping per photon of a band narrower than the one the lines absorb from. On 2026-09-06 the band was merged into the Lyman–Werner interval, which is now 912–1201 Å, and this key was retired: a file that still carries it stops the run with a message asking for the two numbers added into `Stellar LW flux`.
`Stellar FUV B3 flux [erg/cm2/s]:`	real (def. 0)	The same over 1231–1450 Å (band `B3`, branching 0.89/0.11/0.00). Band `B2` is the Lyα line and comes from `Stellar Lya flux`. With `Spectrum type: Load` and `Oxygen chemistry: True`, a run that does not state this key gets the band integrated from the spectrum file over 1231–1450 Å (the exact piecewise-linear integral, as for `Stellar LW flux`; a file that does not reach the band stops the run), and the setup report says so; a stated value, zero included, always wins.
`Stellar FUV B4 flux [erg/cm2/s]:`	real (def. 0)	The same over 1451–2304 Å (band `B4`, branching 1.00/0.00/0.00), integrated from the spectrum file under the same rule when not stated. *Read the caveat in §4.7*: the H_2O cross section falls three decades across this band, so a single band average misrepresents it by a factor of several for a red spectrum.

Table 3, continued

Keyword	Value	Meaning
`Stellar Teff [K]:`	real (K)	Stellar $T_{\rm eff}$. Two consumers: it arms the excited-H model when both this *and* `Stellar radius` are > 0, and it is the temperature of the photosphere under `Spectrum type: Planck`, which is refused without both. Arming the excited-H model makes H(n=2) an absorber of the grid-floor rule, so the photon grid then reaches down to the Balmer head $e_{\rm th,HI}/4 = 3.400\,{\rm eV} = 3647$ Å (§4.5).
`Stellar radius [R_sun]:`	real ($M_\odot$-radius)	Stellar radius: sets the dilution $(R_\star/a)^2$ of the `Planck` field and of the Balmer continuum, and in `EXHALE_transit.py` it normalizes every transit depth and caps the impact-parameter integration at $R_\star/R_p$. Supply it for any run whose spectrum will be synthesized: absent, the transit script falls back to a built-in radius that need not match the system, and the depths scale accordingly.
`Jlya RT file:`	path	Read an external $\bar{J}_{{\rm Ly}\alpha}(r)$ profile to pump H(n=2) (sets `jlya_mode=1`).
`Jlya escape-prob:`	`True`	Compute $\bar{J}_{{\rm Ly}\alpha}$ in-line each step by escape-probability RT (`jlya_mode=2`).
`Stellar Lya flux [erg/cm2/s]:`	real	Incident stellar Lyα flux at the planet (the stellar source for `jlya_mode=2`).
`Lya stellar halfwidth [km/s]:`	real (km/s)	Stellar Lyα line half-width; controls how deep the trapped stellar beam penetrates (default 70).
`Lya stellar boost [-]:`	real	Bounded buildup cap for the trapped stellar Lyα beam (default 5).
`Lya absorbing bottom:`	logical (def. `False`)	Treat the lower boundary of the `jlya_mode=2` closure as a pure absorber (Huang et al. 2017: the H_2 layer beneath the base destroys Lyα through accidental resonances): a downward wing-escape channel $\beta_{\rm bot}$ on the depth to the bottom joins the upward and Sobolev channels, lowering $\bar{J}_{{\rm Ly}\alpha}$ (and n_2) in the bottom few scale heights only. No effect on `jlya_mode` 0/1.

Table 3, continued

Keyword	Value	Meaning
`Deexc heat:`	logical (def. `True`)	Collisional de-excitation of H(n=2) returns 10.2 eV to the electrons. It is not a double count of the H I collisional-excitation cooling, which is the one-way coronal sum over the CHIANTI 11.0.2 H I levels (the same $1s \rightarrow 2s, 2p$ rates the n=2 model uses) with no de-excitation term in it; most of the n=2 population is maintained by Lyα pumping, so the term is absorbed Lyα thermalized by a collision. Active only with the excited-H model; `False` reverts to the one-way ledger.
`Transonic IC:`	`True`	Use a transonic isothermal-wind initial condition (solved from the potential via the Bernoulli integral) instead of the hydrostatic IC; needed for deep-RLOF bases that do not launch from rest. Auto-enables `Force start`.
`Newton solver:`	logical (def. `True`)	Solve the ionization equilibrium with the analytic-Jacobian Newton method (MINPACK `hybrd1` takes over where it fails); `False` reverts to the legacy `hybrd1`. Default is faster ($\sim$25%) and reproduces `hybrd1` to $\sim 10^{-7}$.
`Brent solver:`	logical (def. `True`)	Solve the post-process energy equation with a bracketing (Brent) root-finder that selects the physical root; `False` reverts to legacy `hybrd1` + a 2$\times$-band reject.

Table 3, continued

Keyword	Value	Meaning
`du_th [PLM,WENO3]:`	two reals	Stage thresholds on the cell-centered *du* (e.g. `0.5 1.0e-3`). With `PLM+WENO3` (two-stage) EXHALE marches in PLM until $du <$ first value (or a PLM stall), then switches to WENO3; the second value only has to be smaller than the first (that selects the two-stage mode). **Since 2026-10-09 neither value stops the run**: the marching stops on the face mass-flux spread and the state change (§3.4). The `PLM+WENO3` mode replaces the manual PLM → stop → `Load IC`+WENO3 restart of the official ATES recipe.
`Reconstruction continuation:`	real $\Delta\lambda$ [, real τ] [, `fixed` | `adaptive`] (def. $\Delta\lambda = 0$, i.e. the one-step hand-off)	Walk the PLM → WENO3 hand-off of a two-stage run along the homotopy $R_\lambda = (1-\lambda)R_{\mathrm{PLM}} + \lambda R_{\mathrm{WENO3}}$ instead of changing the discrete operator in one step. $\Delta\lambda$ is the advance in λ per marching step (≤ 0 off; 1.0 reproduces the one-step switch, byte-identically). τ (def. 1.2) is the step control: λ advances while *du* stays within τ times its value at the start of the ramp, and a step that exceeds it halves $\Delta\lambda$ instead of advancing. `fixed` disables that control, `adaptive` is the default; the two optional fields are told apart by content, not position. Requires `Reconstruction scheme: PLM+WENO3` with a non-empty PLM stage, otherwise a warning is printed and the key is ignored. Once λ reaches 1 the continuation disarms and the run continues in pure WENO3, so the JFNK finish and every acceptance gate see the production operator.
`Stall [tol,N]:`	real, int	Stall-detector override (relative *du* change and window; defaults 10^{-6}, 2000).
`CFL:`	real (def. 0.6)	CFL-number override.
`Hot Parker IC:`	real (K)	Warm-seed initial condition: stable hydrostatic density with a warm T/ionization/velocity overlay.

Table 3, continued

Keyword	Value	Meaning
`IC mode:`	word	Initial-condition family: `cold` (default), `transonic`, `hot_parker` (synonyms for the two keys above), `auto`, which picks automatically from the cold sonic-point topology of the actual potential: an interior sonic point (deep-RLOF via the Roche L1 topology, or low-gravity boil-off, $r_c \simeq b_0/2c_0^2$ inside the domain) selects the transonic IC, otherwise the cold hydrostatic IC; or `windae`, which builds the IC in-process with the ported Wind-AE steady-wind solver (§3.10). No tunable threshold for `auto`; the choice and the escape parameter b_0 are logged. Explicit `Transonic IC`/`Hot Parker IC` keys override `auto`. See, for `windae`, `docs/wind_ae_solver.tex`.
`Level tol:`	real	Mass-flux *level*-stability gate: a converged/stalled stop additionally requires the mean $\lvert\rho v r^2\rvert$ over the wind window to change by less than this (relative) across the last `N_stall` steps. Guards against stopping on a transient du dip while the flux level still drifts (≤ 0 = off, default; §3.7).
`Flux spread tol:`	real [real]	FLUX gate (§133): a steady solve is accepted only when the radial spread of $\rho v r^2$ over $r \geq r_{\rm flux}$ is below this tolerance AND the residual is below `Resid tol`. Defaults 2×10^{-5} (`flux_spread_th_default`) and $r_{\rm flux} = 1.2 R_p$; $\leq$ 0 disables it.

Table 3, continued

Keyword	Value	Meaning
`Resid tol:`	real	Residual-based convergence: stop when the steady residual falls below the value (monitored every 500 steps), *replacing* the *du* test (≤ 0 = off, default; the JFNK finish then uses the code's own 10^{-5}). The number compared against it is $\max_k \max_j |R_{j,k}|/s_{j,k}$, each row divided by a bound on the largest term that row itself contains: mass $\max(|F|r^2$ at the two faces$)/\Delta V$, momentum $\max(|\Delta F_2|, |S_2|)$, energy $\max(|\Delta F_3|, |S_3|, \text{heat}, \text{cool})$, plus the operator-split viscous and conduction sources where those are on. The maximum is over cells, so one cell out of tolerance keeps the state out of tolerance; the volume-weighted norm is reported beside it and no longer decides acceptance. *du* alone is blind to the energy imbalance (§3.7); the scales and the reason for each are in `md/input_schema.md` K35.
`Time stepping:`	`Local`	Pseudo-time steps in each cell (steady-state acceleration; not time-accurate). Reaches the deep state fast but leaves residuals in each cell; pair with a Newton finish.
`Energy solver:`	`Explicit`	Revert to the original forward-Euler source update (solver-isolation tests).
`Solver:`	`Newton` [real]	Production steady-state solve: marching warm-up until the FACE mass-flux spread over $r \geq r_{\text{flux}}$ drops below the optional value `newton_flux_switch` (default 0.1; a *du* threshold with default 10^{-2} until 2026-10-10, so a value written for that meaning must be restated), then the JFNK Newton finish to `Resid tol` (default 10^{-5}), then the standard final outputs and post-processing. See §3.7.

Table 3, continued

Keyword	Value	Meaning
`Run mode:`	`init` | `phys` (def. `init`)	What this run is doing, stated rather than inferred. `init`: initialization or continuation. It claims no elapsed time and permits local pseudo-time, pseudo-transient continuation and a chemistry that is not at its root, all as numerical devices; the physical clock `t_phys` stays where it started and no output reports it. `phys`: physical integration, one global *dt* per step, the clock advanced only by a step accepted in full, the temporal error of the step sampled every 20 accepted steps, and the certification refusing a state whose chemistry has no root. Any other word is a fatal `error stop 1`. Refused: `phys` with `Time stepping: Local`, since cells advanced by different intervals do not form one trajectory. The mode sets the default of `Restart intent:` below, is written into the state file's metadata block as `mode` and read back on a restart (a `phys` run loading a file that says `mode=phys` without a finite non-negative `t_phys` is refused), and is echoed in `EXHALE_setup.out` together with whether it was given or defaulted. See `md/input_schema.md` K45.

Table 3, continued

Keyword	Value	Meaning
`Restart intent:`	`trajectory` | `relaxation` | `stationary` [`evaluate` | `equilibrate`] (def. `trajectory` with `Run mode: phys`, else `relaxation`)	What a loaded state is continued as. `trajectory`: the physical clock continues from the `t_phys` the state file records. `relaxation`: march toward stationarity and then hand off to the solver, which is what every restart did before this key existed. `stationary`: rebuild the derived quantities of the loaded state without taking a step, measure the stationary residual and the certification of the *stored pair* (the conserved state and the composition exactly as loaded, with the heating and cooling of that composition in the field it attenuates), then take one composition sweep from it, report how far the composition moved and what the certification reads after the sweep (reported only, never the verdict), and enter the stationary solve from the swept state. The second word `evaluate` stops instead of solving, and `equilibrate` puts the loaded composition on its own fixed point before the measurement (a change of the state, reported when taken). This is the route that certifies a written state without the two CFL steps a marching hand-off takes first. On an open base face (the H_2 partition of the lower ghosts stated by a lower-atmosphere handoff, `base.inp`'s `q_H2_base` or a profile, with `Molecular chemistry: True`) the stationary route first gives the cells of the uniform base region the $n(\mathrm{HI})/n(H_2)$ of ghost 0, each cell keeping its hydrogen nuclei and its mass, so that the seed agrees with its ghost (since 2026-10-09, update log stage 3 section 126; it removed a 57 K base cell on the LHS 1140 b route B entry). If the solve then fails (`info` $\neq 0$) the run exits 3 (§3.7). What `evaluate` writes and which verdict it returns: §4.1. Refused: the key with `Load IC? False`, `trajectory` with `Run mode: init`, `stationary` without `Solver: Newton`.

Table 3, continued

Keyword	Value	Meaning
`Restart option change:`	token list, separated by commas or blanks (def.: no option may differ)	Which switches of the state file's `# options` line are *allowed* to differ between the loaded state and this run. `carrier_newton` is not among the tokens that have to be named: it states the ROUTE and not the equation set, the rows, columns and tolerances being the same either way, so a difference in it is admissible with nothing named, the load says so and the change is recorded as a `# route_change` line (since 2026-09-17). The restart contract otherwise refuses any option difference, which forbids the ladder of settings this code is converged with: converge without an option, restart with it on, converge again. Naming a token (`sec_ion`, `He23S`, `mol_heat`, `base_ir`, `cond`, `interdiff_enth`, ...) permits exactly that token to differ; every other difference still refuses the load and names the token. The loaded state is then a solution of the *former* option set and a starting point for this one, so the change is written into the state this run produces as a `# option_change` line and inherited by the rungs that follow, and `EXHALE_setup.out` states it. Refused: an unknown token; a token that decides how many unknowns the state has (`metals`, `mol`, `oxychem`, `carrier`), whose change makes the rows of the file not the rows of this run and is a cold start and not a restart (`iontrans` is not one of them: the three ionization stages have a column in every state file, and the key changes whether that column is a local root or a transported partition, so it may be named; nor is `he23strans`, the He 2^3S column being in every state file written with the level tracked); a word naming the grid, the reservoir or the constant set; the key with `Load IC? False`. The 26 tokens: §4.1. See `md/input_schema.md` appendix D.2.

Table 3, continued

Keyword	Value	Meaning
`Low-Mach damping:`	real ε_4 [, real $M_{\rm th}$] (def. -1, i.e. off; $M_{\rm th} = 10^{-3}$)	Artificial dissipation of the $2\Delta r$ contact/entropy mode. The HLLC flux carries the middle wave at $S^* \simeq v$, so the dissipation it applies to the entropy family vanishes as the flow stagnates and that mode becomes marginally damped; where metal cooling stalls a shell (HD 209458 b with solar C/N/O, $r \simeq 1.02$, $M \sim 10^{-5}$) it grows to several times the local mean $\lvert v\rvert$. This key adds a gated fourth-difference (Jameson, Schmidt & Turkel 1981) stress $\varepsilon_4\, g(M)\, \rho_f \lambda_f (v_{j+2} - 3v_{j+1} + 3v_j - v_{j-1})$ to the numerical *momentum* flux, and its work $v_f D_p$ to the energy flux, with the gate $g = [\max(0, 1 - M^2/M_{\rm th}^2)]^2$ exactly zero for $M \geq M_{\rm th}$. It enters the numerical flux, so the marching loop and the JFNK steady residual see the *same* equation. Explicit stability requires $\varepsilon_4 < 1/(16\,\mathrm{CFL})$ (0.10 at the default CFL); the classical range $1/64$–$1/32$ works. $\varepsilon_4 \leq 0$ = off, bit-identical. See `src/modules/flux/low_mach_dissipation.f90`.

Table 3, continued

Keyword	Value	Meaning
`Well balanced:`	logical (def. `False`)	Carry the *departure* from the cell's own local hydrostatic equilibrium through the reconstruction, the Riemann jumps and the pressure force, instead of the state itself. Inside cell j the equilibrium is the constant-density one through that cell's own density and pressure, $p_{\mathrm{eq},j}(r) = p_j - \rho_j[\phi(r) - \phi(r_j)]$, so no thermal, entropy or compositional stratification is assumed; the pressure stencil acts on the departure of a neighboring cell's pressure from that equilibrium continued *through* the face between them (this cell's density up to the face, the neighbor's beyond it), which vanishes at a discrete equilibrium, and the momentum row is assembled from the face pressure measured against that equilibrium while the gravitational source term is zero. The equilibrium's flux difference and its source then cancel in the algebra rather than in floating point, and a discrete hydrostatic equilibrium is preserved to rounding (Käppeli & Mishra 2016, A&A 587, A94; the face-pressure-difference form of the source is Käppeli & Mishra 2014, J. Comput. Phys. 259, 199). Exact preservation also needs a numerical flux that resolves a stationary contact, so it holds with `Numerical flux: ROE` and `HLLC` but *not* with `LLF`, and not at a face the positivity repair has replaced with a first-order flux. It applies to both reconstructions and to the PLM $\rightarrow$ WENO3 continuation. Turning it on *moves every result*: the pressure/gravity pair is in every run, and the two discretizations are both second-order consistent with neither a subset of the other. It is stated in `EXHALE_setup.out` and `EXHALE_resolved.out`, and written as the option token `wellbal` in a state file's `# options` line, where `Restart option change:` may name it, since it does not decide how many unknowns a state has. See `src/modules/states/Reconstruction.f90`.

Table 3, continued

Keyword	Value	Meaning
`ATES_photoionization_rate:`	`True`	Revert the He I (1^1S) photoionization cross section to the legacy ATES two-term fit (default: Verner+1996). Does *not* affect the He 2^3S Penning rate, which is always the temperature-dependent form (§9.).
`Legacy_HHe_rates:`	logical (def. `False`)	Revert the H/He case-B recombination and collisional ionization to the legacy ATES fits (Hui & Gnedin 1997 recombination; Abel+1997 / Hui & Gnedin 1997 collisional ionization). Default uses Badnell (2023) RR (+ He II DR) minus the ground capture α_1 for case B, α_1 being the Milne relation on the ground-state photoionization cross section the radiative transfer absorbs with (for H I, He I and He II), and Voronov (1997) collisional ionization (§9.). The ground captures added back by `H_rec_escape` and `He_rec_coupling` are that Milne α_1 in either setting. The free-free Gaunt factor and its ion net-charge weighting are always the van Hoof et al. (2014) form, independent of this switch.

Table 3, continued

Keyword	Value	Meaning

`Atomic rate set:`	`default`
	`Koskinen2022`
	(def. `default`)
		Swap the four atomic H/He rate coefficients for the fits Koskinen et al. (2022) list as R1–R4 of their Table 1, for a like-for-like comparison with their Model A: recombination $4.0 \times 10^{-12}(300/T)^{0.64}$ for H^+ and $4.6 \times 10^{-12}(300/T)^{0.64}$ for He^+ (Storey & Hummer 1995), and collisional ionization $2.91 \times 10^{-8} U^{0.39} e^{-U}/(0.232+U)$ with $U = 13.6\,\mathrm{eV}/kT$ for H and $1.75 \times 10^{-8} U^{0.35} e^{-U}/(0.180+U)$ with $U = 24.6\,\mathrm{eV}/kT$ for He (Voronov 1997). The two collisional fits are the ones EXHALE already uses, so only the two recombination coefficients change value; the branch does pin the Voronov form against `Legacy_HHe_rates`. The default set is the physically preferred one here (case B where the Lyman continuum is optically thick, the escaping ground captures added where it is thin; `H_rec_escape`, `He_rec_coupling`), so the key exists to reproduce their choice and not to replace ours. It swaps *rate* coefficients only, and the recombination cooling follows them: every recombination cooling term is $k_B T\,\alpha(3/2 + \mathrm{d}\ln\alpha/\mathrm{d}\ln T)$ on the coefficient the balance uses, so the electrons removed and the energy removed come from one rate. With `He_rec_coupling` or `H_rec_escape` on, the Milne ground capture α_1 is added to Table 1's coefficient, which has no such split, so the closest like-for-like run also sets `He_rec_coupling: False` and `H_rec_escape: False`. Any other word stops the run (§9.).

Table 3, continued

Keyword	Value	Meaning
`Secondary_ionization:`	`False` | `True` | `Immediate` | `Staged` (def. `True`, from step 0)	Secondary ionization by fast photoelectrons in the main loop: a photoelectron ejected with $E_0 > 30\,\mathrm{eV}$ deposits only $f_{\mathrm{heat}}(x, E_0)$ of its excess as heat and drives further H I / He I / H_2 ionizations, where x is the free electron density over the H+He nuclei. The ionization budget is Shull & van Steenberg's (1985) with the energy dependence of Dalgarno, Yan & Liu (1999); the heating fraction, the H_2 target and the molecular heat returns are Dalgarno et al.'s. `True` and `Immediate` apply the coupling from step 0 (the default since 2026-10-09); `Staged`, the default until then (§3.4), switches it on only once the marching has settled without it, then the run reconverges. `False` restores full photoelectron thermalization, bit-identical to the legacy path.

Table 3, continued

Keyword	Value	Meaning
`Low-energy electron partition:`	`none` | `Furlanetto2010` (def. `none`)	The photoelectrons of $10.2\,\mathrm{eV} \leq E_0 \leq 30\,\mathrm{eV}$, below the range of the partition above, are by default deposited whole as heat. `Furlanetto2010` partitions them by the Monte Carlo tables of Furlanetto & Stoever (2010, MNRAS 404, 1869; the authors' electronic tables, interpolated linearly in E_0 and in x as their `elec_interp.c` does, x clamped to $[10^{-4}, 0.999]$): $f_{\mathrm{heat}} E_0$ is heat, the H I and He I ionizations per primary are added to the secondary ionization rates (their branching renormalized to the cell's neutral densities with the tables' He/H = 0.087 as reference), and the H I line photons it makes leave the cell. Same x as `Secondary_ionization`, applied where and when that coupling is, on the stellar and the on-the-spot recombination photoelectrons, and only in cells with no H_2 (no source covers H_2 at 10–30 eV); the advection-corrected post-process, whose own gas carries no H_2, takes the H_2 of the solved state for that test, so both products apply it in the same cells. The gate is any $n(H_2) > 0$, so the deposition changes discontinuously between a cell without H_2 and one with a trace of it (an open item, `md/TO_BE_DONE.md` (AJ)). The heat share is that of a primordial gas and is not corrected for a helium-rich one. Option token `lowE_partition` of the restart contract (a state written before the token existed reads as `F`); recorded in `EXHALE_resolved.out` as `low_energy_electron_partition Furlanetto2010`, a line written only when the option is on. Any other word stops the run.

Table 3, continued

Keyword	Value	Meaning
`He_rec_coupling:`	logical (def. `True`)	Absorb the ionizing recombination radiation of He II → He I and He III → He II on the spot (§9.). He II: the ground captures (α_1, the Milne relation on the He I cross section) count as recombinations with the local escape weight y, the fraction of their 24.6 eV photons that He I does not re-absorb in the cell, so the He II coefficient is $\alpha_B + y\,\alpha_1$ (case A where the cell is thin, case B where it is thick); the photons that H I, H_2 and the metals take ionize those species; and the case-B captures leave through their own exits, the 584 Å line (scattered by He I and partly converted into the $2\,^1$S two-photon continuum), the 19.8 eV $2\,^3$S line and the two-photon continuum, shared by the network's rates (the H-ionizing photons of a case-B capture are z = 0.966 as $n_e \to 0$ and 0.660 at $n_e = 10^8$ cm^{-3}, at 10^4 K in a cell without He I, as the physics probe `recombination_radiation_on_the_spot` measures them; He I scattering of the 584 Å line lowers z). He III: the ground capture, the direct capture into $n = 2$, He II Lyα and the $2s$ two-photon continuum (Nussbaumer & Schmutz 1984), with the $2s$–$2p$ mixing of He II by ion impact. The photoelectrons are partitioned as the stellar ones are (`Secondary_ionization`, with its staging, and `Low-energy electron partition`), each at the energy of its channel plus the mean capture kinetic energy: above 30 eV (the He III ground capture on H I, He I, H_2 and the low-threshold metals, He II Lyα on the metals below 10.8 eV) into heat (channel `heat_He_recomb`), secondary ionizations and escaping line radiation; with the coupling off all of it is heat. With the He $2\,^3$S network on, case B is split into triplet and singlet captures with Hummer & Storey (1998) Table 5 in both settings of the key. `False` = case B for both recombinations with the cascade photons lost.

Table 3, continued

Keyword	Value	Meaning
`H_rec_escape:`	logical (def. `True`)	The $\mathrm{H\,II} \to \mathrm{H\,I}$ ground captures whose 13.6 eV photons leave the cell or are taken by a metal ion count as recombinations: the H II coefficient is $\alpha_B + y_{\mathrm{HI}}\,\alpha_1$, α_1 the Milne relation on the hydrogenic ground cross section and $y_{\mathrm{HI}} = 1 - n_{\mathrm{HI}}\sigma_{\mathrm{HI}}\,\Delta r(1 - e^{-\tau})/\tau$ the escape weight of the same construction as He II's (τ the cell's optical depth at 13.6 eV over H I and the metal ions), so case A holds in the thin outer wind and case B in the thick base. The recombination cooling uses the same coefficient at the frozen weight; the $n = 2$ sources of the H(n=2) model stay case B. The closure is local: a photon that leaves the cell is dropped, so where the cells are neither thick nor thin the weight depends on the cell width. On by default because it corrects the case-B assumption where it does not hold; `False` = case B everywhere (e.g. for a like-for-like comparison with Koskinen et al. 2022). Written as `use_h_rec_escape` in the parse dump (`EXHALE_PARSE_DUMP=1`).

Table 3, continued

Keyword	Value	Meaning
`He_H_charge_exchange:`	logical (def. `True`)	The He $\leftrightarrow$ H charge exchange: $He^{+}+H^{0} \rightarrow He^{0}+H^{+}$ as the sum of its radiative channel (Stancil, Lepp & Dalgarno 1998, from Zygelman et al. 1989) and its non-radiative one (the Maxwellian average of the H^{+}+He cross section of Loreau, Ryabchenko & Vaeck 2014, turned forward by microscopic reversibility; their calculation starts at 15 eV/u, and since 2026-09-26 the reverse cross section between the threshold $x_{\rm th}$ = 13.655 eV/u and 15 eV/u is taken as $\sigma_r(15)\sqrt{(x-x_{\rm th})/(15-x_{\rm th})}$, i.e. $\sigma_f v$ constant at low forward energy, an assumption no calculation tests; the channel is 3.3% of the radiative one at 10^4 K and 5.9% at 10^3 K); $He^{0}+H^{+} \rightarrow He^{+}+H^{0}$, the detailed-balance reverse of the non-radiative channel (endothermic by 10.99 eV); and the radiative capture $He^{2+}+H^{0} \rightarrow He^{+}+H^{+}+h\nu$ (the Maxwellian average of the cross section of West, Lane & Cohen 1982, 1.6–2.0$\times10^{-13}$ cm^3 s^{-1} at 10^3–3×10^4 K), which has no reverse. The non-radiative exchange heats the gas by 10.99 eV an event and its reverse takes that back; the radiative channels give their energy to the photon (channel `heat_charge_exchange`). Applied by dedicated routines in *every* helium-bearing ionization system, independent of `cx_full` (which gates the metal groups C and D). `False` leaves all three out.
`He_diffusion:`	logical (def. `False`)	Enable diffusive separation of helium: transport the helium *mass fraction* of the two-component H/He gas with advection + molecular-diffusion settling, so He/H falls with altitude. Off = He/H frozen at the input `He/H number ratio` (§4.6).

Table 3, continued

Keyword	Value	Meaning
`He_Kzz:`	real ($cm^2 s^{-1}$, def. 0)	Eddy-diffusion coefficient used with `He_diffusion`: mixes the composition toward uniform below the homopause. Default 0 = pure molecular diffusion; an eddy term belongs to the atmosphere being modeled, so it is stated rather than inherited (Taylor+2025 use 10^9).
`He_ambipolar:`	logical (def. `True`)	Ambipolar-corrected settling: in the ionized wind the polarization field lifts He ions, reducing settling. The field is computed from the local electron-pressure gradient, $eE = -kT\,\partial \ln(n_e T)/\partial r$; `False` = $eE = 0$, i.e. neutral settling $\Delta m = 3$.
`He_alphaT:`	real (def. 0)	Thermal-diffusion factor α_T added to the settling drift ($\alpha_T\,\partial \ln T/\partial r$); 0 = off.
`He_metal_diffusion:`	logical (def. `False`)	Also diffuse each trace metal against the hydrogen background with its own mass, mean charge and stage-resolved diffusion coefficient; heavier elements settle more. Off = metals frozen to H (§4.6).

Table 3, continued

Keyword	Value	Meaning
`Interdiffusion enthalpy flux:`	`True/False` (def. `True`)	Carry the interdiffusion enthalpy flux $q_d = \sum_s h_s J_s$ in the energy equation of the diffusing mixture (Cook 2009, Phys. Fluids 21, 055109, eqs. 11–13): the sensible enthalpy the helium (and, with `He_metal_diffusion`, each metal) carries as it moves against hydrogen, and the sensible enthalpy each transported molecular carrier (H_2; OH, H_2O, CO with the oxygen chemistry) carries relative to the rest of its elements, and, with `Ionization transport: True`, the sensible enthalpy the carried ionization stages move relative to their element (in effect the electrons' $\gamma/(\gamma-1)kT$ times $\sum_k Z_k E_k$, E_k the eddy part of the stage flux), in the marching energy update, in the stationary energy row and in the advection-corrected column alike (§4.6). It acts with `He_diffusion: True` and helium in the gas, or with `Molecular carrier transport: True`; otherwise the key is inert, bit for bit. A restart from a state written without the carrier term names the token `carrier_enth`, and one written without the stage term the token `stage_enth`, on a `Restart option change:` line. **On by default, because the term belongs to the energy equation of any mixture whose components move relative to one another**; `False` exists only to reproduce published escape models that omit it (Koskinen et al. 2013, 2022; Yelle 2004). MEASURED on the LHS 1140 b `He_diffusion` states: 13–26% of the local energy budget at 1.1–5 R_p (`md/interdiffusion_enthalpy_flux_20260926.md`). Written as the option token `interdiff_enth` of a state file's `# options` line; a state written before the key existed reads as `interdiff_enth=F`, so restarting such a `He_diffusion` state under the default needs `Restart option change: interdiff_enth`.

Table 3, continued

Keyword	Value	Meaning
`Molecular chemistry:`	`True/False` (def. `False`)	Add H_2, H_2^+, H_3^+ and HeH^+ to the coupled ionization equilibrium, with the H_2 photoionization opacity and heating and the H_3^+ infrared cooling. Requires `He/H number ratio` > 0. `He_diffusion` is allowed (since 2026-08-26): the element transport closes over the molecular carriers. Turning it on also turns `Molecular base` on (see there). A `metals.inp` may be present: the metal stages are then solved in the same system (§4.7).
`Molecular base:`	`True/False` (def. `False`)	Remove the H_2-bound particles from the base particle count `ntot_bc` using the chemical-equilibrium H_2/H/He partition, giving a lower base pressure and a heavier base μ with the species arrays still atomic (Tier 2a). Forced on by `Molecular chemistry: True`, because an atomic `ntot_bc` under a molecular base inflates the base temperature by the particle-count ratio and dissociates the layer the chemistry just built, so with the chemistry on the key is redundant (§4.7).

Table 3, continued

Keyword	Value	Meaning
`Oxygen chemistry:`	`True`/`False` (def. `False`)	Add OH, H_2O and CO to the same coupled system, with the FUV photolysis of H_2O and OH, so that the code computes its own base H_2/H partition instead of importing one. Requires `Molecular chemistry: True`, He/H $>$ 0 and a non-zero oxygen abundance; refuses `q_H2_base` (it computes that quantity) and `He_metal_diffusion` (that option counts an element over its ion stages alone, and CO carries an oxygen *and* a carbon nucleus, so it cannot follow two element factors at once). `He_diffusion` alone is accepted. Supply the band fluxes with `Stellar LW flux`, `Stellar FUV B3/B4 flux` and `Stellar Lya flux`; with `Spectrum type: Load` the three continuum bands (LW, B3, B4) are integrated from the spectrum file when their keys are absent, and only Lyα has to be stated. Without band fluxes the run returns the chemical equilibrium of the O/OH/H_2O family rather than a photochemical partition, and says so at startup. Writes `output/Oxygen_chemistry.txt` and `output/FUV_bands.txt` and appends `OH H2O CO` to `Ion_species.txt` (§4.7).

Table 3, continued

Keyword	Value	Meaning
`Molecular carrier transport:`	`True/False` (def. `True` whenever `Oxygen chemistry` is on)	Transport the molecular carriers H_2, OH, H_2O and CO vertically instead of closing them by a local steady state: one implicit backward-Euler diffusion-advection step per pass, solved together with the same chemistry rows the local equilibrium uses, with molecular diffusion from Blanc's law over the background carriers, the eddy coefficient K_{zz} of `He_Kzz` or of a lower-atmosphere profile, and settling. `False` restores the local steady state, which isolates the chemistry for testing and is not a model of a base: at a cool base the H_2 chemical time and the flow time are comparable. With $K_{zz} = 0$ everywhere the transport is pure molecular diffusion, the wrong limit for a lower atmosphere, and the startup report says so. It acts only with `Molecular chemistry: True`, not with `Oxygen chemistry`: H_2 is the first carrier and is transported with or without the oxygen cycle, which only adds OH, H_2O and CO to the set, and `Ionization transport` adds the ionization stages H^+, He^+ and He^{2+} (and `He 2^3S transport` the He 2^3S level). With the molecular network off there are no carriers and the key changes nothing; `input_read` reports that at startup rather than refusing the run, the state produced being the correct atomic one (§4.7).
`Oxygen transport:`	*retired*	The former name of `Molecular carrier transport`, retired on 2026-09-02 because the operator transports H_2 whether or not the oxygen cycle is on. A file that still carries it stops the run with a message naming the new key; it is not read as an alias.

Table 3, continued

Keyword	Value	Meaning
`Coupled carrier solve:`	`False` | `True` | `On stall` (def. `False`; a word the key has no meaning for stops the run rather than falling through to `False`)	Solve the transported balances as unknowns of the stationary system itself, $(\rho, \rho v, E, n(\mathrm{H_2})/n_0, \ldots)$ per cell, instead of alternating a wind solve with a fixed-wind carrier relaxation. The alternation was measured not to converge: the two halves are coupled through the particle count, which sets the temperature, so a splitting that evaluates each at the other's old state gets the sign of the H_2 front's motion wrong. **On a molecular configuration this key requires `Molecular carrier transport: True`, and the combination without it is refused at startup.** With the carriers eliminated, $n(H_2)$ is not an unknown but whatever the local-equilibrium sweep returns for the composition it was seeded with, and in the shielded layer the sweep cannot return a content at all: the fast chemistry cycles $H_2 \to H_2^+ \to H_3^+ \to H_2$ without changing the number of H_2 nuclei, so the local balance rows fix the partition among the molecular species and leave their sum where the seed put it. The stationary residual is then not a function of its unknowns (measured seed dependence 3.2×10^5 times `Resid tol` per unit relative seed change, against 9.7×10^{-10} with the H_2 row carried), so it has no root. A run whose only transported balance is an element (`He_diffusion` in an atomic gas) is not affected, and neither is marching. The carrier densities of the coupled system are carried as $\ln n$, so that the positivity of a carrier is a property of the unknown space rather than a bound the step control has to enforce; the floor under $\ln n$ is 10^{-20} of the cell's own element budget, the fraction of its element below which the carrier row scale already calls a row converged (`EXHALE_CARRIER_LOG_UNKNOWN=0` restores the density unknown for measurement).

Table 3, continued

Keyword	Value	Meaning
`Composition update holds:`	`energy` | `pressure` (def. `energy`)	Which thermodynamic variable the composition update of the stationary alternation (§3.7) keeps fixed beside ρ and v while the element and carrier relaxations change the composition at the fixed hydrodynamic state. `energy`: the conserved E, and p and T follow at the new composition. `pressure`: p, with $T = p/(n_{\mathrm{tot}} + n_e)$ of the new composition and E rebuilt from its caloric equation of state, the kinetic part unchanged. A route through the iteration and not an equation set: a stationary state is stationary under either. Any other word stops the run.
`Ionization transport:`	`True`/`False` (def. `False`)	Advect H II, He II and He III as fractions of their own element nuclei, with eddy mixing but no relative ion diffusion. The neutral stages close the element budgets; the lower ghost ionization fractions remain on their local chemical roots. This option applies to atomic and molecular gases. See the Koskinen et al. (2022) comparison in `md/k22_electron_density_excess.md` §7. **Refused at startup**: a mixture without helium; `Molecular chemistry: True` without `Molecular carrier transport: True`; `Coupled carrier solve: True` in an atomic gas; `Coupled carrier solve: On stall`; and the direct steady route `EXHALE_PTC=1`. In a molecular gas, an explicit `Coupled carrier solve: True` uses Newton carrier rows for the ionization stages. Otherwise, `Solver: Newton` uses the partitioned alternation, while marching transports the stages without a stationary solver. Recorded in `EXHALE_resolved.out` as `ionization_transport` and in restart files as `iontrans=T`. It changes ionization-dependent results (§4.7).

Table 3, continued

Keyword	Value	Meaning
`He 2^3S transport:`	`True/False` (def. `False`)	Carry the He 2^3S population with the flow instead of solving it as the local root of its level balance in every cell: $x_3 = n(2^3S)/n_{\mathrm{He}}$ becomes a fourth helium state of the `Ionization transport` operator, riding on the helium nucleus flux with the same face flux, with the level balance of the sweep (`tr_triplet_row`) as its source, and the ground singlet closing the helium simplex $x(\mathrm{He\,II}) + x(\mathrm{He\,III}) + x_3 \leq 1 - n(\mathrm{HeH^+})/n_{\mathrm{He}}$. It is the steady advection balance of Oklopcic & Hirata (2018, Eq. 15) and Lampón et al. (2020, Eq. 11) with the eddy term added. The level moves at the velocity of He I (no drift of 2^3S against ground-state helium). Its 19.82 eV excitation energy is booked by the reactions of the level where they run, so no excitation-energy flux enters the energy row; the certification prints its size against the energy row as a reported-only line. Certification entry `carrier balance He2^3S` (gated at 10^{-5} in the wind); `level balance He 2^3S` is then not applicable. **Requires `Include He23S: True` and `Ionization transport: True`**; each is a fatal refusal at startup. Recorded in `EXHALE_resolved.out` as `he23s_transport` (written only when on), in a restart file's `# coupling:` line as `he23strans=T` and in its `# options` line as `he23strans`, which `Restart option change:` may name in either direction. Default off. Design and validation: `md/He23S_transport_design_20261003.md`.

Table 3, continued

Keyword	Value	Meaning
`Lower atmosphere:`	`none` | `analytic` | `vulcan` [, real (R_J)] (def. `none`)	Lower-atmosphere pre-step run before the wind: generate the base composition over the 1-bar to 1 μbar column, analytically (`src/utils/run_lower.py`) or by calling VULCAN (`src/utils/vulcan_driver.py`, hours on the first run, cached after), and hand it over through `base.inp` (Tier 3). The second token is the 1-bar (transit) radius and is required. An existing `base.inp` is used as is and nothing is regenerated; the generator needs `EXHALE_ROOT` set to the install directory (§4.7).
`Lower column:`	real (R_J)	Tier-1 consistency *report* only: integrate the isothermal-$T_{\rm eq}$ hypsometric column with the chemical-equilibrium H_2/H/He partition from the given 1-bar radius up to 1 μbar and print the derived base radius, base fractions and μ next to the input `Planet radius`. Nothing is overridden (§4.7).
`Lower atmosphere profile:`	path (def. none)	Take the lower atmosphere from a *profile* (the lower model's own solution tabulated over an interval of pressure) instead of the single-level scalars of `base.inp`. The named file supplies the base state at its matching pressure (T_0, R_0, $p_{\rm base}$, the H_2 fraction), the elemental reservoirs (He/H and every element column it carries, overriding `metals.inp`), and the eddy coefficient $K_{zz}(r)$ as a profile, which makes `He_Kzz` inert. The profile also fixes the base *level*: n_0 follows from `p_match_bar` exactly as it does from `p_base` of `base.inp`, and a `Log10 lower boundary number density` line given beside a profile is refused unless the two agree to 1%. A missing file, a malformed table, or a `base.inp` beside it that still states a physics key stops the run (§4.8).

Table 3, continued

Keyword	Value	Meaning
`Base BC:`	`density` | `pressure` [, real (μbar)] (def. `density`)	What the lower boundary anchors. `density` (legacy) fixes the base number density at the n_0 of the core block. `pressure` (CETIMB-style) fixes the base pressure at the optional second token (default 1 μbar) and *derives* n_0 from $n_0 k_B T_0 n_{\mathrm{tot,bc}} = p_{\mathrm{base}}$; the reservoir of the characteristic boundary (§3.6) then holds that pressure and T_0 at the level, the lower face of cell 1, and ρ follows. It is a re-parameterization of the same reservoir, whose physical effect is a much less dense base. Without a lower-atmosphere handoff, `pressure` beside a `Log10 lower boundary number density` key is followed and the density key is ignored, with a WARNING on standard output and in the setup report naming both levels and the one used; with a handoff, both are checked against its level instead.
`Shapiro filter:`	real ε [, int n] (def. -1, i.e. off; $n = 4$)	Apply a weak Shapiro (1970) 1–2–1 low-pass filter u_j += $(\varepsilon/4)(u_{j-1} - 2u_j + u_{j+1})$ to the conservative variables every n steps, damping the gravity-unbalanced sound-wave (base-breathing) mode the same way CETIMB does. $\varepsilon \leq 0$ = off, and off is the default on purpose: with the filter on, an HD209458b cold-IC sweep left the transonic-wind saddle for an infall state ($v < 0$ everywhere) regardless of the velocity BC, while filter-off relaxed to a clean outflow. Enable it only for a case that actually breathes. Unlike `Low-Mach damping` it touches only the marching state, not the numerical flux, so the JFNK residual does not see it.

Table 3, continued

Keyword	Value	Meaning
`Viscosity:`	`True`/`False`, or real μ_0 [, real s] (def. `False`)	`True` adds the radial viscous force $(\nabla\cdot\tau)_r$ and its dissipation $q_\mu = \frac{4}{3}\mu(\partial_r v - v/r)^2$ with the calibrated $\mu(T) = \frac{4}{15}(m_{\rm H}/k_B)\kappa(T)$ (monatomic Chapman–Enskog at Prandtl $2/3$, tied to the Watson et al. 1981 neutral-hydrogen conductivity $4.45\times10^4(T/1000\,{\rm K})^{0.7}$, whatever the composition; `Conduction` uses the mixture conductivity instead): the molecular transport that damps the near-base momentum imbalance the inviscid HLLC scheme lacks. A numeric value is instead a diagnostic power law $\mu = \mu_0 T^s$ in code units ($s = 0.7$ by default), kept for sensitivity scans. Off = byte-identical to the inviscid code.
`Conduction:`	logical (def. `False`)	Add heat conduction $r^{-2}\partial_r(r^2\kappa\,\partial_r T)$ with the conductivity of the mixture $\kappa = [n_e 1.2\times10^{-6}T^{5/2} + n_{\rm HI}379\,T^{0.69} + n_{\rm HeI}299\,T^{0.69} + n_{\rm H_2}k_{\rm H_2}(T)]/n\,{\rm erg\,cm^{-1}s^{-1}K^{-1}}$ (Banks & Kockarts 1973 as Salz et al. 2015 and Sutton et al. 2015 state it; $k_{\rm H_2}$ an Eucken-form fit to Incropera et al. 2007 Table A.4, valid 200–2000 K; n all particles). Off = byte-identical. The base boundary condition is a heat bath at the base ghost centre $r(0)$ (`conduction_base_level_T`, `viscous_conduction.f90`): the temperature of the lower-atmosphere profile at that radius when a profile is in use (§4.8), otherwise the reservoir's hydrostatic isentrope continued below the level; the outer face carries no conductive flux. The heat conducted into the lower atmosphere through the base face is written into `EXHALE_resolved.out` (§5.13), which is what the energy coupling of the elemental-flux closure reads. The same conduction terms enter the advection-corrected temperature (§5.). Option token `cond`; its continuation factor: §4.1.

Table 3, continued

Keyword	Value	Meaning
`Caloric EOS:`	`ladder` | `monatomic` (def. `ladder`)	The caloric equation of state. `ladder`: every atom, ion and electron stores $\frac{3}{2}kT$ and H_2 its rovibrational energy (the Roueff et al. 2019 ladder), so $\gamma_{\rm eff}$ depends on composition and T; in an 80% H_2 layer that is not $5/3$. `monatomic`: H_2 too stores $\frac{3}{2}kT$ and $\gamma = 5/3$ everywhere, kept as a *comparison* option for a published model whose energy equation is $u = c_v T$ with a monatomic c_v. In an atomic gas the two agree exactly. Any other word stops the run.
`Photoelectron heating:`	`excess` | `full` | real $f \in (0, 1]$ (def. `excess`)	What a photoionization deposits as heat. `excess` (physical): the electron energy, $h\nu - I$ for an outer-shell absorption and, for an inner-shell absorption by a metal ion, the photoelectron's $h\nu - E_{{\rm th},s}$ plus the Auger electrons of the vacancy (Kaastra & Mewe 1993; §9.). `full`: the whole photon energy $h\nu$; a number: the fraction f of $h\nu$ (both keep the outer-shell form for the metals). The last two are *comparison* options. Koskinen et al. (2022) Figure 9 shows a stellar heating rate 2.5–3 times what $h\nu - I$ gives at their ionization rate and spectrum, i.e. between the two accountings, and a fraction reproduces their profile without claiming a physics; a run that uses one must say so. Anything else stops the run.
`Resid norm:`	**retired**	Selected between a volume-weighted and a max-over-cells combination of the scaled residual rows. The acceptance norm is now the cellwise maximum and the volume-weighted number is reported beside it every time, so there is nothing left to select. The label is still matched and a file carrying it draws a warning saying so, rather than failing the unknown-key scan.

Table 3, continued

Keyword	Value	Meaning
`Wind-AE seed:`	path, or `grid` | `auto`	Which converged Wind-AE solution `IC mode:` `windae` ramps from; the default is the shipped `inputdata/windae_seed.csv`. `grid`/`auto` instead picks the nearest solution in `inputdata/windae_grid/` to this planet's $(M_p, R_p, F_{\rm XUV})$, which is a shorter and more robust ramp than always starting from the single fiducial seed (§3.10).
`Wind-AE seed out:`	path (def. none)	Write the converged Wind-AE relaxation solution to this file as a reusable seed, e.g. to grow `inputdata/windae_grid/`. Omitted = nothing is written.

What a stationary evaluate writes, and the verdict it returns. `evaluate` writes every product a solved case is read for from the stored pair, with no time update: the two state files and `conserved_state.txt` (whose `certified=` pair is the stored pair's verdict), the heating and cooling breakdowns, the `*_adv` profiles derived from the stored pair and the mass-loss line; its exit status is the stored pair's verdict (0; 2 when the stored pair's certification refuses the file's claim; 7 when that claim stands and the derived `_adv` record is rejected, §5.), and the swept composition is never written (2026-10-04; until then the products and the verdict were of the swept state, and the mass row of the wind, a quantity at its rounding floor, could refuse there a state its solve had certified). An evaluate of an evaluate's output writes the same data again. The same evaluation (`sources_and_ghosts_of_the_held_composition`, `EXHALE_main.f90`: the temperature from u through the caloric map of the composition, the heating, cooling and lower ghost from one sweep that returns the interior composition bit for bit) is also the joint test of every stationary outer pass and the certification of every state a solve writes, so that a certificate stands on the numbers its reload measures and not on the ghost and sources of an earlier iterate (2026-10-04). The `*_adv` files carry a `# derived_from:` provenance line rather than a `# coupling:` header, because they describe a composition the certification was not made on.

The option tokens of a state file, and the continuation factors. The `# options` line of a state file carries 26 tokens (`opt_name`, `load_IC.f90`), the vocabulary of `Restart option change:`: `He23S`, `metals`, `eos_metals`, `mol`, `molbase`, `oxychem`, `carrier`, `carrier_newton`, `iontrans`, `he_diff`, `he_metal_diff`, `sec_ion`, `caloric_mono`, `excH`, `base_ir`, `mol_ir`, `mol_heat`, `visc`, `cond`, `jlya`, `wellbal`, `interdiff_enth`, `carrier_enth`, `lowE_partition`, `stage_enth` and `he23strans`. `metals`, `mol`, `oxychem` and `carrier` decide which columns the state files carry and can never be named; `carrier_newton` names a route and differs without being named; the last five read as `F` in a file written before they existed. `EXHALE_CONDUCTION_SCALE` and `EXHALE_INTERDIFF_ENTH_SCALE` (§3.8) scale the heat conduction and the interdiffusion enthalpy flux by a factor $s \in [0, 1]$. They change the equations, so the

cond and interdiff_enth tokens of the # options line carry them: F without the term, T at $s = 1$, and s itself otherwise, written ES23.16E3 (seventeen significant digits, so two factors are two states and the text reads back as the same double; gfortran and ifx write the same text), and a restart across a change of s names the token on Restart option change:. A token written in the five-decimal F7.5 form (the states of 2026-09-29/30) stands for the interval $[v-5\times10^{-6}, v+5\times10^{-6}]$ and not for one factor. A requested factor inside it is a change of representation: the state loads as a seed with its certification claim dropped (cert_reason=legacy_factor_token), the log says so in one line, and the states this run writes carry a # legacy_factor_token <from> -> <to> at restart of <source> history line. A factor outside it is a physics change and must be named like any other token. A value this code never wrote is refused even when named. The run log and EXHALE_setup.out print the factor in ES13.6, and the report names a legacy-token migration.

4.2 metals.inp: trace-metal abundances

metals.inp is *optional*: if absent, the run is metals-off (H/He only). If present, it lists the total elemental abundance $n_X/n_{\rm H}$ (summed over ionization stages; the equilibrium solver distributes each element over its stages) for any of C, N, O, Mg, Si, Ca, Na, K, S, Fe. Blank lines and # comments are ignored; any non-zero abundance turns metals on. Each quantity may be stated once; a second line for it stops the run. Six control keys are also accepted: pp_metals (metal treatment in the advection post-process: 0=metal-free, 1=frozen at the converged equilibrium (default), 2=re-solve the metal stages inside the post-process iteration, at fixed element totals, after which the electron density, the particle count and the mean molecular weight are formed again; cells the composition class of §5. refuses are skipped; until 2026-10-01 the re-solve never set the cell temperature and every such run stopped in the charge-exchange guard), cx_full (1 adds groups C (metal+He/He^+) and D (metal+metal) of the Huang et al. 2023 Table 4 reaction list to the metal+H/H^+ group A that is always active; it does not gate the $He \leftrightarrow H$ reactions, which have their own input.inp key He_H_charge_exchange, nor the two group E electron captures below), cno_cool (C/N/O line-cooling source: 1=closed-form fits to CHIANTI v11, including N I/N II cooling (default); 0=legacy AIOLOS analytic fits, which have no N cooling and deviate by 40–70% from CHIANTI for O I/O II in the wind-launch region, see cooling_data/cno_cooling_comparison.ipynb), and eos_metals (bulk-gas metal policy: 1=metals contribute their mass to the gas density and base density, their electrons to the free-electron density and the base pressure correction, and their nuclei to the total particle density, a $\sim$1% mass effect at solar abundances, consistent with the charge balance inside the ionization solver (default); 0=legacy trace approximation, H/He-only gas budget), and the two group E scales cx_O2p_H and cx_N2p_H (default 1 each = the published rate). Huang+2023 Table 4 has no doubly ionized oxygen or nitrogen row, so re-sourcing the charge-exchange network to that paper dropped $O^{2+}+H^0 \rightarrow O^+ + H^+$ and $N^{2+}+H^0 \rightarrow N^+ + H^+$. EXHALE carries both separately at the rates of one calculation, Barragán et al. (2006, ApJ 636, 544; their Table 1 k_1 for oxygen, the calculation CHIANTI v11 and Cloudy both adopt, and their Table 2 k_4 for nitrogen, interpolated in log–log and held at the ends of the calculated 10^2–10^5 K). cx_O2p_H <scale> and cx_N2p_H <scale> rescale them, with 0 leaving the row out exactly and other values serving as bounding experiments. They are deliberately *not* controlled by cx_full, which means "all of Table 4". Measured on HD 209458 b, carrying the oxygen reaction suppresses the O III fraction by 4.7 decades at 1.05 R_p, 3.7 at 1.1 and 2.7 at 1.2, and raises O I by 3% below 1.2 R_p and 52% near 2–3 R_p, while $\dot{M}$ moves by 0.01 dex (§103 and §107 of Update_EXHALE_stage1; measured 2026-08-30, before the

charge-exchange corrections of 2026-09-24 to 09-26). The nitrogen capture (9.70×10^{-10} cm^3 s^{-1} at 10^4 K) beats N III recombination wherever $n(\mathrm{HI})/n_e$ exceeds a few 10^{-3}; MEASURED on the `wasp_full` regression case (2026-09-26, `md/charge_exchange_detailed_balance_20260924.md` section 12.6), N III/N falls from 0.444 to 0.0029 at $1.3\,R_p$ and $\log_{10}\dot{M}$ moves by -3×10^{-5} dex. Neither reverse is carried: each is endothermic by more than 16 eV (21.5 eV for oxygen, 16.0 eV for nitrogen), and the nitrogen product $N^+(2s2p^3\,{}^3D^o)$ decays by the 1084 Å multiplet, which leaves the cell, so that event heats the gas by 4.57 eV of its 16.00 eV defect. Solar reference values (Asplund+2009):

```
# metals.inp -- solar n_X/n_H (Asplund+2009)
C 2.69e-4 N 6.76e-5 O 4.90e-4
Mg 3.98e-5 Si 3.24e-5 Ca 2.19e-6
Na 1.74e-6 K 1.07e-7 S 1.32e-5 Fe 3.16e-5
pp_metals 1
```

4.3 `opacity.inp`: opacity model (optional)

`opacity.inp` selects the bound-free opacity model via `OPACITY_MODEL = A|C|P|T`: **A** analytic (default, the standard ATES cross sections); **C** constant (threshold σ times the factors for each species `OPA_CONST_HI/HEI/HEII/HEITR`); **P** pressure-broadened (Robinson & Catling $f(p) = 1 + a(p/p_0)^n$ via `OPA_PB_FACTOR`=a, `OPA_PB_EXPONENT`=n, `OPA_PB_PIVOT`=p_0); **T** tabulated (two-column `.opa` files per species, energy [eV] vs. σ [10^{-18} cm^2], given by `OPA_FILE_HI` etc.). Missing keys keep their defaults, so an absent `opacity.inp` is equivalent to model A.

4.4 Excited hydrogen and the Lyα pumping field

The non-LTE H(n=2) model (the metastable $2s$ + resonance $2p$ levels in the level balance of Christie et al. 2013, with the rate coefficients listed in Table 7: $A(2s \to 1s)$ of Drake 1986, the CHIANTI $1s \to 2s, 2p$ collision rates, the $2s$–$2p$ mixing by electrons of Seaton 1955 and by H^+, He^+ and He^{2+} of Pengelly & Seaton 1964, and the case-B coefficient of the balance split with the $2s$ fraction of Pengelly 1964) feeds two things: in `EXHALE.x` the H(n=2) photoionization adds a Balmer proton source and photoelectric heating to the energy/ionization balance; in `EXHALE_transit.py` the n=2 population sets the Hα/Hβ transmission. The model is *off by default*; it switches on only when both `Stellar Teff` and `Stellar radius` are given, so the standard run is unchanged. Those two keys *arm* the coupling; the Balmer continuum that ionizes and heats n=2 is then the run's own spectrum over $[e_{\mathrm{th,HI}}/4,\ 13.598]$ eV (the power law for a `Power-law` run, the photospheric blackbody for `Planck`, the loaded table for `Load`), integrated against the hydrogenic $\sigma_2(E) = \sigma_2(e_{\mathrm{th,HI}}/4)\,(e_{\mathrm{th,HI}}/4E)^3$, and not a blackbody of its own (§4.5). Arming the coupling also lowers the photon-grid floor to 3.400 eV, so a `Loaded` table that stops above 3647 Å stops the run.

The $2s/2p$ populations are driven by the **Lyα mean intensity** $\bar{J}_{\mathrm{Ly}\alpha}(r)$, which pumps $1s \leftrightarrow 2p$. How $\bar{J}_{\mathrm{Ly}\alpha}$ is obtained is selected by three mutually-exclusive options:

(mode 0) Parameterized: the default with no external input. If neither `Jlya RT file` nor `Jlya escape-prob` is present, $\bar{J}_{\mathrm{Ly}\alpha}$ is the Huang et al. (2017, Eq. 6) estimate computed internally from the run's own state,

$$\bar{J}_{\mathrm{Ly}\alpha}(r) \;=\; \frac{0.1\,F_{\mathrm{LyC}}}{\Delta\nu_D(r)}\,\frac{1}{1+\tau_{\mathrm{Ly}\alpha}(r)},$$

where $\Delta\nu_D = \nu_0\sqrt{2k_BT/\mu}/c$ is the local Doppler width, $\tau_{\mathrm{Ly}\alpha}(r)$ is the *top-down* line-center Lyα optical depth integrated from the outside in through the H I column, and

$$F_{\mathrm{LyC}} = \xi\,\frac{10^{L_{\mathrm{EUV}}}}{4\pi a^2}\left(1-e^{-\sigma_{\mathrm{LyC}}N_{\mathrm{HI}}}\right)$$

is the deposited Lyman-continuum flux (the incident EUV flux times the dayside factor ξ from `2D approximate method` and the absorbed fraction set by the neutral-H column N_{HI}). This needs *no* external file: it is a self-contained, approximate "staging" field, adequate to get the coupling and signs right but not a true radiative-transfer solution.

(mode 1) External RT file: `Jlya RT file: <path>`. Read a tabulated $\bar{J}_{\mathrm{Ly}\alpha}(r)$ profile (two columns, r/R_p and $\bar{J}$ in cgs) produced by a real Lyα radiative-transfer calculation (e.g. a Monte Carlo run); it is interpolated onto the grid and used directly.

(mode 2) In-line escape probability: `Jlya escape-prob: True`. Compute $\bar{J}_{\mathrm{Ly}\alpha}$ every step from the current state by escape-probability radiative transfer: a trapped *internal* field (recombination + collisional sources closed by the escape probability β) plus a trapped *stellar beam* set by `Stellar Lya flux` (with the penetration controlled by `Lya stellar halfwidth` and the buildup capped by `Lya stellar boost`). $\beta = 1/(1+\langle N\rangle)$ with $\langle N\rangle$ the mean number of scatterings before escape of the static plane-parallel damping-wing slab (Neufeld 1990, ApJ 350, 216, eq. 3.27 at zero continuum destruction; for a mid-plane source, Harrington 1973, MNRAS 162, 43, eq. 40, $\langle N\rangle = 0.909316\,\tau$), so β goes as $1/\tau$ in the wings and tends to one when the column thins. By default the lower boundary reflects, which makes the emitting plane a symmetry plane; `Lya absorbing bottom: True` instead puts the planet-ward face of that slab at the bottom of the domain, where the H_2 layer beneath the base absorbs Lyα (a pure absorber, as in the Huang et al. 2017 Monte Carlo), which shortens the random walk and lowers $\bar{J}_{\mathrm{Ly}\alpha}$ in the bottom few scale heights only. This is the most physical of the three but still an approximation; see `Update_EXHALE_stage1` §12 for the full formulae.

`EXHALE_transit.py` rebuilds the Hα/Hβ n=2 population from the profiles with the solver's level balance and the solver's rate coefficients, each transcribed as its own function of `exhale_transit_lib.py` (`n2_populations` agrees with the Fortran to 3.1×10^{-13} relative, MEASURED 2026-09-25, `md/stale_texts_20260925.md`); it takes the run's rate set from `Atomic rate set:` and `Legacy_HHe_rates:` of `input.inp`. What it does not share with the solver is the field, which the profiles do not carry: if a `Jlya_file` is set it is read (mode 1), otherwise the mode-0 parameterized estimate above is used (with F_{LyC} rebuilt from the run's H I column and the stellar EUV), and the Balmer continuum that photoionizes n=2 is a blackbody at `Stellar Teff` diluted by $(R_\star/a)^2$ rather than the run's own spectrum. So a converged run with no Lyα input still yields Hα/Hβ predictions, via the default field.

4.5 The photon grid and the spectrum that fills it

Neutral metals have ionization thresholds *far below* the hydrogen Lyman limit (13.6 eV ≡ 912 Å): the lowest is K I at 4.34 eV (2856 Å), followed by Na I, Ca I, Mg I, Fe I, Si I, S I, and C I (Table 4). So does the He $2\,^3$S metastable ($e_{\mathrm{th,HeTR}} = 4.768$ eV = 2600 Å) and, whenever the excited-hydrogen coupling is armed, hydrogen in $n = 2$ ($e_{\mathrm{th,HI}}/4 =$ 3.400 eV = 3647 Å, the head of the Balmer continuum). To photoionize any of them the photon grid has to reach their

thresholds, and the spectrum has to state a flux there: the EUV/X-ray band alone ($\lambda < 912$ Å) leaves every neutral metal un-ionized and artificially abundant.

One spectrum type builds every band. `Spectrum type:` states the spectrum of the *whole* photon grid, the part below 13.6 eV and the Balmer continuum included. No band is filled from a type the input did not select.

`Power-law`. The power law E^{PLind}, normalized on $[E_{\mathrm{low}}, E_{\mathrm{mid}}]$ and $[E_{\mathrm{mid}}, E_{\mathrm{high}}]$ from L_{EUV} and L_X, evaluated wherever the grid reaches. Below E_{low} that is an *extrapolation* of an EUV fit, which is what the type means: real stars do not follow it into the near-ultraviolet (their flux rises steeply there). Treat power-law metal and He $2\,^3$S photoionization as indicative.

`Planck`. The photospheric blackbody $\pi B_\nu(T_{\mathrm{eff}})(R_\star/a)^2$, from `Stellar Teff [K]` and `Stellar radius [R_sun]`, on the whole grid. Startup refuses the type without both. It is an upper bound in the near-ultraviolet, where line blanketing and the Balmer jump depress a real F or G photosphere below the curve, and it is orders of magnitude *below* the chromospheric and coronal EUV of an active star: a run that needs both bands right needs a measured spectrum.

`Load` (`Spectrum file:`). The tabulated SED, on the whole grid. The file's own rows are the grid points, log–log interpolated where a band asks for a flux between them; no extrapolation.

`Monochromatic`. One energy, `Photon energy [eV]`, and no grid at all ($N_\lambda = 1$).

The grid floor is the lowest active threshold, and a loaded table that does not reach it stops the run. The floor (`photon_grid_floor_eV`) is the lowest ionization threshold over every *active* absorber: the He $2\,^3$S metastable when the triplet is included, the neutral stage of every element `metals.inp` gives a nonzero abundance, and H(n=2) when `Stellar Teff` and `Stellar radius` arm the excited-hydrogen coupling. With none of them active the floor is the H I edge. The two analytic types are evaluated down to it, and a loaded table is read down to it.
A loaded table that *ends above* the band an active absorber needs stops the run at startup. There is no key to continue: a truncated table gives that absorber the photoionization rate of a truncation of the stated spectrum, not of the spectrum. The message names the file, the lowest photon in it, the uncovered band in eV and in Å, every absorber whose threshold lies below that lowest photon (H(n=2), the metastable, each active neutral metal), and the remedies: turn the absorber off (`Include He23S? False`; remove the element from `metals.inp`; remove `Stellar Teff` and `Stellar radius`, which is what arms the excited-hydrogen coupling), or supply an SED that reaches the stated wavelength.

How the grid is partitioned (analytic types). For `Power-law` and `Planck` the code builds the grid itself, from *bin edges*: `e_v` holds the *geometric center* of each bin and `de_v` its exact width, so the bins are contiguous and $\sum$`de_v` is exactly the width of the covered band. Six bands, each with its own budget of bins, put the resolution where the absorbers are rather than spreading it over the X-ray decades:

Band	Bins	Present when
$[e_{\mathrm{th,HI}}/4,\ e_{\mathrm{abs,low}}]$	20	the excited-H coupling is armed
$[e_{\mathrm{abs,low}},\ 13.598]$	89	a metastable or low-IP metal is active
$[13.598,\ 24.587]$	50	always
$[24.587,\ 54.418]$	50	always
$[54.418,\ E_{\mathrm{mid}}]$	50	always
$[E_{\mathrm{mid}},\ E_{\mathrm{high}}]$	50	always

$e_{\mathrm{abs,low}}$ is the lowest threshold of an absorber of the *bulk gas* (the metastable or a low-IP metal), so the first band is the one no active absorber has a cross section in; its 20 bins only have to state it, because the H(n=2) photoionization rate is an integral over its own band and not a sum over these bins. The 89 bins of the second give it the bin ratio of the pure-H I band ($\Delta \ln E = 0.0118$), which is what the neutral-metal cross sections need: every metal photoionization rate the grid forms is then within 5×10^{-4} of the exact integral.

Every ionization threshold of an active absorber is a bin edge. A photoionization rate is the rectangle-rule sum $\sum F\sigma/E\,\mathtt{de_v}$, so a bin *containing* a threshold would charge that absorber's cross section, read at the bin center, over the part of the bin lying below the threshold, where it cannot absorb at all. Each band is therefore cut at every threshold strictly inside it: H I, He I, He II, the metastable, H_2 and its dissociative- and double-ionization channel thresholds when the molecular chemistry is on, and every photo-ionizable stage of an element with a nonzero abundance, and the band's budget is shared over the resulting sub-intervals in proportion to their logarithmic widths, at least one bin each, so the budget is met exactly. A band with no threshold inside it keeps its budget in one piece.

The total is therefore $N_\lambda = 200 + N_{\lambda,\mathrm{TR}}$, where $N_{\lambda,\mathrm{TR}}$ counts the bins below the H I edge: 109 for a run with both sub-13.6 eV bands ($N_\lambda = 309$; He $2\,^3$S or metals plus an armed excited-H coupling), 89 without the $n = 2$ band ($N_\lambda = 289$), and 0 for a pure H/He run with no absorber below the edge ($N_\lambda = 200$). The startup report states the type, what the band below 13.6 eV was built from, the grid floor, and the integrated grid flux $\sum F\,\Delta E$ against the nominal $(10^{L_X} + 10^{L_{\mathrm{EUV}}})/(4\pi a^2)$.

The energy bands set by `[E_low,E_mid,E_high]` and the luminosities L_{EUV}/L_X still describe the *ionizing* (H/He-driving) part of the spectrum above 13.6 eV; the bands below it are in addition to them, which is why the integrated grid flux and the nominal flux need not agree.

4.6 Diffusive separation of helium and metals

By default EXHALE is single-fluid: the composition is re-solved each step by *local* ionization equilibrium, which conserves the element ratios, so $n_{\mathrm{He}}/n_{\mathrm{H}}$ (and each metal/H) stays frozen at the input value at every radius. Setting `He_diffusion: True` adds a genuine transport equation for the helium *mass fraction* $X = \rho_{\mathrm{He}}/\rho$ of the two-component H/He gas: advection at the bulk velocity plus a molecular-diffusion *settling* drift of He relative to H,

$$\Phi_{\mathrm{He}} = -\rho(D + K_{zz})\frac{\partial X}{\partial r} - \rho X(1-X) D\, G, \quad G = \frac{\Delta m_{\mathrm{eff}}\, m_{\mathrm{H}}\, g_{\mathrm{eff}}}{kT}, \quad g_{\mathrm{eff}} = -\frac{1}{\rho}\frac{\partial p}{\partial r},$$

where the settling acceleration g_{eff} is the pressure gradient of the mixture, i.e. gravity plus the acceleration of the wind (g in a hydrostatic column, $g + v\,\partial v/\partial r$ in a steady wind), with eddy diffusion K_{zz} (`He_Kzz`, default 0) acting on

Table 4: Neutral-metal photoionization thresholds (first ionization stage of each modeled element), as $E_{\rm th}$ and the corresponding wavelength $\lambda_{\rm th} = hc/E_{\rm th}$. A loaded SED must extend to at least the longest $\lambda_{\rm th}$ among the *active* metals, or the run stops (§4.5). The He $2\,^3$S metastable (4.768 eV, 2600 Å) and H(n=2) (3.400 eV, 3647 Å) floor the grid the same way. Values are the VFKY96/NIST thresholds used in the code (`species_table.f90`).

Species	$E_{\rm th}$ [eV]	$\lambda_{\rm th}$ [Å]	Note
K I	4.34	2856	lowest, sets the grid floor when K is present
Na I	5.14	2413	
Ca I	6.11	2028	
Mg I	7.65	1622	
Fe I	7.90	1569	
Si I	8.15	1521	
S I	10.36	1197	
C I	11.26	1101	
(H I)	13.60	912	Lyman limit (reference)
O I	13.62	910	$\simeq$ Lyman limit
N I	14.53	853	above the limit (already in the EUV band)

the gradient term only. The binary coefficient D is **stage-resolved**: each element is split into its ionization stages and the friction is built pair by pair, hard-sphere for neutral–neutral pairs (Banks & Kockarts 1973), the induced-dipole (polarization) form of Schunk & Nagy (1980, eq. 17) with the ion charge for non-resonant ion–neutral pairs, and Coulomb for ion–ion pairs, then mixed over the stage fractions (Blanc's law, $1/D_{\rm eff} = \sum_{s,t} y_s y_t / D_{st}$). This matters above the ionization front, where Coulomb friction suppresses the separation by orders of magnitude relative to the neutral hard sphere and keeps the metal mixing ratios flat in the wind. Helium settles out relative to the escaping hydrogen and He/H *declines with altitude*, as in Taylor et al. (2025) and Xing et al. (2023). The equation is solved conservatively (implicit, tridiagonal) and the new split is written back into the composition conserving mass. Physics refinements: `He_ambipolar` (default on) reduces the settling in the ionized wind via the polarization field, computed from the electron-pressure gradient; `He_alphaT` adds thermal diffusion; `He_metal_diffusion` extends the same operator to each trace metal (heavier elements settle more, so they deplete faster than He).

The strength of the separation scales as (settling speed)/(wind speed): it is suppressed by fast escape and is stronger for heavier species. On a gentle escaper (HD 209458b) helium itself separates and the He 10830 line weakens; on a furious escaper (WASP-121b) helium is dragged out almost unfractionated while the heavy metals still settle strongly. Full physics, derivation, and a two-planet before/after comparison: `docs/version_compare.{tex,pdf}` and `md/version_compare.md`; the stage-resolved coefficient is derived and measured in `Update_EXHALE_stage1` §68 and §71. The flags are **default off**, so standard runs are byte-identical to before.

The enthalpy the moving elements carry. Particles that move relative to the mass-averaged velocity carry their enthalpy with them, so the energy equation of a diffusing mixture has the divergence of the interdiffusion enthalpy flux

$$q_d = \sum_s h_s J_s \qquad \text{(Cook 2009, Phys. Fluids 21, 055109, eqs. 11–13),}$$

with J_s the diffusive mass flux of component s and h_s its specific enthalpy (the enthalpy and not the internal energy, because J_s is taken relative to the mass-weighted velocity). In the binary model every species moves with its element and the electrons with their ions, $J_1 = -J_{\rm He}$, so $q_d = J_{\rm He}(h_{\rm He} - h_1)$, plus $\sum_X J_X(h_X - h_{\rm H})$ for each metal with `He_metal_diffusion`. The h_s are the *sensible* enthalpies of the caloric equation of state, $\gamma/(\gamma-1)kT$ a particle with the electrons an ion carries, plus the rovibrational energy of H_2; formation energy is not in them because it already rides with the moved nuclei through the formation-energy reference of the energy update (`binary_element_diffusion.f90`, `component_specific_enthalpies`). J is the operator's whole diffusive flux, eddy term included. Its divergence is taken on the faces and volumes of the element rows, so the energy it moves telescopes over the column, and it enters both routes: the marching update subtracts $\Delta t\,\nabla\cdot q_d$ from the thermal energy after an accepted element step, and the stationary energy row adds $\nabla\cdot q_d$ of the state's own element fluxes. `Interdiffusion enthalpy flux:` (Table 3) is **on by default whenever `He_diffusion` is on**, because the term belongs to the equation; the key exists to turn it off for comparison with published models that omit it. MEASURED on the molecular LHS 1140 b He/H = 1.8 state after 24 outer passes with the term (not yet certified, carrier row 1.6×10^{-5}): T lower by a median 166 K over 1.3–2 R_p (at most 236 K, –4.4%), He/H –3.1%, the He $2\,^3$S column –1.3%, $\log_{10}\dot{M}$ +0.002 (`md/Update_EXHALE_stage3.md` section 29.3).

4.7 Lower-atmosphere connection (Tier 1–3)

Three opt-in layers connect EXHALE's 1-μbar base to the atmosphere below (survey, implementation, examples and figures: `docs/lower_atmosphere_coupling.pdf`). All are **default off**.

Tier 1: analytic lower column (consistency report). Add

```
Lower column: 1.36        # radius of the 1-bar level [R_J], e.g. transit radius
```

and EXHALE integrates the isothermal-$T_{\rm eq}$ hypsometric column with the chemical-equilibrium H_2/H/He partition (Koskinen et al. 2022) from 1 bar to 1 μbar and *reports* the derived base radius (equilibrium value + a fully-atomic bracket), the base H_2/H/He fractions and μ, next to the input "Planet radius". Nothing is overridden. A large base $q_{\rm H_2}$ on a genuinely cool planet signals that the molecular physics below is required.

Tier 2a: molecular base (EOS only). `Molecular base: True` removes the H_2-bound particles from the base particle budget (`ntot_bc`) using the same equilibrium fit: a lower base pressure and heavier base μ, with the species arrays still atomic. Crude on its own, and it is what Tier 2 turns on for itself, so the key is worth setting by hand only for a run that wants the base EOS without the chemistry.

Tier 2: molecular chemistry.

```
Molecular chemistry: True   # H2, H2+, H3+, HeH+ in the coupled equilibrium
```

adds the four molecular species to the coupled ionization equilibrium (the Koskinen et al. 2022 reaction network), the H_2 photoionization opacity and heating (Yan et al. 1998 cross section), and the optically-thin H_3^+ infrared cooling (the LTE cooling function of the MiZATeP line list, Mizus et al. 2017, times the non-LTE factor of Miller et al. 2013; Miller's LTE fit was removed on 2026-10-10). The output `Ion_species*.txt` gains four columns (`H2 H2p H3p HeHp`). It also turns `Molecular base` on automatically, so that key need not be given: with an atomic `ntot_bc` under a molecular base the base temperature is inflated by the particle-count ratio ($\sim$1.8$\times$ for a fully molecular base) and thermally dissociates the layer the chemistry just built, so the parser couples the two rather than leaving the inconsistent combination reachable. Requires He. `He_diffusion` is no longer refused beside it (milestone M4, 2026-08-26): the element transport of `binary_element_diffusion` closes over the molecular carriers, and the `mol_diffusion` regression case runs the combination. A `metals.inp` may be present: the trace metals are then solved in the same system as the molecular network (`System_HeH_mol_metals`), because the two blocks share the free electron density and the metals are the dominant electron donors in the shielded molecular base. H_2 photodissociation in the Lyman–Werner bands is available and opt-in: add `Stellar LW flux [erg/cm2/s]:` with the band flux at the planet. Caveats: without `Molecular carrier transport: True` (its default is on only with `Oxygen chemistry`) the molecular species are closed by their local equilibrium, with no advection; the caloric equation of state is the H_2 ladder of `Caloric EOS` (default), not $\gamma = 5/3$. On hot Jupiters the result is a thin molecular base under a sharp $H_2 \rightarrow$H front with an atomic wind above: the atomic assumption becomes a result rather than an input.

Converging a molecular run takes three keys an atomic run does not need: `Solver: Newton`, `Resid tol: 1.0e-7` and `Max steps: 150000`. (Until 2026-10-10 the first read `Solver: Newton 5.0e-2`, a *du* hand-off threshold raised because the *du* descent of a molecular run is not monotonic; the value is now a face spread, and the face-spread hand-off has not been re-measured on a molecular run.) And $\|R\|$ is set by the two or three cells just above the base, so a run stopped at the default `Resid tol: 1.0e-5` leaves the molecular layer still cooling (its energy residual there is still 0.7$\times$ the local H_3^+ cooling rate). Marching alone never reaches that layer (its H_3^+ cooling time is $\sim 10^7$ CFL steps), so the Newton finish is not optional. The converged layer is much colder than the relaxation snapshot ($\sim$400 K on HD 209458 b, 190–300 K on a hot Uranus) and, without `Base IR field`, that collapse is the model's own steady state rather than a prediction; see `docs/lower_atmosphere_coupling.pdf` §7–§9 and open item (G).

Tier 2b: oxygen chemistry.

```
Oxygen chemistry: True             # OH, H2O, CO in the same coupled system
Molecular carrier transport: True  # the default: carriers transported
Stellar LW flux [erg/cm2/s]:      1248.5  #  912-1201 A at the planet (LW)
Stellar Lya flux [erg/cm2/s]:    14957.2  # 1215.67 A, the Lya line   (B2)
Stellar FUV B3 flux [erg/cm2/s]:  1374.1  # 1231-1450 A               (B3)
Stellar FUV B4 flux [erg/cm2/s]: 46294.8  # 1451-2304 A               (B4)
```

With `Spectrum type: Load` the three continuum lines (LW, B3, B4) may be omitted: each band is then integrated from the spectrum file over its own interval, a trapezoid over the file's rows (the file is already at the planet), and the

setup report names the source of every band. Only the Lyα line is always a key, because a line flux reconstructed from observations is a better number than a trapezoid over the file's rows across 1202–1230 Å. A stated value, zero included, always wins over the file. Tier 2 can build a molecular base but not *decide* how molecular it is: the H_2/H partition is either imposed through the `q_H2_base` handoff or taken from a chemical-equilibrium fit, neither of which is photochemistry. Tier 2b is the layer that computes it, by adding the oxygen carriers that the catalytic cycle runs on. Measured on a photochemical model of the HD 189733 b base, the oxygen family carries 96–99.6% of the net H_2 destruction there, and the whole of it sits in three reactions and four photolysis channels:

$$\mathrm{OH+H_2 \leftrightarrow H_2O+H}, \quad \mathrm{O+H_2 \leftrightarrow OH+H}, \quad \mathrm{O(^1D)+H_2 \rightarrow OH+H},$$
$$\mathrm{H_2O}+h\nu \rightarrow \mathrm{OH+H\,/\,H_2+O(^1D)\,/\,O+H+H}, \quad \mathrm{OH}+h\nu \rightarrow \mathrm{O+H}.$$

Every coefficient is transcribed from a named evaluation with its validity range at the code site, and the reverse of each two-body pair is generated by detailed balance against a NIST-JANAF Shomate table rather than transcribed independently: the pairs run close to cancellation on a hot base, where an inconsistent pair gives an arbitrary net rather than a small error. The audit behind the numbers is a separate audit.

O(^1D) is closed by a local steady state, not carried as a transported species: its only sink in this set is $\mathrm{O(^1D)+H_2} \rightarrow \mathrm{OH+H}$, and that gives it a chemical lifetime of 1.6×10^{-3} s at the HD 189733 b base (shorter than every other time scale in the problem by many decades), so the steady state is the accurate treatment. Its density is reported as a diagnostic column.

CO is an oxygen reservoir, not a reactant. It locks 45–46% of the oxygen at every measured level, so a network that gave the whole oxygen abundance to the water family would over-supply the OH cycle by about a factor two. It has no chemical source in this network: with `Molecular carrier transport: False` it sits at its own $\mathrm{CO} \leftrightarrow \mathrm{C+O}$ chemical equilibrium, which keeps it associated below ∼3000 K and dissociates it by ∼4000 K, above the $\mathrm{H_2} \rightarrow$H front and below the wind, so the carbon and the oxygen of the wind stay atomic. With the transport on it is a transported reservoir, and "inert" would then mean *indestructible*: measured before the ceiling below was added, the wind carried CO out to $1.67\,R_p$ and 2×10^4 K, where it held 55% of the oxygen and switched off the O I and C II cooling of the whole wind. The transported CO is therefore capped at the chemical equilibrium of its own (n, T): a one-sided constraint, inactive through the molecular layer and exact far above the turnover, which over-suppresses *quenched* CO in the narrow interval around 3000–4000 K where the dissociation is neither fast nor negligible. **Neither limit is kinetics** (the truth is quenching, and the audited set has no CO rate to give it), so a run whose *outer* wind is cool can hand the transit tool an O I column short by whatever fraction is still molecular. Every run therefore prints the molecular oxygen fraction at the top of its domain, and the number of cells the ceiling touched.

The FUV arrives as four bands with named fluxes, not as part of the photon energy grid: that grid is floored by the ionization thresholds of the run's active absorbers (§4.5) and carries their opacity, not the H_2 line system or the H_2O and OH continua, and a power-law type below 13.6 eV gives only an XUV fit extrapolated downward rather than a stellar FUV spectrum. The edges are fixed by the H_2O branching ratios, not chosen: the three-body channel $\mathrm{H_2O}+h\nu \rightarrow \mathrm{O+H+H}$ opens only in the interval containing Lyα, where it is 12%. H_2O and OH absorb these bands as *continua*, so the attenuation is $\exp(-\tau)$ on their star-ward columns and no self-shielding function is needed (unlike the H_2 Lyman–Werner band).

One interval, one flux, one beam. Over 912–1201 Å the same photons are absorbed by H_2 in the Lyman and Werner lines and by H_2O and OH in a continuum. That interval is therefore band `LW`, entered once through `Stellar LW flux`: supplying it twice would count its energy twice on the way in, and letting each absorber attenuate a private copy of the beam would count it twice on the way down. Both are closed. The H_2 rate is multiplied by $\exp(-\tau_{\rm cont})$, the H_2O + OH continuum term of Draine & Bertoldi (1996) eq. (40) (identically 1 for a run without this option), and the H_2O and OH rates in that band are multiplied by $1-A$, the fraction of the band the H_2 lines have *not* removed. A is the column integral of the *pump* cross section $\sigma_{\rm diss}/p_{\rm eff}$ of the same level-resolved table the H_2 rate comes from, so the photons the beam loses to the lines and the photons the dissociation rate spends are one number: $p_{\rm eff}$ is by definition the dissociations per photon removed from the beam. A cannot exceed 1, and measured over the whole table it reaches 0.69–1.00 at the top of its column axis. Until 2026-09-06 A was DB96 eq. (39), an equivalent width of the narrower 912–1110 Å band, and the two normalizations disagreed by 45 per cent. The column above a cell is not one slab: A is summed cell by cell along the path, each cell above contributing the table integral across its own column interval at its own $(T, n_{\rm H})$, the same cross section its own H_2 rate is formed from. The split is exact cell by cell, so what the three absorbers take is what the beam loses, and `output/FUV_bands.txt` reports that ledger. A run with the oxygen chemistry on, no spectrum file and `Stellar LW flux` left at 0 is warned that it gets no H_2O or OH photolysis over 912–1201 Å, where those cross sections peak; with a spectrum file that band, and B3 and B4, are integrated from it unless stated.

Three limitations to read before quoting a result. (i) The band average assumes a flat F_λ inside each band. Measured against a real stellar spectrum (the HD 189733 b flux at the planet), that is good to 3% for H_2O and to 18–22% for OH on both halves of `LW`, and to about 35% on `B3`, but **off by a factor 4.6–6.1 on B4**, whose cross section falls three decades across the band; a run in which B4 matters is outside the treatment. Inside `LW` the sensitivity is not uniform: nearly all of it sits shortward of 1110 Å, where H_2O and OH move in *opposite* directions (−60% and +89% for a 3000 K blackbody), while the 1110–1201 Å half is the best-behaved interval of the set for H_2O ($\leq 2\%$). One band average cannot show that; the two half-band numbers of `water_photolysis.f90` can. (ii) The Lyα band uses the *incident* stellar flux, attenuated only by the H_2O and OH continuum: the H I resonance scattering that actually decides how much Lyα reaches a molecular base is not applied, so that band's rate is an upper bound. Note also that setting `Stellar Lya flux` for the H(n=2) pumping therefore drives the photolysis as well; the startup report prints all four band fluxes so this is visible. (iii) With molecular carrier transport enabled, H_2^+, H_3^+, and HeH^+ are transported; O(^1D) remains on its local balance. The band average applies one transmission to a whole band, so the rates are lower bounds deep in the water layer. Every run writes its own chemical, advective and diffusive time scales (`output/Oxygen_chemistry.txt`), which say where the answer is chemistry, where it is flow, and where the two are comparable.

Molecular carrier transport is enabled by default with oxygen chemistry (`Molecular carrier transport: True`) and disabled by default without it. H_2, H_2^+, H_3^+, HeH^+ and, when oxygen chemistry is enabled, OH, H_2O and CO are advanced by an implicit backward-Euler diffusion-advection step solved together with the same chemistry rows the local equilibrium uses, so a rate that changes in one changes in the other. The molecular coefficients come from Blanc's law over the background carriers with the rigid-sphere pair coefficients of Banks & Kockarts (1973) that the element-diffusion operator already carries; the eddy coefficient is read from `He_Kzz` or from a lower-atmosphere profile and from nowhere else; settling uses the same settling acceleration $g_{\rm eff} = -\rho^{-1}\partial p/\partial r$ as the element operator, with no

thermal diffusion. Each cell's *element* totals are unchanged by the step (only the partition among an element's carriers moves), so element conservation holds by construction and ρ is untouched; every positivity or element limiter that fires is counted and reported in the trailer of `output/Oxygen_chemistry.txt`. This operator therefore transports *speciation* and not elements, which is why it cannot be run together with `He_metal_diffusion`: that option transports elements, and CO carries an oxygen and a carbon nucleus and so cannot follow two element factors at once. `Molecular carrier transport: False` restores the earlier local steady state; it isolates the chemistry for testing and is not a model of a base, because at the HD 189733 b 1 μbar level the H_2 chemical and flow times are comparable. With $K_{zz} = 0$ and no profile the transport is pure molecular diffusion (the wrong limit for a lower atmosphere, where eddy mixing is what holds the composition well mixed below the homopause), and the run says so at startup rather than refusing.

The option removes coolants it does not replace. Where the oxygen is in H_2O and CO the [O I] 63/145/44 μm fine-structure lines and the C I/C II line cooling correctly stop (those atoms are not there), but EXHALE carries no H_2O or CO infrared bands to put in their place, and H_3^+ is its only molecular coolant. Measured on the hot-Uranus molecular gate, the layer just above the base cools 15$\times$ more slowly with the option on than with it off, at comparable heating. That is open item (G) arriving through the composition instead of through the temperature, and it is why the resulting molecular-layer temperature is not a prediction. Every oxygen-chemistry run reports how much of its oxygen and carbon is molecular, where, and that those nuclei no longer cool.

The option is **default off** and refuses combinations that would give one quantity two owners: `q_H2_base` (it computes that partition), and `He_metal_diffusion` (the trace-metal transport counts each metal element over its ion stages alone, so it would transport an oxygen reservoir missing everything bound into OH, H_2O and CO (about half the element), and moving the carriers with it needs a rule for CO, which carries an oxygen *and* a carbon nucleus). `He_diffusion: True` on its own is accepted: the H/He element operator scales every H-bearing species by the same factor, so the molecular carriers already move with hydrogen there. A lower-atmosphere *profile* is accepted beside it: the profile owns the region below the matching level and the chemistry the region above it.

Tier 3: `base.inp` handoff (and VULCAN). An optional `base.inp` in the run directory overrides the base state before the derived constants are formed (each override is echoed; absent file = no-op):

```
T_base    1140.0     # [K]      -> overrides T0
r_base    1.402      # [R_J]    -> overrides "Planet radius" (1-ubar radius)
p_base    1.0e-6     # [bar]    -> the level every other value refers to
q_H2_base 0.42       #          -> H2 volume mixing ratio at the base
HeH_base  0.083      #          -> overrides the He/H number ratio
O_H_base  4.90e-4    #          -> O/H nuclei ratio, overrides metals.inp
Kzz_base  1.0e9      # [cm2/s]  -> sets He_Kzz (element + carrier transport)
```

Keys are matched as labels, by the same anchored rule the `input.inp` reader uses, and every key carries a **category** that states what the value is allowed to do to the wind. There are five, and the same table is carried by `read_base_inp`'s header comment and by `md/input_schema.md` §2c:

Key	Category	What it sets, and where the value is consumed
(comments only)	provenance	Nothing. Which code, network and profile produced the file. The one comment that is parsed is `# solution_id`, and only beside a profile.
`T_base`, `r_base`, `p_base`, `q_H2_base`	EOS boundary	`T0`, `R0`, `p_base_bar`, `q_h2_base`: the state of the gas at the matching level. `p_base` also *fixes* that level: $n_0 = p_{\rm base}/(k_{\rm B}T_0 n_{\rm tot,bc})$, so `Log10 lower boundary number density` is redundant beside it and is refused if it disagrees by more than 1%. `q_H2_base` replaces the chemical-equilibrium fit in `comp_ntot_bc`, i.e. it sets the base *particle count*, and it is also imposed on the H_2 partition of the inflowing ghost species, the particle count then coming from that same species state. It is refused above the attainable ceiling $0.5/(0.5+{\rm He/H})$ rather than capped.
`HeH_base`, `<El>_H_base`	elemental reservoir	`HeH`, and `X_<El>` hence `melem_ab`: the nuclei ratio of an element to hydrogen, which the wind then transports.
(none today)	initial guess	A key here would seed a profile the solver is free to move away from. None exists yet.
`Kzz_base`	boundary constraint	`he_kzz`: the eddy coefficient the element-diffusion operator imposes at the base; inert with `He_diffusion` off.
`q_H2O`, `q_CO`, ...	diagnostic	Nothing; metadata written as comments by the generators. Unknown keys are ignored.

The distinction the categories draw is a real one, and `q_H2_base` is where it bites: it is an **EOS anchor**, not a composition pin. Nothing holds H_2 at that value, no H_2 profile is seeded from it, and the molecular network moves away from it in the first cell.

`<El>_H_base` is the elemental form: `C_H_base`, `N_H_base`, `O_H_base`, `S_H_base` and the same spelling for `Mg Si Ca Na K Fe` (any of the ten element symbols of the species table), each the El/H *nuclei* ratio at the handoff level. It overrides `metals.inp` for that element, it turns the metal system on by itself when it is the only nonzero abundance, and on a restart (`Load IC? True`) it renormalizes that element's whole loaded column onto the stated ratio, preserving the ionization split and the radial shape. An element that reached the run from `metals.inp` alone is loaded untouched. With a lower-atmosphere profile in use (§4.8) the EOS boundary, elemental reservoir and boundary constraint keys of this file are **refused** with an `error stop` naming the key and its category: the profile states all of them at the matching level, and a scalar beside it would be the same quantity said twice. Provenance comments and diagnostic keys remain allowed.

The pre-step can be invoked *by EXHALE itself*: with

```
Lower atmosphere: vulcan 1.36     # or: analytic 1.36  (arg = 1-bar radius R_J)
```

in `input.inp`, the code runs the `VULCAN` photochemistry (or the analytic column) at startup when no `base.inp` exists, then proceeds with the wind solve on the generated base, subroutine-style, with the first VULCAN run taking hours

and later runs reusing the cached result in <run_dir>/vulcan_work/. Omitting the key disables the pre-step entirely. The VULCAN/ tree (obtained by setup_vulcan.sh; it ships FastChem inside it) carries its own citation requirements, see §4.9. Two generators are provided in src/utils/:

- run_lower.py <run_dir> --r1bar <R_J> [--guillot]: the analytic column (isothermal $T_{\rm eq}$, or the Guillot 2010 semi-grey $T(p)$); pure Python, no dependencies beyond numpy.
- vulcan_to_base.py <out.vul> <run_dir> --mp <M_J> --r1bar <R_J>: converts a **VULCAN** photochemical-kinetics output into the same base.inp, with the *photochemical* H_2/H dissociation state (on HD 189733 b photochemistry dissociates $\sim$11$\times$ more H at 1 μbar than chemical equilibrium) and a hypsometric base radius using VULCAN's own $\mu(p)$ and $T(p)$.

Running VULCAN is handled automatically by the driver src/utils/vulcan_driver.py, so no manual VULCAN setup is required. When the pre-step is requested it (i) obtains VULCAN if needed (via src/utils/setup_vulcan.sh; see §4.9) and copies the EXHALE/VULCAN/ tree into <run_dir>/vulcan_work/ (once, then reused), (ii) builds a Guillot (2010) semi-grey $T(p)/K_{zz}$ atmosphere for the planet and selects a stellar UV spectrum from the shipped spectra by the host $T_{\rm eff}$ ($>$ 6800 K $\rightarrow$ 51 Eri; 5300–6800 K $\rightarrow$ solar; 4300–5300 K $\rightarrow$ ϵ Eri; cooler $\rightarrow$ GJ 436), (iii) compiles FastChem if its binary is missing, (iv) runs VULCAN to steady state, and (v) converts the result to base.inp. The first run takes hours; the converged <name>-photo.vul is cached and reused on subsequent runs (regenerate with --force). The code root is taken from EXHALE_ROOT when set (for relocated installs). VULCAN's networks are H/C/N/O(/S): it supplies the H_2/H state and molecular mixing ratios, *not* the atomic-metal release (Na/Mg/Ca/Fe): metals.inp stays user-supplied.

4.8 The lower atmosphere as a profile (Lower atmosphere profile:)

The Tier-3 handoff of §4.7 states the lower atmosphere at one level, as a handful of scalars in base.inp. A *profile* states it over an interval of pressure instead: the lower model (photochemistry, with or without a climate solution) is solved on a column that reaches *above* the level where EXHALE places its base, and the whole overlap is handed over as one table. Point EXHALE at it with

```
Lower atmosphere profile: lower_atmosphere_profile.dat
```

in input.inp; the file lives in the run directory and the name is free. Two things ask for the interval rather than the level. The first is eddy mixing: the homopause is where K_{zz} crosses the molecular diffusion coefficient, which rises by nearly five decades between the base and $1.2R_p$, so a single He_Kzz cannot state where it lies (§4.6). The second is the elemental-flux closure below, in which the composition at the base stops being an input and becomes an output.

Schema. A block of # header lines of the form # key value, then a fixed-column table in full double precision running **deep to shallow**: the pressure column must be strictly decreasing, and at least two levels must be present. Three header keys and one schema line are required:

- `solution_id`: a sha256 fingerprint of the lower model's configuration (code and version, mechanism and thermodynamic data, stellar flux and dilution, planet mass and reference radius, elemental abundances, matching pressure, trial fluxes). It is what makes "the same lower-atmosphere solution" a checkable statement.
- `p_match_bar`: the matching pressure, i.e. where EXHALE places its base. It must lie inside the pressures the table covers.
- `p_top_bar`: the shallowest level carried. It may not be larger than `p_match_bar` (a file that stops below the match states nothing at the level the wind starts from); equal is the matching surface as the lower model's top, the convention of the Photochem adapter, above which the wind alone describes the gas. It must not lie above the shallowest tabulated level.
- `# columns:`: the names of the columns, in file order. Every column is found *by name*, because the producers order their element list differently from run to run.

The remaining header keys are optional and are provenance or metadata: `source_code`, `source_version`, `mechanism`, `stellar_flux`, `p_deep_bar`, `trial_flux_H`, `trial_flux_He`, `iteration`, `reached_steady_state` and `notes`, and from the Photochem adapter `trial_flux_H2`, `measured_flux_H2` (the hydrogen H_2 was asked to carry and carries through the top face, $\mathrm{g\,s^{-1}}$), `recombination_heat_flux_cgs` and, only when a steady-state pass of the chemistry reached its steady state on the single retry from a stalled integration (relative tolerance 10^{-4}, the same acceptance checks), `steady_state_retry`, which names the retried passes (`reached_steady_state` stays `T`). A profile with no `iteration` index runs perfectly well; it is simply not drivable by the closure iteration, which the run reports as `lower_profile_iterable F`. Header keys this version has no consumer for are ignored rather than refused.

Nine columns are required: `p` (bar), `r` (R_J), `T` (K), `n_tot` ($\mathrm{cm^{-3}}$), `rho` ($\mathrm{g\,cm^{-3}}$), `Kzz` ($\mathrm{cm^2 s^{-1}}$), `q_H2`, `q_H` and `X_He`. Optional columns are the other elemental ratios `X_C`, `X_O`, `X_Mg`, ...(El/H *nuclei* ratios, summed over every carrier, for any element `species_table` knows), the elemental fluxes `F_H` and `F_He` ($\mathrm{g\,s^{-1}}$, outward positive), and any number of diagnostic mixing ratios such as `q_H2O`. Columns the reader has no consumer for are **kept, not dropped**, so a file survives a round trip through the tools. A data row that does not carry as many numbers as the schema line names stops the run.

Every quantity is interpolated **linearly in** $\log p$ and is never extrapolated; a target that falls exactly on a tabulated level returns that level's value bit for bit, which is what lets a profile with a node at the matching pressure reproduce the scalar handoff exactly. K_{zz} is the one quantity taken as a profile rather than at a point: it is mapped onto the radial grid through the file's own radius column, and above the shallowest level carried each cell takes that level's value, the eddy coefficient being a lower-atmosphere property about which the file makes no statement above its top.

What it supplies, and what `base.inp` may no longer say. At the matching level the profile sets the base temperature T_0 (column `T`), the base radius R_0 (column `r`), the base pressure p_{base} (`p_match_bar`), the base H_2 fraction (column `q_H2`), the He/H ratio (column `X_He`) and each elemental reservoir it carries, which overrides `metals.inp`; the hydrogen partition of the inflowing ghosts is the profile's own census at the match (the atomic share $q_{\mathrm{H}} n_{\mathrm{tot}}/n_{\mathrm{H}}$ of the hydrogen nuclei, $n_{\mathrm{H}}/n_{\mathrm{tot}}$ from the columns `rho`, `n_tot` and `X_<El>`, every other hydrogen nucleus on H_2), and the two ghost cells below the base, for the advection and for the conduction bath alike, carry the profile's temperature at their radius.

It also fixes the base *level*: n_0 follows from `p_match_bar` through $n_0 k_B T_0 n_{\mathrm{tot,bc}} = p_{\mathrm{base}}$, the same relation a `base.inp` carrying `p_base` uses, because the composition, $q_{\mathrm{H_2}}$ and temperature the profile hands over all refer to that one level. A `Log10 lower boundary number density` line given beside a profile is then saying the same thing twice: the two must agree to 1%, and if they do not, startup stops and prints both numbers, the n_0 each implies, $n_{\mathrm{tot,bc}}$, and the two ways to fix it. The columns `n_tot` and `rho` are carried but not imposed: the base particle count and mass follow from T_0, He/H, $q_{\mathrm{H_2}}$ and the elemental ratios exactly as they do on the scalar path.

Because the profile is now the single source of those quantities, a `base.inp` sitting beside it may no longer state them. Three categories of key are **refused**, and each stops the run naming the offending key: the EOS boundary (`T_base`, `r_base`, `p_base`, `q_H2_base`), the elemental reservoirs (`HeH_base` and any `El_H_base`) and the boundary constraint (`Kzz_base`). Provenance comments and diagnostic keys are still allowed, and indeed a `base.inp` whose whole content is provenance is what the adapters write. Such a file must carry a `# solution_id` comment, and it must match the profile's: a `base.inp` beside a profile with no `solution_id` line, or with a different one, stops the run, because two files describing the same gas at the same level from two different lower-atmosphere solutions are not a model of anything.

`He_Kzz:` is treated differently, because it belongs to `input.inp` and not to the handoff: it is not refused, but it is *inert* while a profile is in use (the eddy coefficient comes from the `Kzz` column), and the run prints a warning saying so.

The resolved-configuration report records what was used: `lower_profile_present`, and with a profile in use the file name, the `solution_id`, the source, `p_match_bar`, `p_top_bar`, the trial fluxes, the iteration index and whether the profile is iterable.

Producing a profile. Three tools in `src/utils/` do this, sharing one command line and one schema module (`lower_profile_schema.py`, which also computes the `solution_id` and writes the paired provenance-only `base.inp`). `photochem_to_lower_profile.py` is the production path: it solves a gas-giant photochemistry model to steady state on the column and writes the handoff, with the temperature structure either prescribed (`--tp-file`) or solved for by the radiative–convective climate model (`--climate`), in which case the tropopause, and with it the water cold trap, is part of the solution rather than an input. `vulcan_to_lower_profile.py` converts a finished VULCAN `.vul` output into the same format; it is the cross-check, not a second production path, because VULCAN carries no climate model and so can never be the route away from a prescribed $T(p)$.

Both take the same statement of the eddy coefficient. By default the `Kzz` column is whatever $K_{zz}(p)$ the underlying solution carried; `--kzz-const` K replaces it with one value at every level, and `--kzz-power` α, with its amplitude `--kzz-ref` K_{ref} and the pressure that amplitude is stated at `--kzz-ref-bar` P_{ref} (default 1 bar), replaces it with the saturated gravity-wave form $K_{zz}(p) = K_{\mathrm{ref}}(p/P_{\mathrm{ref}})^{-\alpha}$: $\alpha = 1/2$ being the Lindzen (1981) slope fitted by Parmentier et al. (2013), 0.4 that of Charnay et al. (2015). The two are mutually exclusive, `--kzz-power` without `--kzz-ref` is refused, and the power-law settings enter the `solution_id` only when they are in use, so stating neither leaves a run and its fingerprint exactly as they were. On the Photochem path a stated coefficient is the profile the chemistry is *solved* on and not only the one reported. There is no ceiling on the power law: the file's own top bounds it, and EXHALE gives every cell above the shallowest tabulated level that level's value, so the level the amplitude is anchored at decides what the wind inherits above the profile. Finally `element_flux_closure.py` drives the pair of codes to their fixed point: the trial elemental fluxes of the header are imposed on the chemistry's upper boundary, the wind's own elemental flux

is measured face by face and reduced over a radial window (for hydrogen the hydrogen-nucleus flux `_F_Hnuc_*`, since the chemistry divides `--trial-flux-H` by $m_{\rm H}$; the component-1 flux `_F_H_*` carries the metals too and is used, with the reason logged, only for an EXHALE that writes no `F_Hnuc`), and the two are iterated (under-relaxed) until they agree, at which point the base composition is an output of the coupled problem instead of an input to it.

The energy coupling: the column below the level under the heat the wind conducts. With `Conduction` on, the wind conducts heat into the lower atmosphere through its base face (`base_conductive_flux_cgs` of `EXHALE_resolved.out`, §5.13). On a cold column that heat raises the matching-level temperature the wind is built on, so the two models are also coupled in energy. `photochem_to_lower_profile.py --conducted-heat-flux` F [erg cm^{-2} s^{-1}, positive into the column; default 0, which writes the profile without it] solves the column energy balance below $p_{\rm match}$ (`src/utils/conducted_heat_column_balance.py`),

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}z}\left(r^2\kappa\frac{\mathrm{d}T}{\mathrm{d}z}\right) = C(T,p) - C(T_0(p),p), \qquad \kappa\frac{\mathrm{d}T}{\mathrm{d}z} = F \text{ at } p_{\rm match}, \quad T = T_0 \text{ at the bottom},$$

with κ the mixture conductivity of `Conduction`, C the band emission of CH_4, NH_3, H_2O, C_2H_2, CO and CO_2 from the climate model's own k-distributions and $T_0(p)$ the climate temperature; the climate's radiative heating is held, and the conduction is the total one, so the conduction of T_0's own gradient is a term of the balance and T_0 is not its solution at $F = 0$ where T_0 has a gradient (update log stage 3 section 128; $F = 0$, the default, skips the solve and keeps the climate column), and iterates it with the chemistry (the chemistry re-solved on the new T until $T_{\rm match}$ moves by less than `--column-energy-tolerance`, default 1 K, at most `--column-energy-max-iterations`, default 8). The options:

- `--column-energy-treatment` `nlte` (default; a vibrational band with published V–T rates is a two-level system, the rest in LTE), `nlte_partial` (two-level also for a band whose rate set misses a partner, a lower bound on its emission), `lte` (every band in LTE, an upper bound on the cooling) or `rotational` (rotational bands only, a lower bound). The published rates and their ranges are tabulated, with their sources, in the module's PUBLISHED RATES section.
- `--nh3-h2-rate-analog` `cd4_bending` (default) or `water_bending_overtone`: the analog whose measured $k(H_2)/k(He)$ turns the measured $NH_3\ \nu_2 + He$ rate into the unmeasured $NH_3\ \nu_2 + H_2$ one (`nlte` only).
- `--column-band-transfer` `thin` (default: all of the band emission leaves; where more than 10% of a significant bin's emission excess would be absorbed by the column above, the state is kept and *flagged*, `thin_layer_limit_exceeded T`), `cooling_to_space` (the upward half escapes through the column above with the k-distribution transmission, Dickinson 1972 Eq. 33, refused, `ColumnTransferRefusal`, when the neglected exchange inside the warmed layer moves $T_{\rm match}$ by more than `--column-exchange-shift-max`, default 1 K) or `cooling_to_space_exchange` (the same with that exchange solved, so it has no such bound).
- `--top-pressure-passes` (default 8): the largest number of re-converged solves that move the top face of the Photochem grid to $p_{\rm match}$ (by $H\ln(p_{\rm face}/p_{\rm match})$, halved after a change of sign); the agreement they must reach (10^{-3} in $\ln p$) does not change with it. The model top is the matching level: `--toa` and `--p-top-bar` are refused, and the column energy balance sees no absorber above the match.

- `--trial-flux-H2` [$\mathrm{g\,s^{-1}}$]: the part of the elemental hydrogen flux H_2 carries through the top face, as the wind states it (`base_face_H2_flux_cgs`); atomic H carries the rest, and an inflow of atoms deposits its recombination heat, $D_0/2$ per atom, in the column with the shape of the column's H + H (+M) rate (`recombination_heat_flux_cgs` in the header).
- `--climate-t-deep-guess` (default 400 K): only the starting value of the climate root solve for the deep temperature, raised where the default does not bracket the root (a warm deep column); the root does not depend on it.
- `--face-geometry wind|per_model` (default `wind`, update log stage 3 section 129): with `wind` the chemistry and the climate column take the stellar energy the wind's input states, $\xi_w F$ per unit face area, ξ_w the wind's `dayside_dilution` (1, $1/2$ or $1/4$ by its `2D approximate method`), given by `--wind-dilution` or by the face file of `--wind-face-optical-depth` (a disagreement between the two is refused, and so is `--zenith`, which the convention sets); `per_model` keeps each code's own convention (Photochem $0.5F$ at `--zenith`, the climate column $F/4$), as runs before 2026-10-10 did. `element_flux_closure.py` adds `--face-geometry wind --wind-dilution` ξ_w unless `adapter_args` names a geometry, and refuses a wind run whose `EXHALE_resolved.out` `dayside_dilution` differs.
- `--eddy-heat-transport on|off` (default `off`, sections 130 and 133): off, the chemistry's K_{zz} is a tracer-mixing coefficient that carries no heat; on (needs `--climate`), the climate stratosphere is solved in radiative, molecular and eddy energy balance with the eddy flux $\rho c_p K_{zz}(\mathrm{d}T/\mathrm{d}z + g/c_p)$ (Johnstone et al. 2018, Eq. 44) and the column energy balance carries the same flux. On LHS 1140 b (ξ_w = 1) that reference climate exists only up to $K_{zz} \approx 1.4 \times 10^6\,\mathrm{cm^2\,s^{-1}}$, so on is refused at the closures' 10^9. The stratosphere solve (`stratosphere_energy_balance`, `radiative_convective_column.py`) puts the climate object's solution back on exit, because clima's `TOA_fluxes_dry` leaves its last trial profile in it (section 134); the option off never calls it.

The profile header then carries `conducted_heat_flux_cgs`, `column_energy_treatment` and `column_T_match_K`, with `nh3_h2_rate_analog` for `nlte`, `column_band_transfer` off the default `thin`, and under `thin` the flag `thin_layer_limit_exceeded T|F`, `thin_layer_max_absorbed_fraction` and `thin_layer_max_absorbed_bin`; the column solution is recorded in `conducted_heat_column.txt`. The flux, the treatment, the matching-level temperature and (off `thin`) the transfer enter the `solution_id`; the analog does not. The producer's own elemental fluxes across its top may also appear as `measured_flux_H` / `measured_flux_He`. EXHALE has no consumer for any of these keys and ignores them. In `element_flux_closure.py` the configuration key `energy_coupling` (default false) makes the conducted flux a third fixed-point variable: each iteration's column is solved under $F^{(k+1)} = F^{(k)} + \omega_E(F^{(k)}_{\mathrm{meas}} - F^{(k)})$, ω_E = `energy_coupling_omega` (default 0.5), started from `--fcond0` or from the seed run's `base_conductive_flux_cgs`; the history gains `F_cond_trial`, `F_cond_meas` and `T_match`; and convergence also requires the conducted-flux residual $|F_{\mathrm{meas}} - F|/|F_{\mathrm{meas}}| \leq$ `tol` (0 for two exact zeros) and $|T^{(k)}_{\mathrm{match}} - T^{(k-1)}_{\mathrm{match}}| <$ `T_match_tol` (default 6 K). The hydrogen split at the matching surface is a fourth variable whenever the wind writes `base_face_H2_flux_cgs`: Φ_{H_2} [$\mathrm{g\,s^{-1}}$ of the hydrogen nuclei H_2 carries], passed to the chemistry as `--trial-flux-H2`, takes the same damped update toward the wind's F_{H_2}, the history gains `trial_F_H2`, `meas_F_H2` and `eps_H2` after the energy columns, and convergence

also requires $|F_{H_2} - \Phi_{H_2}|/|F_H| \leq$ `tol`; the first Φ_{H_2} is `--phi0-H2`, or none (the chemistry's own rule) until a wind has stated it. With the chemistry's top at the matching level the overlap window is empty, and the flux is read on the steady window $r \geq r_{esc}$. A `--resume` refuses a last history row without a resolved H and He measurement at the requested tolerance, or one that the saved wind's own measurement does not reproduce. An EXHALE exit status 2 (converged, not certified) continues the closure with the flag; a status 3 (stationary solve failed), a 2 without the certification's closing line (a runtime error) or any other status stops it before anything is measured (§3.7). Each wind solve of the closure is seeded by `src/utils/map_state_to_grid.py --ic`, which carries the previous state onto the new reservoir and radius: the output is an initialization seed (`certified=F cert_reason=mapped_seed`); a target cell centred between the base face and the source's first cell centre takes that cell's values and is never interpolated against the reservoir ghost row; `--reservoir El/H v` rescales an element's stages (for helium HeH^+ as well, the hydrogen it holds taken back from the hydrogen-only species so that n_H of every row is held) and refuses a base row that departs from v by more than 10^{-6} (`BASE_ROW_RATIO_TOL`, the tolerance the loader itself accepts He/H with).

See also. `examples/17_lower_profile/` is a ready-to-run HD209458b case whose `make_example_profile.py` regenerates the synthetic column it ships (§7.3). The authoritative column-by-column schema is `md/input_schema.md` section 2d.

4.9 Obtaining VULCAN and FastChem

The lower-atmosphere pre-step uses the **VULCAN** photochemical-kinetics code, which ships the **FastChem** equilibrium-chemistry code inside it. These are third-party open-source codes and are *not* committed to the EXHALE repository (`VULCAN/` is git-ignored). Fetch and prepare them with

```
src/utils/setup_vulcan.sh
```

which clones VULCAN into `EXHALE/VULCAN/`, applies the two modifications EXHALE needs, builds FastChem, and removes stale run products. EXHALE also runs this automatically the first time the `vulcan` pre-step is used and `VULCAN/` is missing. Obtain updates from, and *cite*, the original sources:

- **VULCAN**: `github.com/exoclime/VULCAN` (mirror `github.com/shami-EEG/VULCAN`). Cite Tsai et al. (2017, ApJS 228, 20) and Tsai et al. (2021, ApJ 923, 264).
- **FastChem**: `github.com/NewStrangeWorlds/FastChem` (shipped inside VULCAN). Cite Stock et al. (2018, MNRAS 479, 865), and Stock et al. (2022, MNRAS 517, 4070) for FastChem 2.

The modifications `setup_vulcan.sh` applies to the cloned VULCAN tree (i.e. the only changes needed if you set it up by hand):

File (in the VULCAN clone)	Modification
`make_chem_funs.py`	add `encoding=None` to the `np.genfromtxt(vulcan_cfg.com_file, ...)` call so the element-conservation check does not crash on Python 3
`vulcan_cfg.py`	set `use_photo = True` and `use_live_plot = False` (this file is the driver's template)
`output/*.vul`	delete (run products; regenerated in `<run_dir>/vulcan_work/`)
`fastchem_vulcan/`	`make` to build the `fastchem` binary

On the EXHALE side (in this repository) the pre-step adds `src/utils/setup_vulcan.sh`, `vulcan_driver.py` and `vulcan_to_base.py`, and the `Lower atmosphere:` key in `input_read.f90/parameters.f90`.

5. Output files

All profiles are written to `output/` as plain text, one row per radial cell, in *cgs* units. Two flavors exist: the equilibrium files (`*.txt`, written during the run) and the advection-corrected files (`*_adv.txt`, written once at the end by the post-processor, which re-solves the ionization balance including advection of the wind). The `*_adv.txt` files are what the analysis tools read by default, and the equilibrium files are what they read when the solved state is asked for (`EXHALE_TRANSIT_STATE=solution`, §6.2) or when a stationary solve wrote no advection-corrected pair. A stationary run that alternates a wind solve with a composition relaxation also keeps the state of its last completed outer pass, the generation `output/pass_state/<state_id>/` named by `output/pass_state/current` (§5.4), a restartable pair that is never a certified state. Every equilibrium pair is accompanied by `conserved_state.txt`, the same state in code units to the last bit, which a restart uses in place of converting the pair when it matches the run (§5.3).

Caveat: the `_adv` profile at a "breathing" base. The advection-corrected (`*_adv.txt`) *temperature and ionization* would otherwise develop a spurious spike / sawtooth in the dense lower atmosphere (typically $r \lesssim 1.1$–$1.2\,R_p$ for strongly Roche-filling planets such as WASP-121b). Two compounding causes, both confined to the base, are responsible:

1. *A breathing, inflowing base (a 1-D limitation).* When the planet nearly fills its Roche lobe the 1-D wind is subsonic / only marginally launched at L_1, so the dense lower atmosphere does not settle into a clean monotonic outflow but into a slowly recirculating state with small *negative* (inflow) velocities and a stagnation point ($v = 0$) near the wind base. This is intrinsic to the spherically-symmetric (1-D) geometry (removing it would require a 2-D/3-D model), and is not a numerical bug.

2. *A spurious hot root in the metal-cooled post-processor.* The post-processor re-solves the energy/ionization balance in each cell *including* advection, assuming an outflow (each cell takes its history from the cells below it, the recursion running outward along the flow). In the dense base the non-monotonic metal line-cooling curve gives the energy equation a second, spurious *hot* root that the Newton (MINPACK) solver can land on; through that recursion one bad cell then cascades outward, producing the sawtooth and an overshoot spike (`post_process_adv.f90`). A guard rejects solves that fall outside a band around the converged equilibrium temperature, but mild spikes *within* that band can survive.

The post-processor guards against cause 2 directly: wherever the flow is *not* a clean outflow ($v \leq 0$, the breathing inflow base) it skips the advection correction and keeps the converged equilibrium ionization and temperature (`post_process_adv.f90`). This removes the spurious spike and sawtooth at the source (it breaks the upwind cascade), while preserving the genuine few-percent `_adv` correction in the outflow above the stagnation point. Cause 1 (the breathing base itself) is intrinsic to the 1-D geometry and remains, but it is now represented there by the smooth, physical equilibrium solution. The shipped `_adv` profiles are therefore clean, and the temperature panel of Figure 6 plots the `_adv` profile directly. (Velocity and density are *identical* in the `eq` and `_adv` files: the post-processor corrects only the ionization and temperature.)

Validity range of the advection correction. The inflow condition above is one of four under which the post-processor keeps the converged equilibrium state instead of correcting it. A cell is left at equilibrium when

1. $v \leq 0$ in the cell or in the one below it (inflow): the step takes its history from the cells below, which are not upstream when the gas moves inward, and the residence time h_j/v_j is negative;
2. $\mathrm{Da} = (h_j/v_j)\,\nu_{\min} > 100$, with $\nu_{\min}$ the smallest total destruction-plus-formation rate of the populations the system solves (H I; with helium also He I, He II and the He $2\,^3$S level, whose radiative rate $A_{31} = 1.272 \times 10^{-4}$ s^{-1} sets the floor): the gas relaxes to local ionization equilibrium many times over while it crosses the cell, so the equilibrium solution already solves the advection ODE and the correction can only add integration error;
3. $x_{\mathrm{H\,II,eq}} < 10^{-6}$: the residuals carry the *neutral* fraction and report the ion density as $(1-x_{\mathrm{H\,I}})n_h$, so the ion fraction is resolved only to the solver's absolute tolerance $\sqrt{\epsilon} = 1.5 \times 10^{-8}$. In a shielded base with $x_{\mathrm{H\,II,eq}} \sim 10^{-13}$ the returned value is numerical noise of arbitrary sign, which the upwind cascade would then carry outward as negative ion densities;
4. the mass flux through the cell changes by too large a fraction of itself. Every equation the post-processor solves is a steady equation integrated along the recorded flow, and the flow it integrates along carries a fractional change of the *face* mass flux across the cell,

   $$m_j = |R_1(j)|/s_1(j),$$

   the mass row of the state divided by the largest term that row itself holds. This is the same operator the stationary certification measures the mass row by (`steady_residual.f90`, `certification.f90`), evaluated by calling those routines on the state handed in rather than by re-deriving it from center-sampled $\rho v r^2$, which is the same quantity in the continuum and a different discrete test. The terms the steady equations drop are the ones the mass divergence puts into them, each of them m_j times a term they keep, so *the correction is first order in m_j*: a cell with m_j at or below `adv_conditional_tol` $= 10^{-2}$ is corrected, and its correction is a *conditional* one, accurate to that fraction of itself in the mass flux; a cell above it keeps the run's own temperature and the ionization equilibrium at that temperature. The value of m_j for every row is written out as the `adv_mass_row` column, so a reader has the condition of each row and not only its verdict.

The fraction above is *not* the tolerance of a certified stationary state. That is `cert_tol_mass` $= 3 \times 10^{-12}$, the number the stationary certification judges the mass row of the *whole* state by, and the verdict it reaches on the input state

travels in the file's own `# adv_input_certified` line. The two are different statements about one measure and the file carries both: the rows of a certified state are corrected, while a corrected row is not thereby certified. Conditions 1 and 2 are properties of the flow, 3 of the representation of the unknown and 4 of the state the equations are integrated on, so none of them depends on whether metal cooling is active. Condition 4 refuses both the temperature and the composition of the cell, condition 1 both as well (the step has no upstream history either way), and conditions 2 and 3 the composition alone. The energy solve additionally applies a *thermal* Damköhler condition of its own, $\mathrm{Da} = (h_j/v_j)|\Gamma - \Lambda|/u_{\rm th} > 1$: where the local radiative balance rather than the flow sets the temperature, the temperature to keep is the run's own, which is the root of that same balance with every channel the run solved. Conditions 1 to 3 trigger in the dense base; condition 4 follows the state, and it is the one that separates a converged wind from a snapshot. The counts are printed at the end of the run, and the verdict and the measure of every row are recorded in the `adv_T_status`, `adv_comp_status` and `adv_mass_row` columns of `Hydro_ioniz_adv.txt` (§5.1). Full account: `md/postprocess_advection_validity.md`.

A last condition concerns the temperature alone. The post-process carries the H/He and trace-metal gas and omits the molecular species, so its energy balance lacks the molecular chemistry heat and the H_3^+ and molecular band cooling the run solved. Wherever those omitted channels, measured on the state handed in as the difference between the run's heating and cooling and the post-process's own at the run's temperature and composition, exceed the channels it carries, the row keeps the run's temperature with `adv_T_status` = 3 (`unsupported`). MEASURED on the LHS 1140 b molecular states: 196 / 389 / 776 rows on 500 / 1000 / 2000 cells, up to 1.134–1.137 R_p, and an He 10830 equivalent width that moves by at most 1.2×10^{-4} of itself.

How the correction is integrated. Every equation the post-process integrates along the flow is an ODE in r, $\mathrm{d}y/\mathrm{d}r = F(y, r)$: the ionization balance $v\,\mathrm{d}x/\mathrm{d}r = R(x)$ and the steady internal-energy equation. Since 2026-09-27 each is advanced cell by cell with the variable-step second-order backward differentiation formula (BDF2),

$$y_j - a_1 y_{j-1} + a_2 y_{j-2} = g\, h_j F(y_j, r_j), \quad h_j = r_j - r_{j-1},\ w = \frac{h_j}{h_{j-1}},\ a_1 = \frac{(1+w)^2}{1+2w},\ a_2 = \frac{w^2}{1+2w},\ g = \frac{1+w}{1+2w},$$

with every rate, heating and cooling term and the velocity of the residence time taken at r_j (`post_process_adv.f90` states the derivation; the form is that of Li & Liao 2022, SIAM J. Numer. Anal. 60, 2253, read as eq. (1.4) of arXiv:2201.00527). It is second order in the cell width and L-stable, so a population that relaxes within a cell settles on its local equilibrium instead of alternating about it as it would under the trapezoid rule. Backward Euler, $y_j - y_{j-1} = h_j F(y_j, r_j)$, takes the first step of the recursion, the first step after a cell it did not integrate, a step whose width ratio exceeds the zero-stability bound $w \le 1 + \sqrt{2}$, and the retake of a BDF2 step whose populations come out negative or whose temperature has no root. The run log counts the BDF2 and backward-Euler steps and the retakes. Until 2026-09-27 every step was backward Euler, a first-order recursion that was the main source of the first-order grid convergence of the equivalent width; MEASURED on nine certified LHS 1140 b states (500, 1000 and 2000 cells), the post-process share of the error falls from +3.0% / +1.0% to below 3×10^{-4}, and the equivalent width itself moves by −3.9% (500 cells) to −0.97% (2000 cells) at He/H = 1.6 (`md/Update_EXHALE_stage3.md` section 29.10). **Equivalent widths quoted from _adv files written before 2026-09-27 are those of the first-order recursion.**

The fractions the steps carry are fractions of one element's nuclei, which move with that element's velocity: with element continuity $\mathrm{div}(n_{\rm el} v_{\rm el}) = 0$ the steady stage balance is $v_{\rm el}\,\mathrm{d}x/\mathrm{d}r = S/n_{\rm el}$. So every hydrogen row is stepped with

the residence time h_j/v_{H} and every helium row with h_j/v_{He},

$$v_{\mathrm{He}} = v + \frac{J}{\rho Y}, \qquad v_{\mathrm{H}} = v - \frac{J}{\rho(1-Y)},$$

J the diffusive helium mass flux of the state (`helium_diffusive_face_flux`, the mean of the two faces of the cell) and Y the helium mass fraction; both equal v without `He_diffusion`, where the product is unchanged bit for bit. The validity conditions above use the same two velocities (inflow of either element refuses the cell; each element's Damköhler number takes its own residence time). The stage's own eddy term $-n_{\mathrm{el}} K\,\mathrm{d}x/\mathrm{d}r$ is not carried; its size is written at the code site (`post_process_adv.f90`). **Every _adv product of a run with `He_diffusion: True` written before 2026-10-04 stepped every element on the bulk velocity** (`md/Update_EXHALE_stage3.md` section 82).

The state of the derived product as a whole. The row fields say what each row is; the `# adv_derived_state:` line of both _adv headers says how the solves that produced the rows ended. Its fields: `verdict` (`rejected` or `unverified`), `chemistry` (`complete` or `cells_failed`), the column energy solve of the transport terms (`column`: `not_run`, `rejected` or `converged_on_residual`; `reason`: `linear_solve`, `nonfinite`, `non_positive_T` or `residual_above_tolerance`; the iterations, the triggering cell, the number of unknowns, and the worst energy row: `column_worst_cell`, `column_worst_residual` in erg cm^{-3} s^{-1} and `column_worst_residual_rel`), the passes whose column was rejected, the cell counts, `cells_marching_unrooted`, and `outer=unverified`; every count is of physical cells, the ghost rows excluded. The column iteration stops when no temperature moves by more than 10^{-10} of itself (or after 100 iterations), and is then judged on its residual: with the transport terms formed again at the final profile, every unknown's energy row $R = q_0 f/(\mu_+\mu_-\Delta r)$ must satisfy $|R| \le a_E + 10^{-6} S_E$, with $a_E = 10\,\epsilon$ times the sum of the magnitudes of the parts the row is assembled from (the grouped terms, the two halves of the advective difference and the three face terms of the conduction divergence; the stagnated $|R|$ is 0.73–0.99 ϵ of that sum on three LHS 1140 b conduction states), S_E the largest of its physically grouped terms (the advective energy divergence as one term, the $p\,\mathrm{div}\,v$ work, heating, cooling, the conduction divergence and the two enthalpy divergences) and 10^{-6} the energy tolerance of the stationary certification. Since 2026-10-01 the column is no longer tested against a temperature band: the 0.5–2 times band of the cell-by-cell Brent solve refused a root of the conduction-coupled column. A cell whose own marching energy step finds no root (its bracket $[0.05, 4]\,T_{\mathrm{in}}$, or that band) is an unknown of the column by default, started from the run's temperature and written `corrected` when the column converges (`cells_marching_unrooted`). A `residual_above_tolerance` rejection prints the energy row of the worst cell term by term. What the acceptance tests is the column subproblem with lagged coefficients: the conductivity and the element and carrier enthalpy divergences are formed on the *input* composition, not the derived one, and the fine-structure escape probabilities are those of the start of the pass, which the cooling written after the last column forms again. Not tested: the full energy residual of the derived state on one composition with its own coefficients, the convergence of the fixed ten outer passes, and the momentum balance of the corrected pressure on the supplied ρ and v. So no record says more than `unverified`. After every composition update of a pass (the H/He sweep, the metal re-solve and its failure branch) the particle count, n_e and the mean molecular weight are formed again, then the ionized fraction of the photoelectron partition, and only then the heating, the energy solve and the output pressure. A rejected column keeps the marching profile of that pass in the numeric columns and writes every one of its unknowns `adv_T_status` = 2 (`failed`); the

verdict is `rejected` when the column of the last pass was rejected or any row is failed. A file without the line has an UNKNOWN derived state, which is not a pass. The `Restart intent: stationary evaluate` route exits 7 when the record is rejected and the input is not refused, 2 when the certification of the stored pair refuses the input (whatever the record says; the record is written either way), and 0 otherwise (7 also when no record was filled); a gfortran runtime error also exits 2, so a 2 is to be read with the log. `EXHALE_transit.py` refuses an `adv` spectrum whose record is rejected, or that has no record (a file written before the record existed; its derived state is unknown), with exit 7 unless `EXHALE_TRANSIT_DIAGNOSTIC=1`, which makes it as diagnostic data and writes the refusal or the unknown verdict into the metadata of every saved curve; its row census refuses a row whose temperature field is not zero or whose composition field is failed.

5.1 `Hydro_ioniz.txt` / `Hydro_ioniz_adv.txt`

Thermodynamic profiles (7 columns, and three more in the `_adv` file). The dimensionless code variables are restored to cgs with the normalizations n_0 (base density), $v_0 = \sqrt{k_B T_0/\mu}$, $p_0 = n_0 \mu v_0^2$, T_0 (base temperature), $q_0 = n_0 \mu v_0^3 / R_p$.

Col	Quantity	Unit	Note
1	r	R_p	cell-center radius (the lower face of cell 1 is the base level $r = 1$)
2	ρ (mass density)	$m_{\rm H}\,{\rm cm}^{-3}$	mass density ($= \rho_{\rm code}\, n_0$)
3	v	${\rm cm\,s}^{-1}$	radial velocity ($\times v_0$)
4	p	${\rm erg\,cm}^{-3}$	pressure ($\times p_0$)
5	T	K	temperature ($\times T_0$)
6	heating	${\rm erg\,cm}^{-3}\,{\rm s}^{-1}$	XUV photoionization heating ($\times q_0$)
7	cooling	${\rm erg\,cm}^{-3}\,{\rm s}^{-1}$	radiative cooling ($\times q_0$)
8	`adv_T_status`	none	`_adv` file only: integer, the validity of the temperature of the row
9	`adv_comp_status`	none	`_adv` file only: integer, the validity of its composition
10	`adv_mass_row`	none	`_adv` file only: m_j, the measure both fields were decided by

Column 2 is a *mass* density expressed in units of $m_{\rm H}$ per ${\rm cm}^3$ (metals included under the `eos_metals` policy), not a number density; multiply by $m_{\rm H}$ to get ${\rm g\,cm}^{-3}$.

Columns 8 to 10: what each row of an _adv file is. Written only in `Hydro_ioniz_adv.txt`; the equilibrium file has seven columns, because its rows are the run's own solution by construction. `Ion_species_adv.txt` carries the two status fields as its last two columns, so a tool reading the species file alone sees the same verdicts.

The advection post-process solves the steady ionization and energy equations along the recorded flow. The temperature of a row and its composition are *two* statements: a row whose temperature was corrected can carry the run's own composition and the reverse, so each has its own field and neither is the first refusal of a single mixed integer. Both take the same five values.

Value	Name	Meaning
0	`corrected`	the steady equation was solved and adopted for this quantity
1	`retained`	the run's own value was kept: m_j above the fraction below, the local radiative balance rather than the flow sets T (thermal Damköhler > 1), or the gas enters the cell
2	`failed`	the cell solve did not converge or its root was out of band, or (temperature) the cell is an unknown of a rejected column energy solve, whose temperature is then the marching profile of the pass
3	`unsupported`	the closure does not cover this cell class. Composition: the omitted molecular and oxygen carriers hold more of the particle count than the species the reconstruction carries, or one population the advection systems solve (H^+, He^+, He 2^3S) loses faster to H_2 than to the channels those molecule-free systems carry, on the state handed in (the composition class; both fields are then 3). Temperature: the heating and cooling channels the molecule-free post-process omits (molecular chemistry heat, H_3^+ and molecular band cooling) exceed the ones it carries, so the run's own T is kept (below about 1.13 R_p on the LHS 1140 b molecular states)
4	`not_evaluated`	the post-process did not reach this row (a ghost row below the first cell at which its upwind difference can be taken)

A refused field is the run's own value *exactly*, not to a tolerance: a composition field that is not `corrected` carries the handed-in species and metal columns, a temperature field that is retained, unsupported or not evaluated the handed-in temperature, and a row whose composition is refused and whose temperature is refused but not failed carries the handed-in pressure, heating and cooling as well. The exception is the temperature of a `failed` row of a rejected column, the marching profile of the pass. A row with a corrected temperature and a refused composition is therefore not the run's solution as a whole. The composition class compares two losses of one population with no free parameter; passing it does not bound the omitted H_2 loss to a small error, only says that it is not the larger one. A `corrected` row is a *conditional* correction, accurate to `adv_conditional_tol` of itself in the mass flux, and column 10 carries the m_j it was judged by, so a reader can weigh a corrected row rather than only count it.

The file's own # header states all of it: the schema version, the legend of both fields, an `# adv_status_counts` line with the number of rows carrying each value in each field, the `# adv_stationarity_operator` the measure comes from, the `# adv_conditional_tol` line with the fraction a corrected row is accurate to, the `# adv_input_certified` verdict of the stationary certification on the whole input state (a separate statement from the fraction), and two declarations of what the product is not: `# adv_product`, that this is a *one-way* correction on a fixed density and velocity field, so a converged temperature solve does not make the row a self-consistent solution and a profile assembled from corrected and retained rows need not satisfy any interface balance, and `# adv_model_restrictions`, that the molecular and oxygen columns are the equilibrium solution, that those species are absent from the particle and electron counts every equation here is solved with, and that the H_3^+ infrared cooling is absent from its energy balance; and the `# adv_derived_state:` record of the product as a whole (§5.). A file written before the schema carries no `# adv_schema` line, and the validity of its rows is UNKNOWN, not corrected.

`exhale_io.load_hydro` returns the three columns under their own names together with the header's statements,

and exhale_io.Run exposes them as adv_T_status, adv_comp_status, adv_mass_row and adv_conditional_tol (None for an equilibrium file); EXHALE_transit.py prints the census of the rows its chords sample with the fraction and the largest measure among them, and copies both into the comment metadata of every saved curve. The two lowest rows are ghost rows below the first computational cell, whose temperature the energy loop never corrects whatever their status.

Header lines. The file opens with # comment lines that every loader skips: the schema tag, the # columns list, the # rows line saying which rows are ghost cells, and

```
# coupling: sec_ion=T sec_ion_step=2402 recon=WENO3 certified=T mode=init
```

which records the run state the profile was produced under: every switch that changes *while* a run converges rather than being fixed by input.inp. Whether the SvS85 photoelectron secondary ionization was being applied and at which step it was armed; the discrete reconstruction operator in force, which is not rec_method alone while the PLM $\rightarrow$ WENO3 continuation is armed; whether the state passed the stationary certification, with cert_reason= beside it whenever the verdict is qualified (no_stationary_claim for a run that ended on a cap, certified_in_wind for a state whose species rows were gated at $r \geq 1.20$ only; the states written certified=F on purpose name what they are: marching_snapshot_s<step> for the periodic write of a marching run, measured by nothing at that step, pass_snapshot_p<k> for a pass state, projected_seed, option_change_seed after a named Restart option change:, legacy_factor_token, molecular_seed, and mapped_seed for the output of map_state_to_grid.py); and the Run mode the run declared. The coupling line also carries iontrans=T and he23strans=T when those transports are on. Two fields that used to follow sec_ion_step, valve and fluxconst, were the switches the old base closure chose for itself and are gone with it (§3.6); a reader of an older file still finds them there, and the parser takes the fields it recognizes one at a time.

This matters because the file is also the restart input: copied to Hydro_ioniz_IC.txt and read back under Load IC? True, its sec_ion field tells the new run to start the coupling where the old one left it, instead of re-running the staging on a state that no longer needs it (§144 of md/Update_EXHALE_stage1.md). A file written before this line existed simply has no such line, and such a restart starts staged exactly as it did before. The other fields are reported, never imposed: input.inp states them for the new run.

Provenance. Three further # lines say what produced the file, so that a profile found on disk in a year can be tied to an executable and an input without asking anyone:

```
# provenance: git=6d07d48afd41 tree=dirty run=2026-09-03T20:19:09
# provenance: ck_input=1260237361560544917 ck_base=-1 ck_metals=4583919659595562347
# provenance: recon=WENO3 base_bc=isothermal carrier=none restart_schema=3 resid_def=145 N=500
```

git and tree are the revision the *executable* was built from and whether the working tree carried uncommitted changes at that moment; the Makefile stamps them in and the run never calls git. No build time is compiled in: a timestamp in a constant would make two builds of one source different files, and the md5 of the binary is what ties a bitwise regression verdict to a revision, so the time shown is the *run* time, from date_and_time. The ck_ fields are checksums

of the input files as they were read (−1 means the file was absent), and the last line records the discretization, the lower boundary model, the molecular carrier model, the version of the restart layout, the version of the residual definition the acceptance gates used, and the grid size.

The lower boundary the state was written with. Two further # lines carry the boundary itself, because a state read back a year later has to say which boundary it is a solution of:

```
# boundary_model characteristic_face_ps_reservoir_C_minus_contact_upwind_
    ghost_fixed_point_seed_reservoir_row_face_at_level_
    discrete_equilibrium_v5
# boundary_reservoir version 1 p[p0] T[T0] nhat[n0/rho0] r_level[Rp] . . .
```

(each is one line in the file). The first names the model that built the face state (§3.6); the identity changes when the model does, v2 being the contact upwind without a stated ghost composition, v3 adding the ghost seed and the fixed point the ghost composition is held to, v4 (2026-09-27) putting the lower face on the base level, so that the reservoir is the face state it owns with no transport along an isentrope, and v5 (`. . . _face_at_level_discrete_equilibrium_v5`, 2026-09-29) building the interior pressure at the face and the ghost cells from the well-balanced reconstruction's own discrete equilibrium. A run with `Well balanced: True` states v5; a run without it builds the v4 boundary unchanged and states v4. The second states the prescribed reservoir with the version of its prescription: the pressure and temperature at the base level, the particles per unit mass of the reservoir gas and the radius the three are stated at, each written at round-trip precision, so the four numbers on the line are the numbers the run held and not a rendering of them. The entropy is not among them: it is carried implicitly, as the isentrope through (p, T) at the base composition. The reservoir travels with the state because it is the only boundary input the physical column cannot reconstruct; the ghost rows the file holds are the boundary's own output and are rebuilt at the read from the column and this run's reservoir. A restart written under an older boundary model is reported, not refused on that account: the loader names both models and says that the state's residual is not the residual it was written with. A v3 or older state, however, was written on a grid whose cell centers lie lower (by half a cell on the `Mixed` grid), so the grid check refuses it; it reaches a v4 run only as a seed mapped with `src/utils/map_state_to_grid.py --ic` (§3.6). Both lines are # comments, so no numeric parse and no regression comparison changes.

5.2 `Ion_species.txt` / `Ion_species_adv.txt`

Ion number densities [cm^{-3}] (34 columns). Column 1 is r; columns 2–7 are the H/He ions; columns 8–34 are the 27 metal-ion stages, in this order:

```
r ; HI HII HeI HeII HeIII HeI* ; CI CII CIII OI OII OIII NI NII NIII MgI MgII MgIII SiI SiII SiIII CaI
                        CaII CaIII NaI NaII KI KII SI SII FeI FeII FeIII
```

(HeI* is the metastable $2\,^3$S triplet.) The ionization fraction of any element is the stage density divided by the sum over that element's stages. With `Molecular chemistry: True` four more columns (`H2 H2p H3p HeHp`, cm^{-3}) are appended after the metals, and with `Oxygen chemistry: True` three more after those (`OH H2O CO`, cm^{-3}); the # `columns` header always names the layout. In `Ion_species_adv.txt` two further columns follow, `adv_T_status` and

adv_comp_status, the validity of the row's temperature and of its composition (§5.1), with the same header block as the hydrodynamic file, so a tool reading the species file alone sees the same verdicts; the measure they were decided by is a column of that other file. Note that with the oxygen chemistry on the OI column means *free atomic* neutral oxygen: the element total is $n(\mathrm{O\,I})+n(\mathrm{O\,II})+n(\mathrm{O\,III})+n(\mathrm{OH})+n(\mathrm{H_2O})+n(\mathrm{CO})$, and the same holds for carbon with CO.

5.3 conserved_state.txt: the exact code-unit state

What: the conserved variables (ρ, ρv, E) and every species fraction f_{sp} of every cell, ghosts included, in code units, written as ES25.17E3 text; 17 significant digits reproduce a binary64 value exactly on reading. *When*: with every equilibrium pair the run writes (the periodic and final states of every route, the EXHALE_DUMP_IC=1 dump, and each pass-state generation, §5.4), after the two dimensional files are closed. *Why*: a restart from the dimensional pair converts physical units back to code units, recovers the species fractions from densities and rebuilds the energy from the pressure through the caloric equation of state, which moves the last digits of the state (on a 20-step wasp_he23off state, 146 of the 500 conserved rows and all 500 composition rows of the physical cells); from this file the restart starts from the bits the run ended on.

Restart: copy it with the pair, to output/conserved_state_IC.txt, and run with Load IC? True. After load_IC has accepted the pair, the file is used only when every recorded line agrees with the run: the format version; the shape (N, N_g, number of species); the rolling checksums of the two dimensional files it was written with (PAIR_CHECKSUM); the reservoir and options fields of the restart metadata (CONFIG); the base boundary model (BOUNDARY); the species order (SPECIES); the normalization constants and planetary parameters (NORMALIZATION); checksums of the physical input files (SOURCE_CHECKSUM: the spectrum file, the lower-atmosphere profile or base.inp, metals.inp, the opacity model and coefficients, the opacity tables) and of the gravitational potential and irradiation field (FIELD_CHECKSUM); the cell and face radii (R, E rows); the data rows (U, F) with their checksum (DATA_CHECKSUM); and the end marker. The checksums are rolling sums for detecting accidental mismatches, not cryptographic digests. A mismatch, a truncated or damaged file, a nonfinite value, a nonpositive density or internal energy in a physical cell, a negative species fraction, or a load that applied a Restart option change:, migrated a legacy factor token or read a pair without restart metadata, leaves the file unused: a notice framed by asterisks names the reason and the restart proceeds from the dimensional pair, which load_IC has already accepted and which remains the state of record. A molecular-seed conversion (its atomic state is read from the seed directory) and a seed whose lower layer is projected with EXHALE_SEED_HYDROSTATIC_CELLS never read it. What is restored: the physical cells and the upper ghosts; the lower ghosts are rebuilt by the boundary condition from the restored interior and the run's reservoir, and their composition is the one load_IC formed. MEASURED: two successive write–reload–write cycles through EXHALE_DUMP_IC=1 reproduce u and f_{sp} of all 500 physical cells of that state bitwise.

Writing never changes the exit status. A state that cannot be written exactly (a nonfinite value, a nonpositive density or internal energy in a physical cell, a negative species fraction in any row, an unreadable input file, an I/O failure) is not written; any older file at that path is removed, because it would describe another state, and a warning names the reason. The dimensional pair is then the restart. A write of a pair without the conserved state, the molecular-seed conversion, removes an older conserved_state_IC.txt beside the pair it writes. The graphical interface copies conserved_state.txt with the pair and deletes a stale conserved_state_IC.txt when the state has

none; src/utils/map_state_to_grid.py refuses a destination that already holds such a file, since a mapped state is a new state.

5.4 pass_state/: the state of every outer pass

The state of the last completed outer pass of the stationary route (§3.7), written since 2026-09-28 so that a run stopped from outside keeps the passes it made, and published as generations since 2026-09-29 (until then it was the pair Hydro_ioniz_last_pass.txt / Ion_species_last_pass.txt, overwritten half by half, md/Update_EXHALE_stage3.md section 33). *When*: after every outer pass after which the iteration goes on, i.e. only where the route alternates a wind solve with a composition relaxation (He_diffusion, or molecular carriers without Coupled carrier solve: True). A pass that ends the iteration publishes nothing, because the run then writes its final Hydro_ioniz.txt / Ion_species.txt, and the last pass of the budget takes no composition update, so its state is that final state; the generation left beside a finished run is one pass older than it. *What*: the state the next pass starts from, i.e. the accepted composition update of that pass (or the entry state of the coupled block where the route handed over to it), written by the same writers and with the same header lines as the final state files, so the two files have exactly the columns of Hydro_ioniz.txt and Ion_species.txt. Writing them changes nothing the run reads afterwards; the final state files of a run are bitwise those it wrote before the snapshot existed (MEASURED on the regression matrix, 2026-09-28, and again for the generations on iontrans_metals and hp_front, 2026-09-29).
Layout: one directory for each published pass,

```
output/pass_state/<state_id>/Hydro_ioniz.txt
output/pass_state/<state_id>/Ion_species.txt
output/pass_state/<state_id>/conserved_state.txt
output/pass_state/<state_id>/manifest.txt
output/pass_state/current
```

with <state_id> = r<run identity>_p<n>. The run identity is formed once at the start of the run from the wall clock to the millisecond and the processor clock count (yyyymmddThhmmss.sss_c<count>), so two runs writing the same pass number never name one generation. manifest.txt states, one key=value a line, the state_id, the run_identity, the pass, the previous_state_id (the generation current named when this one was published), the two file names with their sizes in bytes, conserved_state=conserved_state.txt when the exact state of the pass was written (§5.3), and the certification pair. current is one line, the state_id of the generation a restart takes.
The order of publication: (1) the files are written and closed in output/pass_state/.<state_id>.tmp/; (2) that directory is renamed onto output/pass_state/<state_id>; (3) output/pass_state/current.part is written with the new state_id and renamed onto output/pass_state/current, the single rename that publishes the generation; (4) the generation current named before is kept, and the one it had replaced (its manifest's previous_state_id) is removed, file by file under its name. An interruption at any point therefore leaves current naming a complete generation, or, before the first publication in a directory, no current. An interruption before step 2 leaves a .<state_id>.tmp directory, one between steps 2 and 3 a complete generation no pointer names; neither is read, and neither is removed by a later run.

Headers: both halves carry the `# coupling:` line, with `certified=F cert_reason=pass_snapshot_p<n>` for pass *n* and, as its last field, `state_id=<state_id>`, and a line

```
# pass_snapshot pass=<n> carrier_movement_bound=<b> element_omega=<w> (the state after the composition update of this pass; not certified)
```

A snapshot is *never* a certified state: the certification of a pass is taken before its composition update and the snapshot is the state after it, so no tool may read it as certified. `load_IC` refuses a pair whose halves state two different `state_id` values, and a pair in which one half states one and the other does not (a generation file beside a final state file); a pair in which neither states one (every final state, every file written before the field existed) is read as before. The final state files of a run carry no `state_id`.

Restarting from them: copy the pair of the generation `current` names to `output/Hydro_ioniz_IC.txt` and `output/Ion_species_IC.txt`,

```
G=$(cat output/pass_state/current)
cp output/pass_state/$G/Hydro_ioniz.txt output/Hydro_ioniz_IC.txt
cp output/pass_state/$G/Ion_species.txt output/Ion_species_IC.txt
cp output/pass_state/$G/conserved_state.txt output/conserved_state_IC.txt
```

(the last line where the generation holds the file; otherwise remove an older `output/conserved_state_IC.txt`, which the restart would ignore with a notice) and run with `Load IC? True` as from any state file (the restart contract of §4. applies unchanged). The carrier movement bound and the element under-relaxation factor the run had reached are *not* restored by a restart; to continue at the values of the `# pass_snapshot` line, pass them as `EXHALE_CARRIER_TRUST=<b>` and `EXHALE_DIFF_OMEGA=<w>` (Table 1). MEASURED: a run killed during pass 3 left the complete pass-2 pair, which restarted and continued as the uninterrupted run did (`md/Update_EXHALE_stage3.md` section 29.12, with the pair of that time).

5.5 `Cooling_breakdown.txt`

Radiative cooling by channel [erg cm^{-3} s^{-1}] vs. radius. Columns: `1`=r, `2`=T [K], `3`=n_e [cm^{-3}], `4`=total cooling; `5--10`= the six atomic channels: recombination (the kinetic energy $k_B T\,\alpha\,(3/2 + \mathrm{d}\ln\alpha/\mathrm{d}\ln T)$ of the captured electrons of H II, He II, He III *and* of every recombining metal ion), collisional ionization (the ionization potential of every H, He and metal ionization), then collisional excitation split by absorber into H I (the Lyα-dominated H-line cooling), He I and He II, and finally free–free emission of every ion by its net charge. With the He $2\,^3$S tracked, the He I column carries the metastable's own terms: the net $1\,^1\mathrm{S} \leftrightarrow 2\,^3\mathrm{S}$ exchange, which is *negative* where the superelastic $2\,^3\mathrm{S} \rightarrow 1\,^1\mathrm{S}$ collisions heat the gas, the triplet cascade that radiates 10830 Å and the higher triplet lines, and the $2\,^3\mathrm{S} \rightarrow$ singlet excitations; `11`= the H_3^+ infrared cooling (zero unless the run tracks the molecular network); `12--14`= the H_2, H_2O and CO infrared bands, which are NET rates (emission less absorption of the field from below, so a negative value is that band heating the gas) and are identically zero unless `Molecular IR bands` is on; `15--41`= the 27 metal-line cooling channels (same ion order as `Ion_species`). The channels sum to column 4. Useful for identifying the dominant coolant vs. altitude (cf. Huang+2023 Fig. 10), and the H_3^+ column is the coolant of a molecular base. The `# col` header line always names the layout.

5.6 Heating_breakdown.txt

Volumetric heating by channel [erg cm^{-3} s^{-1}] vs. radius, the companion of `Cooling_breakdown.txt` and written unconditionally at the end of the run. 24 columns: `1`=r [R_p], `2`=T [K], `3`=n_e [cm^{-3}], `4`=total heating; `5--10`= the photoionization heating split by *absorber*: H I, He I, He II, He 2^3S, H_2, and the sum over the metal stages (whose electrons include the Auger electrons of an inner-shell vacancy); `11`= the photoelectric heating of the H(n=2) population (the excited-H model); `12`= the collisional de-excitation of H(n=2), i.e. absorbed Lyα thermalized by a collision; `13`= `heat_He_recomb`, the photoelectrons of the H I, He I, H_2 and metal ionizations driven by the recombination radiation absorbed on the spot (He II and He III recombinations, and the H II ground captures taken by a metal); `14`= the Penning branch of He 2^3S + H; `15`= its associative branch He 2^3S + H $\rightarrow$ HeH$^+$ + e; `16`= He 2^3S + H_2 Penning ionization; `17`= the fragment kinetic energy of the H_2 Lyman–Werner photodissociation; `18`= the collisionally de-excited Lyman–Werner fluorescence; `19`= the collisional reactions of the H_2/He network; `20`= the FUV photolysis of H_2O and OH; `21`= the collisional reactions of the oxygen network including the O(^1D) sink; `22`= He$^+$ + CO $\rightarrow$ C$^+$ + O + He, the charge-transfer destruction of CO; `23`= the fragment kinetic energy of the CO photodissociation; `24`= `heat_charge_exchange`, the energy defects of the charge-exchange reactions the run applies (metal+H, metal+He, metal+metal and He $\leftrightarrow$ H), each less the excitation of a product state that leaves as radiation (an allowed decay, the photon of a radiative transfer, and, below the critical densities, the forbidden lines of an N I, S II or C I product formed out of statistical equilibrium), negative for an endothermic reaction. A channel of physics the run does not carry is identically zero.

One state per output file. These columns are not recomputed for the dump: they are the channel array the ionization sweep filled for the state written beside them, through the one heating assembly `heating_of_composition` (`src/modules/radiation/util_ion_eq.f90`), whose total is the running sum of its channels and is the `heat` column of `Hydro_ioniz.txt`. Column 4 therefore reproduces that column to round-off: measured at 1.7×10^{-16} over the physical cells of the `mol_base_handoff` case (2026-09-06). The header names the layout and is built from the channel list, so a deposit added to the energy equation appears here by construction.

Rebuilding the rates on the written state instead, which earlier versions of this file did, is a different state: the rates a sweep contracts belong to the composition that *entered* it, and cannot be recovered from the composition it returned. That put a lag of one sweep between the two columns, measured at 1.2×10^{-5} in the median cell of the same case.

5.7 Lyman_Werner.txt

H_2 photodissociation diagnostic, written when a molecular run carries a `Stellar LW flux`. 8 columns: `1` r [R_p], `2` T [K], `3` $x_{H_2} = 2n_{H_2}/n_H$, `4` n_{H_2} [cm^{-3}], `5` the star-ward H_2 column N_{H_2} [cm^{-2}], `6` the level-resolved self-shielding factor f_{shield}, a diagnostic, $\sigma_{diss}(N_{H_2})/\sigma_{diss}$ at the bottom of the tabulated column axis, evaluated at that cell's own column, `7` the dissociation rate k_{LW} [s^{-1}], which is the cell mean of the tabulated rate between the cell's two faces and is therefore *not* f_{shield} times a thin-limit rate, and `8` the photodissociation heating [erg cm^{-3} s^{-1}]. It shows directly how deep the band penetrates: $f_{shield} \rightarrow 1$ in the thin wind above the $H_2 \rightarrow$H front, collapsing by orders of magnitude inside the self-shielded molecular base.

5.8 `Oxygen_chemistry.txt`

The solved oxygen partition, written for the equilibrium state of a run with `Oxygen chemistry: True`. 18 columns: 1 r [R_p], 2 T [K], 3 n(O) (*free atomic* neutral oxygen), 4 n(O II), 5 n(O III), 6 n(OH), 7 $n(H_2O)$, 8 n(CO), 9 $n(O(^1D))$ from its local steady state, 10 $x_{H_2} = 2n_{H_2}/n_H$, 11 τ_{chem} [s], 12 τ_{adv} [s], 13 the Damköhler number τ_{adv}/τ_{chem}, 14--15 the H_2O and OH photodissociation rates summed over the bands [s^{-1}], 16 $D(H_2)$ [$cm^2\ s^{-1}$], 17 K_{zz} [$cm^2\ s^{-1}$], and 18 $\tau_{diff} = \Delta r^2/(D_{H_2} + K_{zz})$ [s]. Columns 3--8 sum to the oxygen element total.

$\tau_{chem} = 1/(k_{O1}n_{OH} + k_{O2}n_O)$ is the H_2 lifetime against the oxygen cycle and $\tau_{adv} = r/|v|$ the radial flow time. These are a *required* output rather than a diagnostic. With `Molecular carrier transport: True` a cell whose chemistry is slow is solved rather than assumed away, and the time scales say which part of the profile the reaction set is responsible for and which part the flow is; with `Molecular carrier transport: False` they say something stronger, namely where that limit is simply wrong.

With the transport on, a trailer at the end of the file records the last transport step: `newton_steps`, the relative residual `newton_resid` it stopped at, `limited_cells` (cells where a carrier would have taken more nuclei than its element has), `worst_overshoot`, and `co_ceiling_cells`, the cells in which the transported CO was cut back to the $CO \leftrightarrow C + O$ equilibrium of their own (n, T). A limiter count that is not small is a statement about the step and is reported rather than absorbed.

Every run also closes the file with the **net H_2 loss budget at the base cell**, decomposed into eleven channels from the same coefficients the residual uses and net of its own reverse where a channel has one (so a negative entry is a net H_2 *source* and a share may exceed 1). The option exists to make the oxygen cycle set the base partition, and whether it does is a property of the run: on a 2331 K base the thermal channel $H_2 + M \rightarrow H + H + M$ carries 70–97% of the net and the oxygen cycle a few percent, while on an 864 K base the oxygen family carries 96–99.6%. Read that block before reading the partition.

5.9 `FUV_bands.txt`

How deep each FUV band penetrates, and the band energy ledger. Written for an `Oxygen chemistry` run. 16 columns: 1 r [R_p], 2--3 the star-ward H_2O and OH columns [cm^{-2}], 4--7 the four band optical depths (`LW`, `B2`, `B3`, `B4`), 8--11 the H_2O photodissociation rate in each band [s^{-1}], 12--15 the same for OH, and 16 the total FUV photolysis heating [$erg\ cm^{-3}\ s^{-1}$]. A header line above the data names the flux and the H_2O and OH band cross sections of each band.

The trailer of the file carries the energy ledger. For a beam attenuated as $\exp(-\tau)$ the photons absorbed over a column are exactly $N_b(1 - e^{-\tau_b})$ whatever the density profile, and the file reports the grid sum against that closed form band by band. They agree to round-off at any grid spacing, because each cell's rate is the mean over the cell and not its face value; a residual there means the rates and the columns have come apart. It also reports the absorbed energy split into deposited heat and bond energy, against the incident band flux: the absorbed energy can never exceed it, which is the property that keeps the band average from over-heating.

The last block is the shared beam of the `LW` band. Over 912–1201 Å the same photons are absorbed by H_2 in the Lyman–Werner lines and by H_2O and OH in a continuum, so the closed form the band's sum is checked against is $N_b[1 - (1 - A)e^{-\tau}]$ with A the line-removed fraction, and the H_2 share belongs inside it: the H_2 photon count is the

cell mean of the *pump* cross section $\sigma_{\rm diss}/p_{\rm eff}$, because the pumps that fluoresce back also take a photon out of the band. Since A is the column integral of that same cross section, the two sides are one normalization of one absorption and the `LW` row is a closure residual like `B3` and `B4`. The block reports `LW_H2_absorbed_en`, `LW_cont_absorbed_en`, `LW_incident_en` and `LW_line_removed_frac`. The run warns if the absorbers together take more energy out of the interval than it carries. A separate warning fires where the star-ward column passes the top of the self-shielding table's column axis; above it the table is clamped at its edge cross section, so A goes on growing past 1 and is capped there, which is the one place the `LW` closure residual does not go to round-off.

5.10 `OI_levels.txt` / `OI_levels_adv.txt`

Populations of the three fine-structure levels of the O I $2p^4\,{}^3P$ ground term, written whenever the run carries metals (no key: the presence of `metals.inp`, or of a lower-atmosphere profile with an oxygen reservoir, is enough). 11 columns: `1` r [R_p], `2` T [K], `3` n_e, `4` $n_{\rm HI}$, `5` n(O I), `6--8` the fractional populations $f(^3P_2)$, $f(^3P_1)$, $f(^3P_0)$, and `9--11` the same three as densities [cm^{-3}], which sum to column 5 by construction.

The term is inverted, so $^3P_2$ is the lowest level. These are the lower levels of the O I $\lambda\lambda$1302.168, 1304.858, 1306.029 resonance triplet, which is why they are an output field: a transit forward model that applied the total O I density to all three components would count the same atoms three times. They are not a second calculation – the file is filled by the same three-level statistical equilibrium, with the same electron and H-atom collisional rates and the same line escape probabilities and incident infrared field, that the [O I] 63.2/145.5/44.1 μm cooling is assembled from (`Cool_coeff.f90`). In the dense base the solution is the Boltzmann one to round-off; it departs from it in the thin outer wind, where the levels drain radiatively toward $^3P_2$.

5.11 `Excited_H.txt`

H(n=2) diagnostic (written when the excited-H model is on, i.e. stellar $T_{\rm eff}$ and radius are set). 16 columns: `1` r, `2` T, `3` $n_{\rm HI}$, `4` n_e, `5` $\bar{J}_{{\rm Ly}\alpha}$, `6` n_{2s}, `7` n_{2p}, `8` Balmer proton source $S_{\rm p}$, `9` photoelectric heating, `10` de-excitation heating, `11` $\tau_{{\rm Ly}\alpha}$, `12--14` ground-state HI photoionization / collisional-ionization / recombination rates (for the proton-budget comparison), `15--16` the internal and stellar contributions to $\bar{J}_{{\rm Ly}\alpha}$.

5.12 `EXHALE_setup.out` and the mass-loss rate

`EXHALE_setup.out` is a human-readable setup report (planet/star parameters, the Jeans parameter, composition, numerics) plus the final result. The **steady-state mass-loss rate** is $\dot{M} = 4\pi\rho v r^2$ evaluated near the outer boundary, corrected by the 2-D-approximation factor from `2D approximate method` (`Rate/2 + Mdot/2`$\Rightarrow \times\frac{1}{2}$, `Mdot/4`$\Rightarrow \times\frac{1}{4}$). It is reported as $\log_{10}\dot{M}$ [g s^{-1}]; divide by M_p and multiply by 1 Gyr to express it in M_p Gyr^{-1}. The functions `exhale_io.mdot_log10` / `mdot_Mp_per_Gyr` reproduce this.

5.13 `EXHALE_resolved.out`: the configuration the wind actually used

A machine-readable record written in the run directory (next to `input.inp`, not inside `output/`) by `write_setup_report.f90` once `input.inp`, `metals.inp` and any `base.inp` / lower-atmosphere-profile handoff have been resolved.

Format: # comment lines, then one `key value` pair per line, reals at full double precision. It carries the base radius, planet mass, base temperature, He/H, orbital distance and stellar mass in effect; `dayside_dilution`, the factor ξ_w on every stellar beam set by `2D approximate method` (1, $1/2$ or $1/4$; the elemental-flux closure compares it with the geometry it handed the column models and refuses a mismatch), and `newton_flux_switch`, the JFNK hand-off threshold (§3.7; both keys since update log stage 3 section 135); `base_inp_present` and `he_diffusion`; the EOS factors `mass_per_H_amu` and `ntot_bc_per_H`; one `abundance_<El>` line per element; and, with a profile in use (§4.8), the profile's provenance and the measured elemental fluxes (`lower_profile_flux_state`, the overlap window `lower_profile_flux_r_lo_Rp / _r_hi_Rp`, `lower_profile_F_H_median`, `_F_Hnuc_median`, `_F_He_median`, `_Mdot_median` with their spreads, and the same set over the steady window `steady_*`). $F_{\rm H}$ is the mass flux of component 1 of the binary split, the hydrogen carriers together with the trace metals and the C and O bound in molecules ($F_{\rm H}+F_{\rm He}=\dot{M}$); $F_{\rm H,nuc}=s_{\rm H}F_{\rm H}$ is its hydrogen-nucleus part, $s_{\rm H}=m_{\rm H}n_{\rm H}/m_1$ in the upwind cell of each face, and is the hydrogen flux the closure hands the chemistry (update log stage 3 section 134). The transport switches are recorded as `carrier_transport`, `carrier_in_newton`, `carrier_newton_on_stall` and `ionization_transport`; `he23s_transport` and `low_energy_electron_partition Furlanetto2010` are written only when those options are on, so a run without them writes the record it wrote before. With `Conduction` on it also records the heat the wind conducts into the lower atmosphere through the base face, measured by the certification on the state the run writes: `base_conduction_state` (`unmeasured` before the wind exists, then `measured`), `base_conductive_flux_cgs` [erg cm^{-2} s^{-1}], $q=\kappa_{1/2}(T_1-T_{\rm bath})/(r_1-r_0)$, *positive for heat leaving the wind into the lower atmosphere*, and the two temperatures of that face formula, `base_conduction_T_bath_K` (the bath at the ghost centre $r(0)$) and `base_conduction_T_cell1_K`; the energy coupling of the elemental-flux closure reads the flux from here. Where the carrier operator transports H_2 against a stated base partition (`Molecular carrier transport` with a `base.inp` `q_H2_base` or a profile) it records the ghost partition `base_ghost_h2_nuclei_fraction` and the H_2 the wind exchanges with the lower atmosphere through the base face, on the certified state: `base_face_carrier_state` (`unmeasured`, then `measured`), `base_face_H2_flux_cgs` [molecules cm^{-2} s^{-1}, positive outward], its advective and diffusive halves `base_face_H2_flux_adv_cgs` and `base_face_H2_flux_dif_cgs`, and `base_face_radius_cm`; the closure hands it to the top of the chemistry as that column's H_2 flux. It exists so that consumers do not re-parse `input.inp` and inherit a value the handoff has since moved: `EXHALE_transit.py` reads it in preference to `input.inp` (§6.2), and `src/utils/element_budget.py` and `src/utils/element_flux_closure.py` read it as the authority on the resolved abundances.

5.14 `element_flux_profile.txt`

Face-by-face elemental fluxes carried by the element-diffusion operator, written to `output/` whenever a lower-atmosphere profile is in use (where they are the quantity the two models must agree on, not a diagnostic), and on demand with `EXHALE_DIFFUSION_CHECK=1`. Columns (named by the file's own `# columns:` line): the face index and radius, the face helium mass fraction, the diffusive and advective fluxes, then $F_{\rm He}=4\pi r^2(\rho X v+J)$, $F_{\rm H}=4\pi r^2(\rho(1-X)v-J)$ (component 1, metals included; the hydrogen-nucleus flux $F_{\rm H,nuc}$ is reduced over the windows of `EXHALE_resolved.out` only) and $\dot{M}_{\rm face}=4\pi r^2\rho v$ in g s^{-1}, the face $\Delta m_{\rm eff}$, the effective and neutral diffusion coefficients with their ratio, and the dominant friction pair. Replaced at every diffusion step, so it always describes the

last one.

6. Analysis tools

6.1 `EXHALE_plots.py`: live profile viewer

`python3 EXHALE_plots.py [--live N]` draws an 8-panel figure from `output/`: density, velocity, pressure, temperature, momentum $\log_{10}(\rho v r^2)$, the ion densities, and the H and He ionization fractions. With `--live N` it refreshes every N seconds (watch a run converge). If the `*_adv.txt` files exist they are overplotted (dashed) against the equilibrium profiles (solid). It also prints $\log_{10}\dot{M}$.

6.2 `EXHALE_transit.py`: transmission spectra

`EXHALE_transit.py` post-processes a converged run into transit transmission spectra and reports the line-center absorption depths (theoretical, instrument-convolved, and instrument+rotation-convolved) for **He 10830**, **Lyα**, **Hα/Hβ** (from the non-LTE n=2 population, the level balance of Christie et al. 2013 with the solver's own rate coefficients, pumped by the Lyα field of §4.4), and the metal resonance doublets **Mg II h&k**, **Ca II H&K**, **Na I D** and the **O I 1302/1304/1306** triplet: the metals get the same full treatment as the H/He lines (all components in one window, theoretical / instrument / +rotation spectra, written under the same `tpm_<line>` names), plus the validated single-component line-center and 4 Å-band depth table; a metals-off run skips them automatically.

Which state the spectrum is built from (`EXHALE_TRANSIT_STATE`). The tool reads one of two state pairs, and the selection is a physical choice rather than a filename convenience. `adv`, the default, reads `Hydro_ioniz_adv.txt` and `Ion_species_adv.txt`: a post-process of a marching state, a steady advective correction cell by cell, second order in the cell width (variable-step BDF2 along the flow, each element's stages on the velocity of that element's nuclei, which is the bulk velocity without `He_diffusion`; §5.), with no stage eddy term and no ion–neutral drift, refused in a cell where an assumption of the steady equations fails. It is an independent discretization of the same column and no equation of the run was solved for it. `EXHALE_TRANSIT_STATE=solution` reads `Hydro_ioniz.txt` and `Ion_species.txt`: the state the solver produced and the certification judged. With `Ionization transport` on, the composition of that state is the partition the stage transport itself produced, so an advective correction laid on top of it would be a second and different transport approximation of the same stages.
A stationary run writes no `_adv` products, and none are to be manufactured for one: the default selection then refuses by name, says why, and names the other selection instead of quietly reading it. The two files of a selection are read together, including the O I ground-term level populations below, because the temperature of one state with the composition of the other solves neither set of equations. Every saved curve carries the selection, the path, md5 and size of each file it read, and the state's own provenance, coupling, boundary model and reservoir lines.

When an `adv` profile is refused (`EXHALE_TRANSIT_DIAGNOSTIC`). The tool prints the `# adv_derived_state:` record of the profile (§5.) and makes no spectrum, exiting with status 7, when the record says `verdict=rejected`

or when the profile has no record (a file written before the record existed, whose derived state is unknown). `EXHALE_TRANSIT_DIAGNOSTIC=1` makes the spectrum anyway, as diagnostic data: every saved curve then carries the record and an `adv_derived_state_refusal` or `adv_derived_state_unknown` line in its metadata. `verdict=unverified` is the best a record says, and such a profile is read. Inside it the row census refuses a row whose temperature field is not zero or whose composition field is `failed`; a refused row enters the spectrum with the run's own numbers, and the share of the line-center depth it carries is printed as a contribution, not an error bar. An equivalent width integrated from these curves, excess absorption in per cent over wavelength in Å (as `LHS1140b/he10830_equivalent_width.py` does), is in per cent Å, not mÅ. The solution selection carries no such record and is not tested against one.

On a marching case that carries both states the choice is small but not negligible. MEASURED on the `wasp_full` configuration (before the atomic-data corrections of 2026-09-24 to 09-26 and the solver's rates in the transit tool's n=2 model; not re-measured since), as the peak transit depth relative to itself: He I 10830 −0.64 per cent, Hα −1.28, Hβ −3.48, Lyα unchanged at its saturated ceiling, and the optically thin metal lines +0.1 to +0.8. The metastable helium column is 12 per cent *larger* in the solved state while its line is shallower, because the advection correction moves that column outward (column-weighted mean radius 1.2941 against $1.3112\,R_p$) and a saturated line's disk-averaged depth follows how far the absorbing shell reaches; Hβ moves most because it is the line still on the linear part of the curve of growth.

O I 1302: the one line whose components do not share a lower level. The three components of the O I resonance triplet absorb out of the three *different* fine-structure levels of the $2p^4\,{}^3P$ ground term, so the script reads their populations from the `OI_levels` file of the selected state (§5.10), `OI_levels_adv.txt` under `adv` and `OI_levels.txt` under `solution`, rather than applying the total O I density three times over, which at the near-statistical populations of these winds would triple the column. A run whose output carries no such file, like a metals-off run, simply skips the line. Because each absorption component owns its own lower-level density, the component descriptor is `(lambda0, f, A21, n_lower)` for every line, not only for this one.

The O I line is also the only one compared here against a published measurement, and the comparison is written into `tpm_OI_band_depths.txt` together with the conventions that make it legitimate: the depth is *band-integrated* over the observers' band (1300–1310 Å for Vidal-Madjar et al. 2004, 1299–1310 Å for Ben-Jaffel & Hosseini 2010), never at line center; the band weight is the stellar emission-line profile those authors measured (0.2 Å FWHM, peak ratios 1:1.5:1.17), because what was measured is a flux ratio and essentially all the flux in that band is in the three narrow chromospheric lines; the interstellar absorption is already inside those measured peak ratios and is therefore *not* applied a second time; the instrument LSF is not applied either, because a Gaussian LSF conserves the integral and cancels from a ratio over a band many LSF widths wide; and the published values, which are $(R_{\rm abs}/R_\star)^2$ and include the opaque planetary disk, have each paper's own continuum band subtracted before comparison. Overrides: `EXHALE_TRANSIT_RES_OI` (default 1000, HST/STIS G140L), `_OI_WEIGHT` (`star` | `flat`), `_OI_STAR_FWHM_A`, `_OI_STAR_PEAKS`, and `_OI_NSCALE`, which multiplies all three level densities at fixed ratios so the same code path can be driven up and down the curve of growth. It supports two geometries via `geometry = 'spherical'` (Phase 5a) or `'triaxial'` (Phase 5b, the 3-D Roche-equipotential reconstruction in `roche_recon.py`).

Where the system parameters come from. Every planet and star parameter is taken from the run directory, so the spectra always describe the same system as the simulation. The labels are matched exactly as `input_read.f90` does, so line order is irrelevant. Two sources are read, in this order:

`EXHALE_TRANSIT_*` environment override > `./EXHALE_resolved.out` > `./input.inp` > built-in default,

so an explicit override still wins for an experiment. The middle step matters whenever a `base.inp` or a lower-atmosphere profile moved the base: `EXHALE_resolved.out` (§5.13) is written by the wind solver *after* those overrides are applied, so it carries the R_p, M_p, T_0, a, $M_\star$ and He/H the wind actually used, while `input.inp` alone would be stale. When the file is present the script says so at startup and takes those six from it; when it is absent it says that too and falls back on `input.inp`. A present-but-unreadable file is a hard error, never silently replaced by `input.inp`. The stellar radius and $T_{\rm eff}$ are not in that file and come from `input.inp` (or their environment overrides) either way.

Quantity	`input.inp` label	Override
R_p	`Planet radius [R_J]`	-
M_p	`Planet mass [M_J]`	-
$T_{\rm eq}$	`Equilibrium temperature [K]`	-
a	`Orbital distance [AU]`	-
$M_\star$	`Parent star mass [M_sun]`	-
$L_{\rm EUV}$	`Log10 of EUV luminosity`	-
ξ (day–night)	`2D approximate method`	`xi_override`
$R_\star$	`Stellar radius [R_sun]`	`EXHALE_TRANSIT_RSTAR_RSUN`
$T_\star$	`Stellar Teff [K]`	`EXHALE_TRANSIT_TSTAR`
$P_{\rm spin}$	from a, $M_\star$+M_p (tidal locking)	`EXHALE_TRANSIT_ROTP`

The planet spin period has no `input.inp` key: a close-in giant is tidally locked, so the default is the Keplerian orbital period $P_{\rm orb} = 2\pi\sqrt{a^3/[G(M_\star+M_p)]}$ built from the orbital distance and the masses already in `input.inp`. Only the instrument resolving powers, the wavelength windows, and the geometry switch remain script/environment settings: they describe the observation, not the system. At startup the script prints every resolved parameter together with its source (`input.inp`, `env ...`, or built-in default), and warns when `Stellar radius`/`Stellar Teff` are missing from `input.inp`; read that block first when a spectrum looks wrong. A missing `Stellar radius` is the one worth checking before anything else: transit depths are normalized by $\pi R_\star^2$ and the chord integration stops at $R_\star/R_p$, so the built-in default radius shifts every depth, and not by a single factor. Measured on the tutorial planet (a $1.13\,M_\odot$ star, so $R_\star \simeq 1.155\,R_\odot$), the built-in default returned a He 10830 depth of 22.8% where the correct radius gives 4.0%.

Outputs. Every line carries one key (`He10830`, `Lya`, `Halpha`, `Hbeta`, `MgII`, `CaII`, `NaI`, `OI`), and both products of a line are named from that key and land in the run directory: the model curve `tpm_<line>.txt` (always written) and, when `EXHALE_TRANSIT_FIG_PREFIX` is set, the figure `<prefix><line>.png` together with its vector `.pdf`. A curve file has the columns `lambda[A]`, T_λ (theoretical), T_λ (instrument-convolved), T_λ (+rotation), so the excess absorption in percent is $(1-T_\lambda)\times 100$; a metals-off run simply writes no metal curves. `EXHALE_TRANSIT_SAVE_PREFIX` decorates

the curve name in the same way the figure prefix decorates the figure name; both are resolved against the run directory, so an absolute prefix writes elsewhere.

Step-by-step usage. `EXHALE_transit.py` reads the state pair of the selection in force from `path`, which is the current directory unless `EXHALE_TRANSIT_PATH` names the run directory: `output/Hydro_ioniz_adv.txt` and `output/Ion_species_adv.txt` by default, or `output/Hydro_ioniz.txt` and `output/Ion_species.txt` under `EXHALE_TRANSIT_STATE=solution`:

1. *Choose the state, and make sure its two files are there.* A stationary solve writes no `_adv` products, and the solution selection is how such a run is read; do not manufacture `_adv` files for one. A marching run writes both pairs automatically. To give an existing state the advection-corrected pair as well, copy `Hydro_ioniz.txt` → `Hydro_ioniz_IC.txt` and likewise for `Ion_species`, set `Load IC? True` and `Restart intent: stationary evaluate` in `input.inp`, and run `./EXHALE.x` once (seconds): that route loads the state, certifies the stored pair with the sources of its own composition, and writes every product from that pair without entering the time integration (the composition sweep it then takes is measured and reported, not written).

2. *Set what describes the observation*, if the defaults do not: the instrument resolutions and wavelength windows (`EXHALE_TRANSIT_RES_*`, `_HE_LMIN/LMAX/N`), the azimuthal sampling of the rotation integral, and `geometry` for a Roche run. The star and the spin period need nothing: they come from `input.inp`.

3. *Run* `MPLBACKEND=Agg EXHALE_TRANSIT_PATH=<run dir> python3 EXHALE_transit.py` (the `Agg` backend suppresses the interactive windows on headless machines). The script writes the `tpm_<line>.txt` curves into the run directory and prints the line-center absorption depths (theoretical / instrument-convolved / +rotation) and the metal-line summary; add `EXHALE_TRANSIT_FIG_PREFIX=tpm_` to save the figures as well.

Worked examples (the converged Newton solutions of WASP-121b and HD189733b, with all four H/He spectra and the metal lines): `backup/regression/tpm_wasp/` and `backup/regression/tpm_hd189/`, collected in the notebook `backup/regression/steady_solver_results.ipynb`. Figure 2 shows the four H/He spectra `EXHALE_transit.py` saves for the WASP-121b Newton solution of `backup/regression/tpm_wasp/` (run with `Include He23S? True`, so the He I 10830 line is populated).

Transit-depth conventions (important). Huang et al. (2023) quote transit depths as the *effective transit radius* $R_{\rm eff}/R_\star$, not a percent absorption; convert from the code's line absorption fraction h via $R_{\rm eff}/R_\star = \sqrt{(R_p/R_\star)^2 + h}$. The metal NUV lines (Mg II) are quoted in a 4 Å bin, while the optical lines (Ca II, Hα, Na) are line-center; line-center vs. 4 Å can differ by $\sim 3\times$ for an optically-thick line, so always compare in the same definition.

Comparing a model line with an observed profile: the line width and the equivalent width. The 1-D model has no velocity field across the line of sight beyond its radial outflow, and its lines are narrower than the observed ones (He 10830 of LHS 1140 b: model FWHM about 0.27 Å against the observed 0.841 Å). For every planet and every line compared with an observed transit profile the convention is (user decision 2026-10-04): the excess absorption

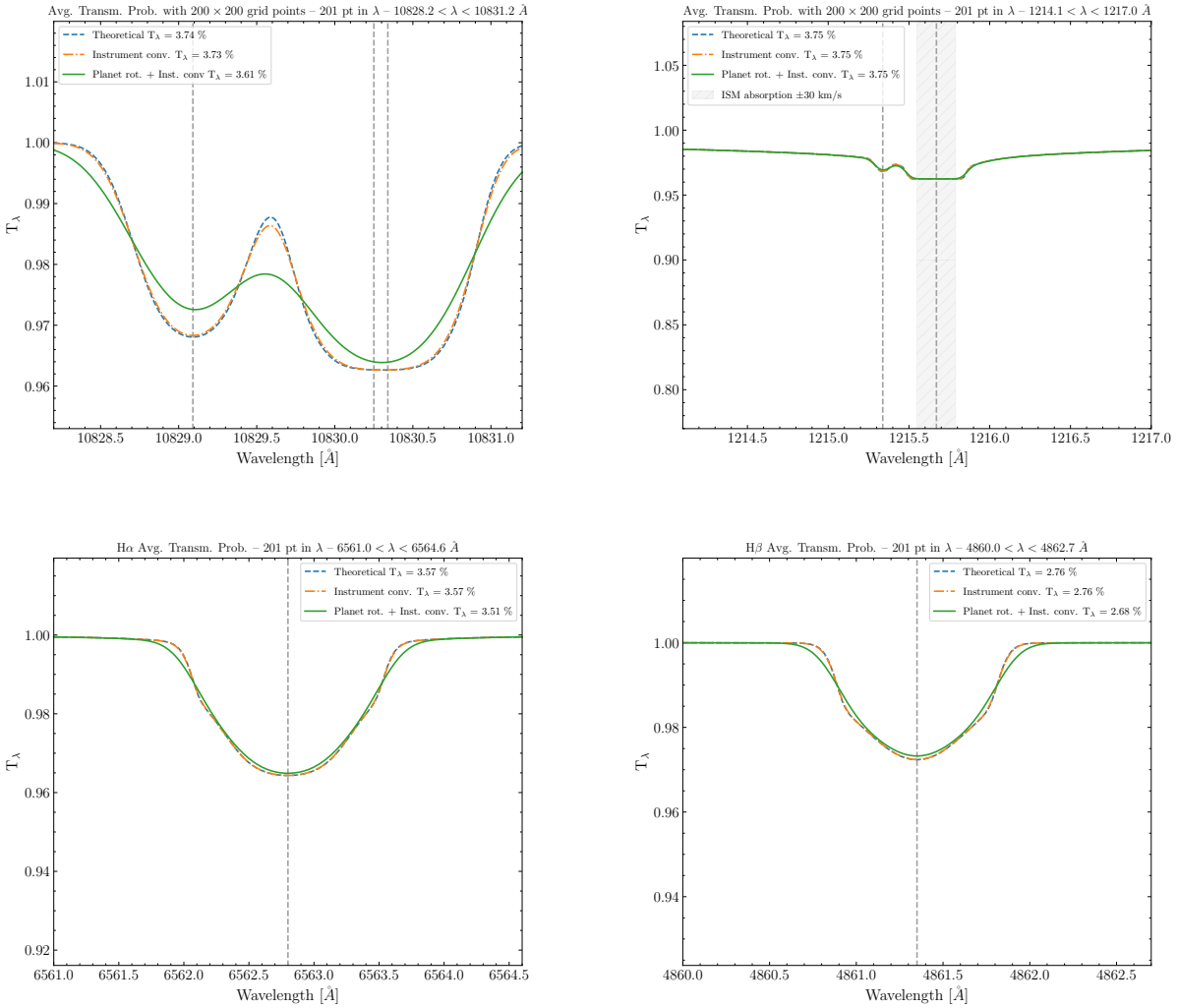


Figure 2: `EXHALE_transit.py` output spectra for WASP-121b (Roche, Newton solution with He 2^3S on; `backup/regression/tpm_wasp/`): He I 10830 (top left), Lyα (top right), Hα (bottom left), and Hβ (bottom right). Each panel shows the theoretical line, the instrument-convolved line, and the instrument+rotation-convolved line as produced by the script.

$(\max T - T)/\max T$ of the planet-rotation and instrument-convolved curve (column 4 of `tpm_<line>.txt`, at the instrument's own resolving power; the instrument-only column 3 until 2026-10-06) is convolved with a Gaussian of FWHM f in velocity, planet rest frame; f is chosen for each model so that the extractor used on that planet's data returns the observed FWHM of the same feature; and the model EW is the window EW of that broadened curve through the same operator as the observed one (interpolated onto the observation's samples in its window, trapezoid weights). Any fit of a planet parameter to the observed EW is judged on this width-matched EW, and every result that uses it reports f. The kernel stands for a velocity field the 1-D model does not contain; it is not a computed physics term. For LHS 1140 b He 10830 the code is `LHS1140b/he10830_equivalent_width.py` (`broaden_excess`, `width_matching_fwhm_kms`, `width_matched_product`; the command line `--width-matched <tpm_He10830.txt> <out> [f]` writes the broadened curve with its record), the window vacuum 10832.60–10834.20 Å and the three-Gaussian extractor `he_line_metrics.fit_metrics`; and `LHS1140b/models/.P1/fiducial_conduction/ew_of_state.sh <state dir> <work dir> <binary>` runs the stationary evaluate of a stored state (with its exact `conserved_state.txt` staged) and both spectra, and writes `EW.txt`: the provenance of what it consumed, the evaluate exit status, `exact_state_read`, the `adv_derived_state` record, `written_state certified=T|F` (the stored pair's own verdict) and one line for each product; with `EW_NONTHERMAL_KERNEL=width_match` (or a FWHM in $\mathrm{km\,s^{-1}}$) each product line gains `kernel: fwhm_kms=. . . EW_kernel=. . .`, while `EW=` stays the unbroadened EW. A model shifted in wavelength on a figure (LHS 1140 b: the fitted $v_{\rm wind} = +2.26\,\mathrm{km\,s^{-1}}$) is shifted for the overlay only; every EW, metric and χ^2 is computed from the unshifted spectrum. The full statement, with the EW operator and its error: `docs/transmission_spectrum.pdf`, section "Comparison with an observed line profile".

6.3 `exhale_io.py`: named, physical-unit loaders

`examples/exhale_io.py` loads every EXHALE output file into named arrays in physical units, so analysis code need not remember column orders (Table 5). The central object is the `Run` container: `load_run(outdir, inputfile)` fills it with the hydro profiles (`r` [R_p], `n` [$\mathrm{cm^{-3}}$], `v` [$\mathrm{cm\,s^{-1}}$] with a `v_kms` shortcut, `p`, `T`, `heat`, `cool`), the full ion-density dictionary `run.ion` (the 6 H/He plus 27 metal stages of §5.), and the parsed `input.inp` in `run.inp`. By default the `_adv` (advection-corrected) files are read; pass `adv=False` for the equilibrium snapshot. Typical use:

```
import exhale_io as aio
run = aio.load_run('output', 'input.inp')     # _adv profiles by default
print(aio.mdot_log10(run))                    # log10 Mdot [g/s], 2D factor incl.
print(aio.mdot_Mp_per_Gyr(run))               # same in M_p/Gyr
xh = run.x_ion(['HI', 'HII'])                 # stage fractions of one element
plt.plot(run.r, run.T)
plt.plot(run.r, xh['HII'])
cool = aio.load_cooling('output/Cooling_breakdown.txt')
plt.semilogy(cool['r'], cool['chan']['FeII']) # one cooling channel
```

The notebook `examples/EXHALE_analysis.ipynb` uses these loaders to plot the profiles and the parameter-difference comparisons of §7., and `examples/make_figures.py` (which generates every figure in this manual) is a further worked example.

Function	Returns
`load_run(outdir, inputfile, adv=True)`	a `Run`: hydro profiles + `run.ion` dict + parsed input.
`Run.x_ion([stages])`	ionization fractions of one element, e.g. `x_ion(['HI','HII'])['HII']` = $n_{\rm HII}/(n_{\rm HI}+n_{\rm HII})$.
`load_hydro(path)`	one `Hydro_ioniz(_adv).txt` as a dict (`r,n,v,p,T,heat,cool`).
`load_ions(path)`	`(r, {ion: density})` from `Ion_species(_adv).txt` (all 33 stages).
`load_cooling(path)`	`r,T,ne,cool_total` + `chan` dict of cooling in each channel (6 H/He channels `rec/coll_ion/coex_HI/coex_HeI/coex_HeII/brems`, the `H3p` infrared channel, + 27 metal-line channels).
`load_excited_H(path)`	`Excited_H.txt` as a dict (`n2s, n2p, Jlya, tau_lya, Sproton, Hpe, . . .`).
`load_lyman_werner(path)`	`Lyman_Werner.txt` as a dict (`x_H2, NH2, f_shield, k_LW, heat_LW`).
`read_input(path)`	`input.inp` as a dict (`Rp_RJ, Mp_MJ, a_AU, HeH, LX, LEUV, . . .` + raw lines).
`mdot_log10(run, j_from_top=20)`	$\log_{10}\dot{M}$ [g/s]: $4\pi\rho v r^2$ near the outer boundary *including* the `2D approximate method` factor (mirrors `EXHALE_main.f90`; see §5.).
`mdot_Mp_per_Gyr(run)`	the same in units of $M_p\,\mathrm{Gyr}^{-1}$.
constants	`RJ_CM` (and its alias `RJ`), `MJ`, `Msun`, `AU`, `mu` (= $m_{\rm H}$), `GYR` (cgs). These are the one Python definition of each; the tools import them from here rather than writing the number down again.

Table 5: `examples/exhale_io.py` public interface. Run analysis scripts either from `examples/` or after adding it to the Python path (`sys.path.insert(0, '<repo>/examples')`).

6.4 `exhale_to_lart.py`: LaRT Lyα input from an EXHALE run

`EXHALE_transit.py` estimates the Lyα mean intensity $\bar{J}_{\mathrm{Ly}\alpha}$ with the simple Huang et al. (2017) prescription (§4.4). For a proper treatment of the resonant Lyα radiative transfer that sets the n=2 hydrogen population (and hence Hα), the converged atmosphere can be post-processed with the Monte Carlo code **LaRT** (Seon & Kim 2020; Seon et al. 2022). `examples/exhale_to_lart.py` builds a complete LaRT spherical-illumination input set directly from an EXHALE run.

Given a run directory it loads the `_adv` profiles (via `exhale_io.py`; `--eq` takes the equilibrium pair instead, which is what a stationary solve leaves) and `input.inp`, and writes

- `dens_profile.txt`: $r\,[R_{\mathrm{p}}]$, $n_{\mathrm{HI}}\,[\mathrm{cm}^{-3}]$ (the Lyα scatterer is ground-state H);
- `temp_profile.txt`: $r\,[R_{\mathrm{p}}]$, $T\,[\mathrm{K}]$;
- `velo_profile.txt`: $r\,[R_{\mathrm{p}}]$, $v_r\,[\mathrm{km\,s}^{-1}]$;
- `line_profile.txt`: the incident stellar Lyα profile, a double-Gaussian with peaks at $\pm m$ and Gaussian width s (defaults $m = 74$, $s = 49\,\mathrm{km\,s}^{-1}$, the Huang et al. 2017 / Yan et al. 2022 values; pass `--fwhm` to read s as FWHM);
- `<name>_lya.in`: the LaRT namelist (`geometry='spherical_atmosphere'`, `source_geometry='stellar_illumination'`);
- `run.sh`: a multi-host `mpirun` launcher.

The length unit is set to R_{p} (`distance2cm` $= R_{\mathrm{p}}$ in cm), so all LaRT radii are in R_{p} and match the EXHALE grid; the star–planet geometry (`distance_star_to_planet` $= a/R_{\mathrm{p}}$, `stellar_radius` $= R_\star/R_{\mathrm{p}}$) is read from `input.inp`. Because the stellar illumination breaks spherical symmetry, the scattering rate is requested as a cylindrical 2-D grid (`geometry_JPa = 2`): $P_\alpha(\rho, z)$ is axisymmetric about the star–planet axis, which is the form needed to build the 2-D $n_{2p}(\rho, z)$ population for the Hα transit. To populate P_α, build LaRT with `make CALCPnew=1` (producing `LaRT_calcP.x`).

```
# build the LaRT input from a converged EXHALE run, then run it
python examples/exhale_to_lart.py WASP-52b/fxuv0p25_he98 \
       -o WASP-52b/LaRT_lya --name WASP-52b \
       --m 74 --s 49 --rmax 10 --ngrid 201 --nphotons 1e7 \
       --hosts lart4,lart3,lart2
cd WASP-52b/LaRT_lya && ./run.sh
```

It is also importable: `import exhale_to_lart as e2l; e2l.build(run_dir, outdir, m_kms=74, s_kms=49, geometry_jpa=2, . . . )`.

7. Worked examples

7.1 A first run: a generic hot Jupiter

The directory `examples/tutorial/` contains a minimal, self-contained model of a generic HD 209458b-like hot Jupiter: a spherical domain out to $10\,R_p$, solar metals, EUV+X-ray power-law SED. The full `input.inp` is:

```
Planet name: demoHJ
Log10 lower boundary number density [cm^-3]: 12.70
Planet radius [R_J]: 1.38
Planet mass [M_J]: 0.69
Equilibrium temperature [K]: 1320.0
Orbital distance [AU]: 0.047
Escape radius [R_p]: 2.00
He/H number ratio: 0.083
2D approximate method: Rate/2 + Mdot/2
Parent star mass [M_sun]: 1.13
Spectrum type: Power-law
Power-law index: -1.0
Use only EUV? False
[E_low,E_mid,E_high] = [ 13.60 , 123.98 , 1.24e3 ]
Log10 of X-ray luminosity [erg/s]: 27.0
Log10 of EUV luminosity [erg/s]: 29.30
Grid type: Mixed
Numerical flux: HLLC
Reconstruction scheme: PLM+WENO3
Include He23S? True
Load IC? False
Do only PP: False
Force start: False
Domain mode: Spherical
Outer radius [R_p]: 10.0
du_th [PLM,WENO3]: 0.5 1.0e-3
Solver: Newton
Stellar radius [R_sun]: 1.155
```

The `Reconstruction scheme: PLM+WENO3` line enables the two-stage reconstruction (with the two `du_th` values as the PLM and WENO3 thresholds), and the `Solver: Newton` line adds the steady-state Newton finish (§3.7), so the run converges to the true steady state without manual restarts. With `metals.inp` present (solar abundances, §4.), run `./EXHALE.x` in that directory. Figure 3 shows the converged profiles read back with `exhale_io`: the wind heats to $\sim 10^4$ K, the velocity rises monotonically outward (transonic outflow), and hydrogen ionizes across a front near 1.2–1.4R_p.

7.2 Turning physics on and off

The same model is re-run with single-line changes to illustrate each option: *metals off* by removing `metals.inp`; *Roche/RLOF* by removing the `Domain mode`/`Outer radius` lines (the default truncates at L1); the *atomic-helium limit* by setting `Include He23S? False` (the triplet is on by default, so this is the change that removes it); the *Hα/Balmer*

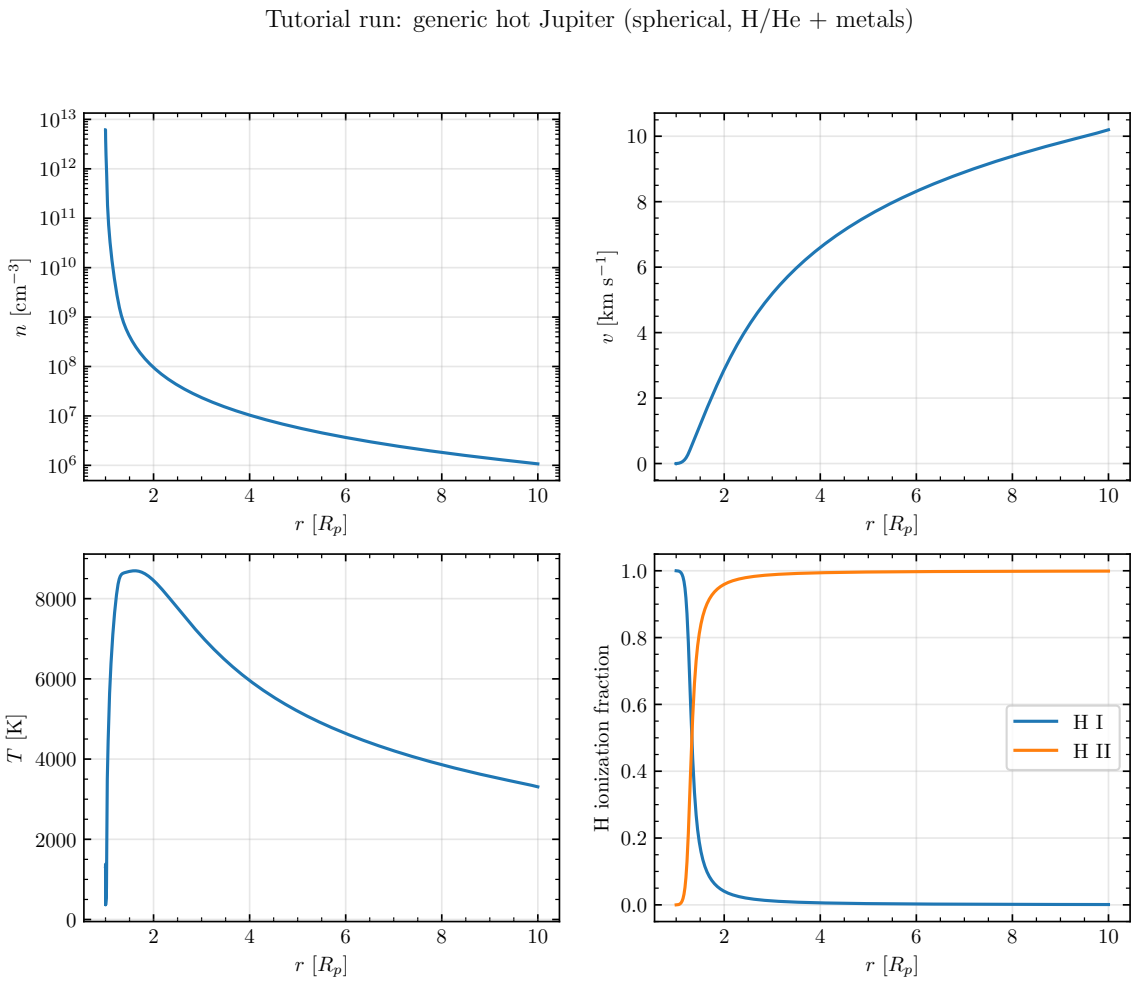


Figure 3: First-run output of the generic hot-Jupiter tutorial: density, velocity, temperature, and the H ionization fraction vs. radius.

physics by adding `Stellar Teff/Stellar radius`; and the *transonic IC* by adding `Transonic IC: True` for a near-unbound (deep-RLOF) base; and *diffusive He/H separation* by adding `He_diffusion: True` (extend it to the metals with `He_metal_diffusion: True`; §4.6). The comparison studies below quantify the effect of the first two.

7.3 Ready-made example configurations (`examples/`)

For every solver/physics combination there is a ready-to-run folder under `examples/` (paths relative to the `EXHALE` root). Folders `01`–`11` are the *same* planet (HD189733b) so that the effect of each option can be isolated; `12` and `14`–`17` use HD209458b (where the Wind-AE warm start, diffusion, molecular chemistry with and without metals, and the lower-atmosphere profile handoff are validated), and `13` spans four planets. Each folder is self-contained: `cd` into it and run the repo-root binary (`../../EXHALE.x`); outputs land in the folder's own `output/`. The "Added / changed lines" column of Table 6 is relative to each example's natural base: the legacy `01_legacy_marching/input.inp` for the solver examples `02`–`04` (and Wind-AE `11`/`12`), and the recommended-solver `03_newton/input.inp` (two-stage `PLM+WENO3` + `Solver: Newton`) for the physics examples `05`–`10`, which all inherit that solver and add only their signature option on top. A `diff` against the appropriate base shows exactly what an option adds; all configurations have been smoke-tested. The metastable helium triplet is on by default in the code, but the ladder holds `Include He23S? False` on purpose: in both bases and in every rung whose signature option is something else (`02`, `04`, `05`, `09`–`12`), so that atomic helium is the baseline and `06_he23s` remains a one-line difference. Outside the ladder (`tutorial/`, `13`–`19`, the planet folders) the triplet is on. `11_windae_ic` (HD189733b) exercises the self-consistent-BC continuation: its Wind-AE ramp converges and writes the IC, though EXHALE's own time-integration of HD189733b is limited by a separate base-breathing instability (§3.10). `12_windae_ic_hd209` is the seed-adjacent HD209458b case. `14_diffusion` and `15_molecular` turn on the He/metal diffusive separation (§4.6) and the full molecular network (§4.7) respectively, on HD209458b, and `16_molecular_metals` adds a `metals.inp` to the latter. `17_lower_profile` replaces the scalar `base.inp` handoff by the profile of §4.8. `18_oxygen_chemistry` adds the oxygen chemistry to `16` and `19_molecular_ir_bands` the molecular infrared bands to `18`.

Table 6: Ready-made example configurations under `examples/` (`01`–`11` are HD189733b; `12`, `14`–`19` are HD209458b; `13` spans four planets). The Wind-AE IC ramp converges for both of the planets that use it (HD 189733 b (`11`) and HD 209458 b (`12`)), and `11`, being far from the seed, additionally exercises the self-consistent-BC continuation and the molecular-layer turn-off. Metals on/off is a runtime switch: any folder becomes metals-on by copying a `metals.inp` into it, molecular folders included: that is what `16_molecular_metals` is.

`examples/...`	Demonstrates	Added / changed lines
`01_legacy_marching`	classic ATES v2 baseline: PLM marching, H/He only, Roche domain, *du*-based stop	(none, baseline)
`02_two_stage`	two-stage PLM→WENO3 marching (§3.4)	`Reconstruction scheme: PLM+WENO3`, `du_th [PLM,WENO3]: 0.5 1.0e-3`

Table 6, continued

examples/...	Demonstrates	Added / changed lines
03_newton	**recommended default**: two-stage warm-up + Newton finish (§3.7)	+ Solver: Newton
04_newton_from_state	Newton re-convergence of an existing state (no marching; needs output/*_IC.txt, see the folder README)	Load IC? True, Resid tol: 1.0e-5; run via the EXHALE_PTC hook
05_metals	trace-metal cooling, solar C/N/O/Mg/Ca/Na/Fe (§4.)	metals.inp present
06_he23s	He 2^3S metastable level (He I 10830); the one rung that leaves the ladder's atomic-helium baseline	Include He23S? True
07_balmer_lya	non-LTE H(n=2) + Lyα pumping (Hα/Hβ, §4.4)	+ Stellar Teff/radius, Deexc heat, Jlya escape-prob, Stellar Lya flux/halfwidth/boost
08_full	everything on (= the HD189733b/ planet folder + Newton); feeds EXHALE_transit.py	07 + metals.inp
09_spherical	spherical domain instead of the default Roche/L1 truncation (§2.3)	Domain mode: Spherical, Outer radius [R_p]: 10.0
10_warm_seed_ic	warm-seed (hot Parker overlay) initial condition	Hot Parker IC: 10000
11_windae_ic	in-process Wind-AE IC for HD189733b (§3.10): far from the seed, so the self-consistent-BC continuation (base-BC re-convergence + molecular-layer turn-off) is exercised; the Wind-AE ramp converges and writes the IC. EXHALE's own HD189733b base-breathing instability (separate from the IC) then limits the warm start	+ IC mode: windae, Solver: Newton; symlinks inputdata/ for the shipped seed
12_windae_ic_hd209	in-process Wind-AE warm-start IC that *works*: HD 209458b (**not** HD189733b) is close to the shipped seed, so the ramp converges and EXHALE warm-starts cleanly (spherical 10 R_p)	HD209458b params + Domain mode: Spherical, IC mode: windae, Solver: Newton; symlinks inputdata/

Table 6, continued

examples/...	Demonstrates	Added / changed lines
13_lower_atmosphere	lower-atmosphere connection for four planets (HD209458b, HD189733b, WASP-121b, WASP-52b; §4.7): an analytic 1 μbar base column plus a base.inp hand-off (isothermal or Guillot $T(p)$ generator); see the folder README	Lower column: <R_1bar>; driver-generated base_iso.inp / base_guillot.inp
14_diffusion	diffusive separation of He and metals (HD209458b; §4.6): the He/H ratio declines with altitude and each trace metal settles independently, reshaping He I 10830	HD209458b params + Include He23S? True, He_diffusion: True, He_metal_diffusion: True, He_Kzz: 1.0e9, He_alphaT: 0.0; metals.inp present
15_molecular	full molecular chemistry (HD209458b; §4.7): $H_2/H_2^+/H_3^+/HeH^+$ in the coupled ionization equilibrium; a sharp $H_2 \rightarrow$H front forms above a thin molecular base, the wind above it essentially atomic. Newton-converged from a cold start, re-run 2026-08-15 with the current code ($\|R\| = 6.105 \times 10^{-6}$ in 89 outer iterations): the front sits at $r = 1.0097\,R_p$ and the molecular layer below it collapses to $\sim$400 K	HD209458b params + Molecular chemistry: True, Solver: Newton, Resid tol: 1.0e-7, Max steps: 150000 (no metals.inp here)

Table 6, continued

examples/...	Demonstrates	Added / changed lines
16_molecular_metals	molecular chemistry *and* trace metals solved in one system (HD209458b): the $H_2/H_2^+/H_3^+/HeH^+$ network and the metal ionization stages share the free electron density, which the metals dominate in the shielded molecular base. Converges only from a warm restart, and the seed matters: seed it with a 15 converged by *the same binary* (a state left by an older binary is a different seed and fails, info = 2), then expect info = 0 at Resid tol: 1.0e-7 and $\log_{10}\dot{M} = 9.65$–9.67. The cold path is closed: with metals on, the JFNK line search bottoms near $\|R\| \sim 10^{-3}$ (worst cell $r = 1.021$, momentum) and falls back to marching at every Low-Mach damping coefficient tried: off, 5×10^{-3}, 10^{-2}, 2×10^{-2} (measured 2026-08-15)	as shipped, 15_molecular + metals.inp (solar C/N/O) without 15's three convergence keys. For the warm restart put the seed run's Hydro_ioniz.txt/Ion_species.txt in output/*_IC.txt and use 15's three keys + Load IC? True + Low-Mach damping: 5.0e-3 (run 1.0e-2 as well and compare, see the folder README)
17_lower_profile	lower-atmosphere *profile* handoff (HD209458b; §4.8): the lower atmosphere handed over as a table over an interval of pressure instead of the scalars of base.inp. The base state (T_0, R_0, p_{base}, q_{H_2}), the elemental reservoirs (He/H and solar C/N/O) and $K_{zz}(r)$ all come from the one file; make_example_profile.py regenerates it	14_diffusion + Lower atmosphere profile: lower_atmosphere_profile.dat, *minus* He_Kzz (the profile carries K_{zz}) and *minus* metals.inp (the profile carries C/N/O)

Table 6, continued

examples/...	Demonstrates	Added / changed lines
18_oxygen_chemistry	oxygen chemistry (HD209458b; §4.7): OH, H_2O and CO in the coupled system with the molecular network, the FUV photolysis of H_2O and OH, and the molecular carriers transported (the default with the oxygen cycle), so the base H_2/H partition is computed rather than imported. HD 209458 b's hot base is not the regime the option exists for (there collisional dissociation sets the partition); it is the example because it is a molecular configuration that converges	16_molecular_metals + Oxygen chemistry: True, Stellar FUV B3/B4 flux, Stellar Lya flux, Stellar LW flux: 480.9
19_molecular_ir_bands	molecular infrared bands (HD209458b): 18 plus the H_2 quadrupole and magnetic-dipole lines and the H_2O and CO vibration–rotation bands, each the net exchange with the diluted $B_\nu(T_0)$ that Base IR field supplies; writes the H2_IR, H2O_IR and CO_IR columns of Cooling_breakdown.txt	18 + Base IR field: True, Molecular IR bands: True

8. Comparison studies

8.1 Effect of metal-line cooling (metals on vs. off)

Trace metals add strong line cooling (and some photoionization heating and free electrons). Figure 4 compares the tutorial planet with and without metals.inp: metal-line cooling lowers the peak temperature by ~ 1800 K (acting as a thermostat that prevents the H/He-only over-heating), while the H ionization structure is barely changed. The effect is far stronger for hotter, more irradiated planets (e.g. WASP-121b, where metal cooling is essential to match the observed temperatures); the detailed cooling by channel budget is in Figure 5.

8.2 Spherical vs. Roche (RLOF) and the mass-loss rate

Removing the Domain mode: Spherical line switches on the full Roche tidal potential, truncated at the L1 radius. Figure 6 compares a spherical model (Case A) and a Roche model (Case B) of WASP-121b: the tidal potential lowers

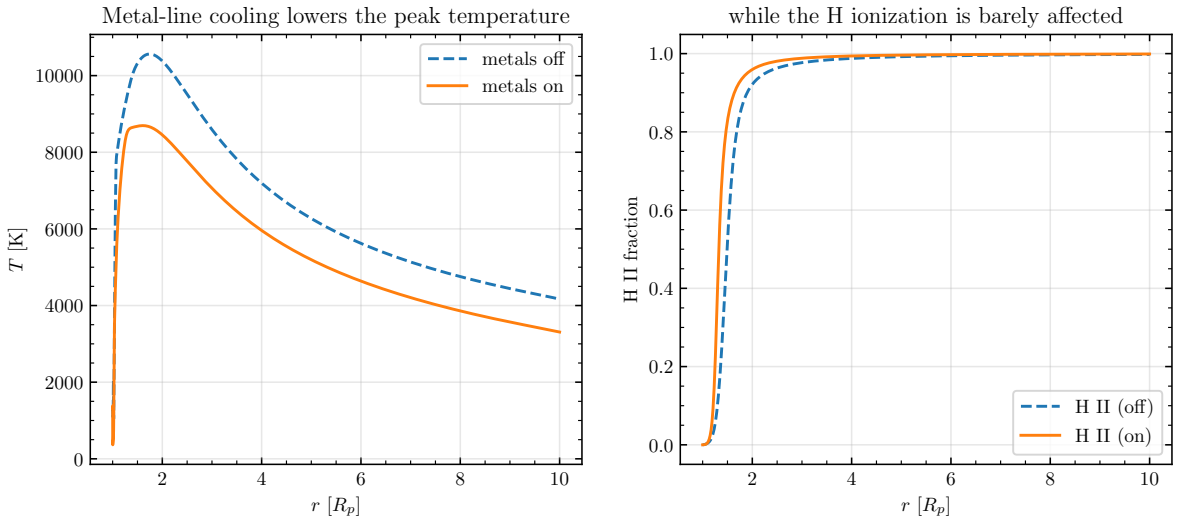


Figure 4: Metals on vs. off for the tutorial planet: metal-line cooling lowers the peak temperature (left) while the H ionization fraction is nearly unchanged (right).

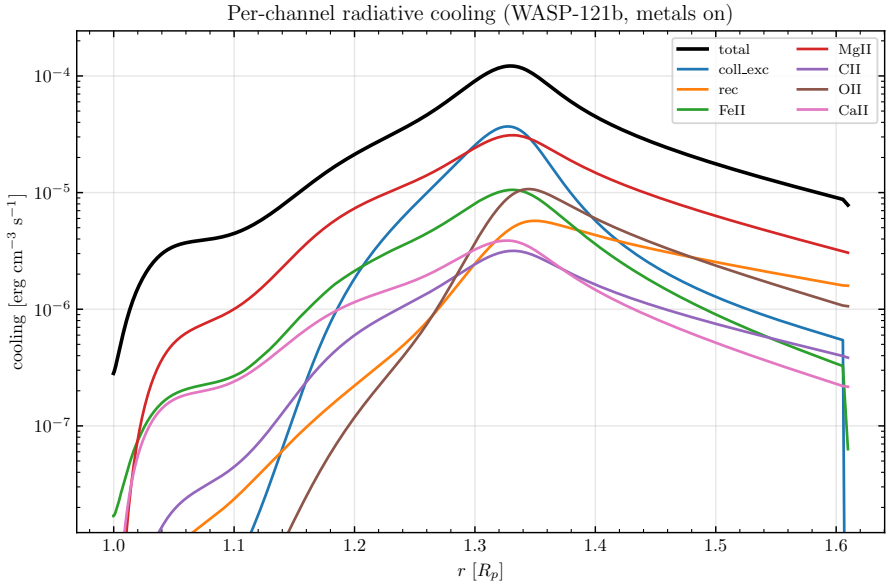


Figure 5: Radiative cooling by channel vs. radius (WASP-121b, metals on; from `Cooling_breakdown.txt`). Below $\sim 1.4 R_p$ the metal lines (e.g. Fe II, Mg II, C/O) compete with or dominate the H/He collisional channels.

the effective gravity toward the star, so the Roche wind accelerates far faster and the mass-loss rate rises by an order of magnitude ($\dot{M}$ from ~ 0.05 to $\sim 0.34\,M_p$ Gyr^{-1} here), the "Roche-lobe overflow" enhancement.

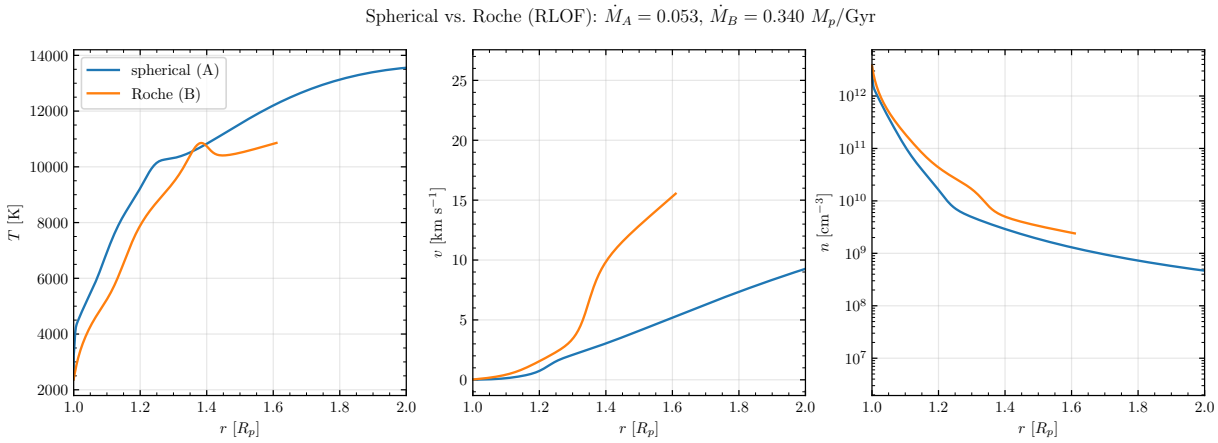


Figure 6: Spherical (Case A) vs. Roche (Case B) WASP-121b: temperature, velocity, and density. The Roche wind reaches L1 ($\approx 1.6R_p$) and accelerates much faster; $\dot{M}$ is annotated. The temperature panel is the _adv profile; its base is smooth because the post-processor falls back to the equilibrium solution where the flow is not a clean outflow (see the caveat in §5.). The Roche velocity (middle) dips slightly negative near the base: the genuine, physical "breathing" inflow of a near-Roche-filling planet.

8.3 Transmission spectra

`EXHALE_transit.py` turns a converged model into transit absorption. The metal resonance lines are the key NUV/optical diagnostics; Figure 7 compares the modeled effective transit radii $R_{\rm eff}/R_\star$ (spherical Case A) against Huang et al. (2023). Recall the conventions of §5.2: Mg II is a 4 Å bin, Ca II/Na are line-center, and $R_{\rm eff}/R_\star = \sqrt{(R_p/R_\star)^2 + h}$.

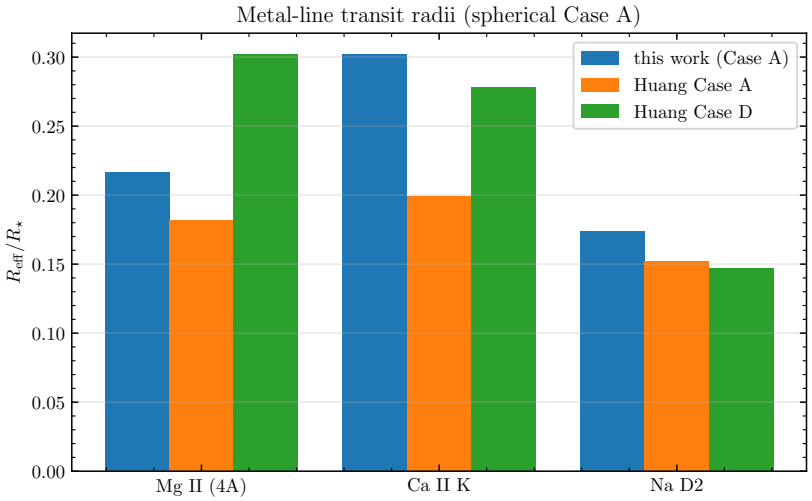


Figure 7: Metal-line transit radii $R_{\rm eff}/R_\star$ for spherical Case A, compared with Huang et al. (2023) Case A and Case D.

9. Atomic and molecular data

By default EXHALE is an *atomic/ionic* code: the modeled species are H, He (including the He $2\,^3$S metastable triplet), and ten trace metals (C, N, O, Mg, Si, Ca, Na, K, S, Fe) across their first two–three ionization stages. The optional molecular network (`Molecular chemistry: True`, §4.7) adds H_2, H_2^+, H_3^+ and HeH^+ to the coupled equilibrium, with H_2 photoionization opacity (Yan et al. 1998) and H_3^+ infrared cooling (MiZATeP, Mizus et al. 2017, with the non-LTE factor of Miller et al. 2013); `Oxygen chemistry: True` adds OH, H_2O and CO (§4.7), and `Molecular IR bands: True` the H_2 line spectrum (Roueff et al. 2019) and the H_2O and CO bands (HITEMP) as infrared coolants. There is no H^-, and the only molecular opacities are those of H_2 (its ionizing continuum and the Lyman–Werner lines) and, with the oxygen chemistry, the FUV photodissociation of H_2O, OH and CO. Table 7 collects the provenance of the atomic rates and cross sections, grouped by physical process. Of the H/He data of the original ATES (Caldiroli et al. 2021, 2022) the code keeps the hydrogenic cross sections of H I and He II and, as comparison options, the legacy fits (`ATES_photoionization_rate`, `Legacy_HHe_rates`); the default recombination, collisional and cooling coefficients of H and He were replaced after the atomic-data audits of 2026-09-24 to 09-26 (`md/Update_EXHALE_stage3.md` sections 17, 19 and 22), and the metal, excited-hydrogen, and charge-exchange data were added in this work.

Table 7: Atomic data sources, by process. Source files: `cross_sec.f90` (cross sections); `Cool_coeff.f90` (collisional rates, recombination, cooling); `charge_exchange.f90`; `excited_hydrogen.f90`; `EXHALE_transit.py` (transmission line data).

Process	Species	Source
Elemental abundances		
Solar reference	10 metals (`metals.inp` template)	Asplund et al. (2009)
Helium	He/H ratio	input parameter (`He/H number ratio`)
Photoionization cross sections		
Bound–free $\sigma(E)$	H I, He II	hydrogenic (Kramers) analytic form on the measured thresholds (orig. ATES)
	He I	Verner, Ferland, Korista & Yakovlev (1996) default; legacy ATES 2-term fit via `ATES_photoionization_rate`
	He I $2\,^3$S (metastable)	two VFKY96 (Verner) wings + power-law bridge, fitted to TOPbase/Opacity-Project (extends the orig. Norcross 1971 broken power law past 60 eV)

Table 7, continued

Process	Species	Source
	C, N, O, Mg, Si, Ca, Na, S, Fe (+ ions)	Verner, Ferland, Korista & Yakovlev (1996) outer-shell fits below the first inner-shell edge, Verner & Yakovlev (1995) above it; the inner subshells (Verner & Yakovlev 1995, their `phfit2`) are in the sum, so σ is the whole photoabsorption of the ion
	K I	Verner & Yakovlev (1995) at every energy, outer shell and subshells
Inner-shell relaxation	the 17 photo-ionizable metal ions	a vacancy below the outer shell relaxes by Auger decay (one or more further electrons, the ion jumping two or more stages) or radiatively, with the probabilities of Kaastra & Mewe (1993, A&AS 97, 443) Table 2 and the fluorescence of their Table 3, the vacancy energy closed exactly on the code's thresholds ($E_{\mathrm{th},s} = I_i$+ further ionization energies + Auger energy + fluorescence); the photoelectron and the Auger electrons are the electron energy, the fluorescence leaves the cell. A neutral of C, N, O, Mg, Si, Ca, Fe that loses two or more electrons goes to its third stage in the balance; a jump beyond the carried stages is counted in the top carried stage
Opacity model (runtime)	all (via `opacity.inp`)	A=analytic above (default); C=constant above threshold, σ = `OPA_CONST_`X times the analytic value at the edge; P=that constant base times the pressure-broadening factor (Robinson & Catling); T=tabulated files
Collisional (electron-impact) ionization		
Rate $C(T)$	H I, He I, He II	Voronov (1997, ADNDT 65, 1) Table I default; legacy ATES fits (Abel+1997 / Hui & Gnedin 1997) via `Legacy_HHe_rates`
	C–Fe, all stages	Voronov (1997) Table I, each row with Voronov's own fitted ΔE; the fits are stated for 1 eV–20 keV (3 eV He II, 2 eV N II, O II) and are evaluated below that range with their exact e^{-U} threshold factor and no accuracy claim

Table 7, continued

Process	Species	Source
	He $2\,^3$S	Black (1981, MNRAS 197, 553) eq. 12, $8.38 \times 10^{-10} T^{1/2} e^{-55338/T}$
Recombination (radiative RR + dielectronic DR)		
Ground capture α_1	H I, He I, He II	the Milne relation on the ground-state photoionization cross section the transfer uses, so the capture and the absorption of its photon are one process (0.998 of Hummer 1994 for H I, 1.006–1.011 of Hummer & Storey 1998 for He I, as the code comment records them)
Case B	H II, He II, He III	Badnell (2023) RR (+ Badnell DR for He II) minus that α_1 default; legacy Hui & Gnedin (1997) fits via `Legacy_HHe_rates`; Koskinen et al. (2022) Table 1 R1, R2 via `Atomic rate set`
Case A where thin (def. on, `H_rec_escape`, `He_rec_coupling`)	H II, He II, He III	the balance coefficient is $\alpha_B + y\,\alpha_1$, y the fraction of the ground-capture photons the recombined species does not re-absorb in the cell (case A in a thin cell, case B in a thick one); `False` = case B
Recombination radiation absorbed on the spot (def. on, `He_rec_coupling`)	He II $\rightarrow$ He I, He III $\rightarrow$ He II	He II: the case-B captures leave through their own exits (the 584 Å line, the 19.8 eV $2\,^3$S line, the $2\,^1$S two-photon continuum of Drake, Victor & Dalgarno 1969), shared by the network's rates as in Draine (2011, sect. 14.3.2, 15.5) and Osterbrock & Ferland (2006, sect. 2.4); He III: the ground capture, the direct capture into $n = 2$ (Seaton 1959), He II Lyα and the $2s$ two-photon continuum (Nussbaumer & Schmutz 1984; $A(2s) = 526.823\ \mathrm{s}^{-1}$ of Drake 1986), with the $2s$ share of Pengelly (1964) and the $2s$–$2p$ ion mixing of Pengelly & Seaton (1964). Every photon is shared among H I, He I, He II, H_2 and the metal ions by their opacity in the cell; what leaves the cell is dropped

Table 7, continued

Process	Species	Source
He I 584 Å resonance line	He I $2\,^1\mathrm{P} \rightarrow 1\,^1\mathrm{S}$	scattered by He I in the cell and converted into the $2\,^1\mathrm{S}$ two-photon continuum with probability $1/912.0$ per scattering (NIST A values), a flight ending in continuum absorption, line scattering or escape, after Wood, Mathis & Ercolano (2004, sect. 4.1); in a cell thick in line and continuum the continuum share tends to their eq. (17) with every continuum absorber in place of H I. Doppler mean cross section, no frequency redistribution, and no electron-collisional $2\,^1\mathrm{P} \rightarrow 2\,^1\mathrm{S}$ transfer (validity stated at the code)
Metastable	He II $\rightarrow 2\,^3\mathrm{S}$, He II $\rightarrow$ excited singlets	case B split into triplet and singlet captures with the triplet share of Hummer & Storey (1998) Table 5 (0.771 at 10^4 K), so ground + singlets + triplets = the Badnell α_A
Electron collisions	He $1\,^1\mathrm{S} \leftrightarrow 2\,^3\mathrm{S}$, $2\,^3\mathrm{S} \rightarrow 2\,^1\mathrm{S}$, $2\,^1\mathrm{P}$	$8.629132 \times 10^{-6}\,\Upsilon\, e^{-E/kT}/(g\sqrt{T})$ with the Bray et al. (2000) effective collision strengths (their table inside $10^{3.75}$–$10^{5.75}$ K, the CHIANTI 11.0.2 he_1 temperature dependence joined continuously outside it) and NIST level energies; the $2\,^3\mathrm{S} \rightarrow 1\,^1\mathrm{S}$ de-excitation q_{31g} is the detailed-balance reverse of the excitation
	He $1\,^1\mathrm{S} \rightarrow n\,^3L$, $2\,^3\mathrm{S} \rightarrow n\,^1L$ ($n \geq 3$)	the excitations into the higher triplets cascade into $2\,^3\mathrm{S}$ and are added to q_{13} (0.019, 0.107, 0.331 of it at 5×10^3, 10^4, 2×10^4 K); the excitations of $2\,^3\mathrm{S}$ into the singlets above $2\,^1\mathrm{P}$ destroy it and are added to q_{31b}; both are coronal sums over the Bray et al. (2000) terms $n \leq 5$
Penning ionization	He $2\,^3\mathrm{S} + \mathrm{H}$	García Muñoz (2025) Maxwell average of the Movre & Meyer (1997) cross sections (always on); the total quenches the metastable, 0.9 of it is Penning ($\mathrm{He} + \mathrm{H}^+ + e^-$) and 0.1 associative ($\mathrm{HeH}^+ + e^-$)

Table 7, continued

Process	Species	Source
	He $2\,^3S + H_2$	García Muñoz (2025) network rows (Cohen & Lane 1977 cross sections; 500–10 000 K), split 0.9 Penning (H_2^+) : 0.1 associative ($H + HeH^+$). Active only when a molecular run also tracks the triplet
RR + DR	C, N, O, Mg, Si, Na, K, S, Ca II (+ stages)	Badnell (2006, ApJS 167, 334) RR + Badnell adf48 DR (TAMOC clist_K)
RR only	Ca I (Ca II+e)	Shull & Van Steenberg (1982); no DR
RR + DR	Fe I, Fe II	Huang et al. (2023) analytic fit; the Badnell DR project does not cover these isoelectronic sequences (see the note below the table)
Charge exchange (charge transfer)		
metal + H/H^+ (default)	group A: the 23 rows of Huang et al. (2023, ApJ 951, 123) Table 4, 19 carried	the reaction list of that table, each rate from the original source where it could be read (Kingdon & Ferland 1996 and the calculations behind it: Allan et al. 1988, Butler & Dalgarno 1979, 1980, Neufeld & Dalgarno 1987, Gargaud et al. 1982; Kimura et al. 1996; Stancil et al. 1998, 1999; Zhao et al. 2005; Rutherford & Vroom 1972; Dutta et al. 2001; Watanabe et al. 2002). The direction the source calculated or measured is carried as published; the other is computed from it by detailed balance with NIST internal partition functions. Four reverses (Mg^+, Fe^+, Si^+, $Na^+ + H$) are not carried, their forward ending in an excited or radiative product
density-dependent reverses	$N + H^+$, $S^+ + H$, $C + S^+$	for the three forwards whose sources name their product levels, the reverse is the thermal one times the product populations of a five-level statistical equilibrium of N I, S II and C I at the cell's (T, n_e) (CHIANTI 11.0.2 A values and electron collision strengths); it equals the thermal reverse at the collisional limit

Table 7, continued

Process	Species	Source
He + H/H$^+$ (default on, `He_H_charge_exchange`)	group B and He^{2+} + H	He$^+$ + H: radiative (Stancil, Lepp & Dalgarno 1998, Table 1 row 19, $1.20 \times 10^{-15}(T/300)^{0.25}$, from Zygelman et al. 1989) + non-radiative (Loreau, Ryabchenko & Vaeck 2014, by reciprocity, with $\sigma_r \propto \sqrt{(x-x_{\rm th})/(15-x_{\rm th})}$ between the threshold $x_{\rm th}$ = 13.655 and 15 eV/u where their calculation starts); He + H$^+$: detailed balance of the non-radiative channel; He^{2+} + H $\rightarrow$ He$^+$ + H$^+$ + $h\nu$: West, Lane & Cohen (1982), Maxwellian average of their cross section, no reverse. Applied by dedicated routines in every helium-bearing system, *not* gated by `cx_full`
full network (`cx_full`)	groups C, D: metal+He/He$^+$, metal+metal	Table 4 reaction list with the originals where read (Kimura et al. 1993; Zhao et al. 2004; Satta et al. 2013; Prasad & Huntress 1980; Rutherford & Vroom 1972; Chenel et al. 2010; Kwolek et al. 2019; Lavvas et al. 2014), reverses by detailed balance; C$^+$, O$^+$, Si$^+$ + He and Fe$^+$ + O, Fe$^+$ + N, Na$^+$ + Ca are not carried; C + Si$^+$ as printed
O^{2+}, N^{2+} + H (default on, `cx_O2p_H`, `cx_N2p_H`)	group E, not in Table 4	Barragán et al. (2006) Tables 1 and 2, interpolated over 10^2–10^5 K; no reverse; not gated by `cx_full`
heat (channel `heat_charge_exchange`)	every applied reaction	the energy defect less the excitation of a product state that leaves as radiation (product states from the sources read; Fe$^+$ products in the near-resonant z^{4}F/z^{4}D terms)
Radiative cooling		
recombination	H II, He II, He III, metal ions	the kinetic energy of the captured electrons, $kT\alpha(3/2 + {\rm d}\ln\alpha/{\rm d}\ln T)$, with the coefficient α the ionization balance uses (for H II, He II and He III the net coefficient $\alpha_B + y\alpha_1$, at the frozen escape weight; for the metals their RR+DR coefficient); checked against Hummer (1994)

Table 7, continued

Process	Species	Source
coll. excitation	H I	coronal sum over the CHIANTI 11.0.2 h_1 levels (Anderson et al. 2000 collision strengths), observed energies, the $1s \rightarrow 2s, 2p$ rates the H(n=2) model uses
	He I	coronal sum over the 28 terms $n \leq 5$ of Bray et al. (2000); with the He $2\,^3$S tracked, the $1\,^1\mathrm{S} \rightarrow 2\,^3$S channel is charged with the network's own excitation and its detailed-balance reverse (a net that turns negative where the metastable is overpopulated), plus the triplet cascade cooling of $2\,^3$S (Bray terms $n \leq 5$) and the thresholds of its excitations into the singlets
	He II	coronal sum over the CHIANTI 11.0.2 he_2 levels (Ballance 2003)
coll. ionization, free–free	H, He, metals	the measured ionization potential per event (He $2\,^3$S and the metal ions included); free–free bremsstrahlung with the van Hoof et al. (2014) thermal Gaunt factor at $\gamma^2 = Z^2\,\mathrm{Ry}/kT$, Ry the infinite-mass Rydberg (their eq. 21 and sect. 2; $\mathrm{Ry}/k = 157887.5$ K), weighted by ion net charge
CHIANTI descaling	every CHIANTI-based coefficient	the Burgess–Tully scaled collision strengths descaled with the natural cubic spline of CHIANTI's `descale_scups.pro`, as CHIANTI prescribes

Table 7, continued

Process	Species	Source
H_3^+ infrared (molecular runs)	H_3^+	LTE emission per molecule from the MiZATeP cooling function (Mizus et al. 2017; ExoMol `1H3_p__MiZATeP.cf`, 20–5000 K at 1 K nodes, $\ln W$ linear in $1/T$) times the Table 6 non-LTE departure factor $s(T, n_{H_2})$ of Miller et al. (2013), interpolated as a power law in $n(H_2)$ between the columns and linearly in T; optically thin. The Miller et al. Table 5 LTE fit, 1.1–2.1 times above MiZATeP at 900–3000 K, was removed on 2026-10-10. The dominant coolant of a molecular base, where every atomic channel above is exponentially suppressed
metal line cooling	C I/II, N I/II, O I/II	the ground term in statistical equilibrium (row below) plus the channels that leave it from (T, n_e) statistical-equilibrium tables of CHIANTI 11.0.2 (default); legacy AIOLOS fits via `cno_cool 0` (no N cooling)
	Mg II	closed-form fit (0.2%) to the coronal sum over the 32 bound levels of the CHIANTI 11.0.2 model with the observed level energies
	Mg I, Ca II	(T, n_e) statistical-equilibrium tables of CHIANTI 11.0.2 carrying every channel of the ion
	Na I D	Van Regemorter with effective $\bar{g} = 0.2$ (not in CHIANTI)
	Fe I	NIST f-values + Van Regemorter, lower levels Boltzmann-populated: the high-density (LTE) limit, an upper limit at most 2.0 times the low-density value
	Fe II	(T, n_e) multilevel statistical-equilibrium table (CHIANTI)
	all (T, n_e) tables	above their top density the last slope in $\log n_e$ is carried on, clipped to $[-1, 0]$

Table 7, continued

Process	Species	Source
ground-term fine structure (statistical equilibrium)	C I $2p^2\,{}^3P$, C II $2p^2P$, N II $2p^2\,{}^3P$, O I $2p^4\,{}^3P$	the split ground term is solved in exact statistical equilibrium at the local $(n_e, n_{\rm H\,I})$ instead of the low-density coronal limit, which overstates these lines by 2.9–8.3 decades at the base. Electron collision strengths from CHIANTI v11.0.2; H-atom rates from Yan & Babb (2023) for C I/N II, Abrahamsson, Krems & Dalgarno (2007) for O I, Barinovs et al. (2005) for C II. The complete set: the other coolants have a single ground level, are built from permitted lines (Fe I), or already use a statistical-equilibrium table (Fe II)
line trapping	the eight ground-term fine-structure lines	line-center escape probability from the column above the cell, Hollenbach & McKee (1979) / de Jong, Boland & Dalgarno (1980) plane-parallel Doppler form renormalized to $\beta(0) = 1$; applied as $A_{ul} \to \beta A_{ul}$ *inside* the statistical-equilibrium solution, the only placement correct in both the low- and high-density limits. Other lines, including Mg II h&k, are treated as optically thin
validity floor of the coronal fits	all CHIANTI-derived coefficients	the coronal part is cut off smoothly below 10^3 K (the lower end of the fitted range), continuous in value and dT-slope and bit-identical above it; only the explicitly solved ground-term fine-structure terms survive below. The roll-off is $\exp(-x^2)$ with $x = (T_{\rm floor}/T - 1)/w$ and $w = 0.1$ by default, settable as `Coronal cutoff width:`. With the C/N/O ground-term fine-structure floors saturated no coronal cooling survives below the floor for the guard to remove, and the base-cell balance temperature of all four paper planets is identical over $w = 0.02$–1.2. Not applied to the legacy AIOLOS branch (`cno_cool 0`)
Photoelectric heating		

Table 7, continued

Process	Species	Source
secondary ionization (def. on, `Secondary_ionization`)	H I, He I, H_2	Shull & van Steenberg (1985) $f_{\rm ion}(x)$ fits with the Dalgarno, Yan & Liu (1999) energy dependence, and Dalgarno et al.'s heating fraction $f_{\rm heat}(x, E_0)$, H_2 target channel and molecular heat returns; photoelectrons with $E_0 > 30\,$eV drive secondary H I / He I / H_2 ionizations, heating scaled to $f_{\rm heat}$; the He 2^3S photoelectron and the metal photoelectrons and Auger electrons feed the same heating and secondary-source budget, and so do the photoelectrons of the recombination radiation absorbed on the spot
photoelectrons of 10.2–30 eV (opt-in, `Low-energy electron partition`)	H I, He I	Furlanetto & Stoever (2010) Monte Carlo tables of $f_{\rm heat}$ and of the H I and He I ionizations per primary, primordial He/H; the H I line photons leave the cell; stellar and recombination photoelectrons; cells whose solved state holds no H_2 only; default: deposited whole as heat
Excited hydrogen H(n=2) and Lyα		
$2s/2p$ rate equilibrium	H(n=2)	the level balance of Christie, Arras & Li (2013, eqs. 12–13); $A(2p \to 1s) = 6.2649 \times 10^8\,{\rm s}^{-1}$ (NIST), $A(2s \to 1s) = 8.22461\,{\rm s}^{-1}$ (Drake 1986, eq. 27); $1s \leftrightarrow 2s, 2p$ collisions from the CHIANTI 11.0.2 h_1 strengths and their detailed-balance reverses; $2s$–$2p$ mixing by electrons, Seaton (1955) eq. (55), and by H^+, He^+ and He^{2+}, Pengelly & Seaton (1964) with their cut-off; in a molecular run the H_2^+ and HeH^+ dissociative recombinations as an n=2 source
recombination cascade to n=2	H	the case-B coefficient of the balance, split with the $2s$ fraction of Pengelly (1964, Table I)
Balmer continuum	H(n=2)	$\sigma_2 = 1.4 \times 10^{-17}\,{\rm cm}^2$ at the $n = 2$ edge (Seaton 1959, 1.378×10^{-17} with his Gaunt factor), falling as E^{-3}, over the run's own spectrum from $e_{\rm th,HI}/4$ to $e_{\rm th,HI}$

Table 7, continued

Process	Species	Source
Lyα pumping field (mode 0)	H	Huang et al. (2017) Eq. 6
Transmission line data (`EXHALE_transit.py`)		
λ, f-value, A_{ki}	He 10830, Lyα, Hα, Hβ, Mg II, Ca II H&K, Na I D, O I 1302/1304/1306	NIST ASD
line profile	all	Voigt (Faddeeva, `scipy.special.wofz`)

Note on Fe. Iron recombination uses the Huang et al. (2023) analytic RR+DR fit while every other metal uses the Badnell tabulation, and this is not a deferred item: no Badnell fit exists for either stage. Fe I is formed by recombining Fe II (Mn-like, 25 electrons) and Fe II by Fe III (Cr-like, 24), whereas the Badnell dielectronic-recombination project reaches the phosphorus isoelectronic sequence (15 electrons) as of its paper XVI (2022). Ca I has the same problem one sequence lower (K-like, 19 electrons) and likewise keeps a non-Badnell rate. **Note on completeness.** The metal grid carries the stages needed for the XUV-driven ionization balance and the observed NUV/optical lines: three stages of C, N, O, Mg, Si, Ca, Fe and two of Na, K, S. Twelve of the 27 carry a line-cooling channel (C I, C II, N I, N II, O I, O II, Mg I, Mg II, Ca II, Na I, Fe I, Fe II); the others (the doubly ionized stages, Si I, Si II, Ca I, Na II, K I, K II, S I, S II) cool only through recombination, collisional ionization and free–free emission, where each applies. Higher stages and additional elements are not modeled.